WHAT DO TECHNOLOGY GRANTS DO?*

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Abstract

We study how EU technology grants affect labor demand and skill bias in Finnish small and medium-sized manufacturing firms from 1994 to 2018. The subsidies funded new machinery, including robots. Comparing close grant winners and losers, we find that subsidized investments raised employment by 23 percent, with no detectable change in the skill mix or the labor share. We develop a method to match firms using machine learning on grant evaluation texts. Analyzing firms' stated plans and later outcomes, we observe that the technology grants primarily supported expansion—new products and markets—rather than automating work. Outside the program, IT is strongly associated with skill upgrading, while machinery is not.

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I Introduction

Industrial policies that support technology investment are common but controversial. One concern is that technology subsidies encourage automation, which could displace less-skilled workers and increase inequality.[1] This concern is widespread—the view that robots and other new technologies displace workers has received significant attention (The Economist, 2022). But not all capital investments are alike. A contrasting view is that these subsidies may help firms expand—scaling up, entering new markets, launching new products—and that worker outcomes depend on how that expansion unfolds. Despite substantial public spending on technology adoption, evidence on how subsidized investments affect workers and skill demand remains remarkably limited, with a few notable exceptions (Becker et al., 2010; Criscuolo et al., 2019; Curtis et al., 2022).

This paper assesses the effects of technology subsidies on labor demand and skill bias in the context of one of the world's largest industrial policies, the European Structural and Investment Funds. We focus on the ELY program, which subsidizes technology investments in Finnish small and medium-sized manufacturing firms. This program funds new machinery—for example, robots and computer numerically controlled (CNC) machines. We present direct evidence on labor demand effects at the firm level from 1994 to 2018 and explore mechanisms that help explain what happens, what does not happen, and why. Comparing close grant winners and losers, we find that subsidies increased employment by 23 percent with no detectable change in the skill mix. Our micro-level evidence and text analysis suggest that, even when the grants were used to acquire advanced manufacturing technologies, they primarily supported expansion rather than directly automating work.

The ELY Center technology subsidies, the policy we study, provide direct grants to firms investing in technology. These grants lower the cost of new technology investments but let firms decide which technologies to adopt. We develop a new method using the text of the grant evaluations to determine "close winners" and "close losers" of the grant. These firms had similar assessments, but only some received the subsidy. We use machine learning to turn the evaluation texts into firm-level propensity scores for winning the grant, measuring how close each firm came to approval. We also contrast winners to matched non-applicant control firms.

We merge administrative records, surveys, and field observations, creating a high-quality dataset on technologies and workers. These data track firms' technology choices and workforce composition before, during, and after applying for subsidies. For example, we document firms' adoption of robots, CNC machines, and IT, as well as employees' education, cognitive skills, and personality. Text data from the subsidy program enable us to observe why firms applied and how they planned to use the investments. Field visits and interviews with CEOs, managers, shop-floor workers, and grant officials clarify the context. This combination of administrative data and text analysis provides a direct view of how technology subsidies work and whom they benefit.

The first part reports our core findings on employment and skill mix. Receiving a technology grant led to a 23 percent increase in employment, but we find no detectable effects on the skill

[1] This concern reflects the ideas of labor replacement and skill-biased technological change (Acemoglu and Restrepo, 2020; Katz and Murphy, 1992). Existing research suggests that new technologies, such as computers and IT, have replaced workers' tasks and favored skilled labor (Autor et al., 1998; Akerman et al., 2015).

mix—measured by the share of highly educated workers, years of education, or the share of production workers. We also find no effect on the labor share. These results contrast sharply with the concern that technology subsidies would reduce employment or lead to skill-biased changes. In practice, trained machinists and skilled welders were needed before and after the subsidy. The firms did not employ relatively more educated workers after the subsidy-induced investments. Looking more closely at detailed skill measures—education and occupation groups, cognitive performance, and personality—we generally find no effects.

Several observations support the validity of our findings in this context. The subsidy program induced a strong first stage: firms experienced a sharp rise in technology investments after receiving subsidies. Treated and control firms followed similar pre-trends in investment, employment, and skill mix before applying. The results are also consistent across two alternative designs: a regression discontinuity (RD) design based on changes in the criteria for small firms, and an event-study design of investment spikes outside the subsidy program.

The second part explores firm-level mechanisms behind these effects. Our fieldwork suggests two stylized views of how firms could respond to subsidies. One is that subsidies push firms to automate, replacing human tasks, lowering costs, and raising productivity. Previous work argues that this could displace workers, reduce labor shares, and shift the skill mix if low-skill or routine tasks are automated (Acemoglu and Restrepo, 2018a,b; Autor et al., 2003). The other is that subsidies help firms expand—scaling up or shifting into new activities—without necessarily increasing productivity or fundamentally changing the production process. If firms use the subsidies to expand, the effects on workers are uncertain, but this could result in increased revenues, new products, and even new tasks and work (Acemoglu and Restrepo, 2018b; Autor et al., 2024). In short, the micro-level effects of technology subsidies are not obvious from theory alone. Firms may choose to use subsidies to automate, expand, both, or neither.

Next, we present evidence on firm performance. We find strong positive effects on revenue but no detectable changes in productivity, profit margin, or capital intensity from the subsidy program. This evidence is consistent with the idea that these firms tended to expand rather than automate tasks. Zooming in, our detailed evidence shows that subsidies led firms to export, introduce new products, discontinue old ones, and raise prices and marketing expenditures. These new products and markets were as skill-intensive as the existing ones.

Text data from subsidy applications reveal how firms planned to use the grants. We hand-code the texts into categories that reflect their stated motivations. We find that 74 percent of firms cited expansion, while only 14 percent mentioned work-related motives. Common motivations included introducing new products, improving quality, and increasing capabilities or flexibility. Among the few firms that aimed to automate, we find some evidence of increased demand for skills. Fieldwork provides context for these observations. For example, one piston manufacturer invested in a robot and a CNC machine to produce a new piston variant, not to replace workers. Our sample firms mainly used the subsidies to invest in machinery rather than IT.

Our findings are grounded in a specific policy and context, raising the question: What local average treatment effect (LATE) do our estimates capture? One view is that financial frictions

limit firms' ability to adopt new technologies, and EU subsidies lower these barriers, enabling large investments. Another view is that these firms already have sufficient resources, and subsidies fund incremental investments with limited productivity impact. Our evidence is more consistent with the latter view. The average grant size (EUR 80K) points to discrete machinery investments rather than a wholesale redesign of production. These firms did not move from no technology to full automation—they already had some in place. Our estimates capture local effects for firms close to the margin of investment—those induced by the subsidy—but not all firms (Zwick and Mahon, 2017; Lileeva and Trefler, 2010). We find limited evidence that credit constraints, employment biases, or signaling drive our results.

We focus on flexible manufacturing, where small and medium-sized enterprises (SMEs) produce specialized, low-volume products to meet shifting demand. This setting stands apart from mass production. Although automation and skill bias are central to the literature, not all technology investments displace labor tasks or favor more skilled workers (Goldin and Katz, 1998; Autor et al., 2024). Piore and Sabel (1984) argue that in flexible manufacturing, new technologies often support expansion and product variety alongside skilled labor, rather than automation (Milgrom and Roberts, 1990). Focusing on this context is both a strength and a limitation: the results may not generalize to all settings, but flexible manufacturing represents a substantial share of industrial activity (Berger, 2020).

The third part steps outside our quasi-experimental setting to place the findings in a broader context. Why did subsidized machinery investments not lead to the skill-biased effects documented in prior work (Autor et al., 1998; Akerman et al., 2015)? We explore three possibilities: (1) the effects operate between firms rather than within them, (2) machinery differs from IT, and (3) skill demand in Finland differs from contexts studied before. The evidence points to a sharp distinction: IT is consistently associated with skill upgrading, while machinery is not.

At the macro level, subsidies could raise skill demand within firms or through compositional changes across them. Comparing subsidy applicants to non-applicants, we find that applicant firms had lower labor shares and somewhat more educated workers before applying, and grew faster afterward. This evidence is consistent with compositional effects that would increase aggregate skill levels and reduce labor shares. However, the skill differences are modest—less than 0.1 years—limiting the scope for these effects.

Expanding the analysis beyond our subsidy sample to all Finnish manufacturing firms, we find that machinery investments are moderately—but not strongly—more common in firms whose workforces are more educated. At the industry level, industries investing more in machinery show no faster skill upgrading.

But IT tells a different story. In firms outside our subsidy sample, IT is concentrated in more educated firms. At the industry level, industries that adopted more IT have significantly increased their share of skilled workers. These findings align with Autor et al. (1998) and Berman et al. (1994).

Taken together, the skill-neutral effects of technology subsidies may reflect the types of investments they supported. The program primarily funded machinery, with limited investment in IT. This contrast could help reconcile our findings with earlier studies documenting skill-biased effects

from IT. However, because the program focuses on machinery rather than IT, there is limited scope to compare the two within our design.

The broader backdrop is a shift toward greater skill demand in Finnish manufacturing, reflected in the rising share of educated workers. This trend matches global patterns, such as in the US (Acemoglu and Autor, 2011).

Our work contributes to the literature on firm subsidies (Becker et al., 2010; Criscuolo et al., 2019; Curtis et al., 2022).[2] The key question is how subsidized technology investment affects labor demand (quantities) and skill demand (upgrading/downgrading), and why it has these effects. Our distinct contribution is direct firm-level evidence on how technology subsidies affect employment and skill mix, and we explore the mechanisms that generate these effects. To this end, we develop a new method to match program participants using machine learning on evaluation texts. We assemble linked data on firms, technologies, and workers, and strengthen our analysis with an RD design and an event-study of investment spikes outside the program. This micro-level evidence and text analysis enable us to document how firms actually use these grants. We find that subsidy-induced machinery investments increased employment without shifting the skill mix. The overall evidence is consistent with the view that these firms used the grants primarily for expansion rather than automation.

This research also relates to the literature on technology and work. A central idea in this literature is that technology shapes skill demand (Autor et al., 2024). Our findings—that technology grants increased employment without shifting the skill mix—are consistent with recent evidence on subsidized capital investments in the UK and US (Criscuolo et al., 2019; Curtis et al., 2022), earlier results on factory-floor machinery (Doms et al., 1997), and recent work on manufacturing capital (Aghion et al., 2025). They differ, however, from studies showing skill-biased effects from IT adoption (Akerman et al., 2015; Gaggl and Wright, 2017). While several factors may reconcile this contrast, our evidence points to one observable difference: the program funded machinery rather than IT. Our results also contrast with studies where automation directly replaced workers' tasks (Feigenbaum and Gross, 2024; Bessen et al., 2025). The evidence indicates that automation of this kind was uncommon under this subsidy program. Our research provides direct evidence on a prominent industrial policy and informs the design of future policies that tax or subsidize technology adoption, such as robot taxes (Acemoglu et al., 2020a; Costinot and Werning, 2023).

[2]Beyond these papers, our research relates to the literature on business and capital taxes (House and Shapiro, 2008), R&D subsidies (Howell, 2017; Einiö, 2014, for Finland), historical cases of industrial policy (Lane, 2025), and place-based policies (Kline and Moretti, 2014).

II Moore's Law for Pistons

We conducted fieldwork that led us to an industrial piston manufacturer. Pistons move up and down inside combustion engines, converting fuel into motion.

The company invested in a new CNC machine, a robotic arm, a measurement device, and computer-aided design (CAD) software. When we asked why it adopted these technologies, the CEO emphasized a central theme in piston manufacturing: constant quality improvement. "With the old technologies, we couldn't make these pistons," he said. Figure G1 illustrates the rise in piston quality over the last 100 years. The firm called this trend "Moore's Law for pistons."

The new technology enabled the production of larger, lighter, and more efficient pistons, helping the firm stay competitive and expand its revenue and employment. The company operates in niche markets where demand for each specific product is limited. Because it cannot significantly expand by increasing production of existing products, it attempts to grow by introducing new ones—often incremental variations of its current offerings. The firm's primary motivation for adopting technology was to make new pistons, not to replace workers.

The technology investment led to "small but important changes" in production processes, and the robotic arm automated certain tasks. The adoption required workers to acquire new skills: production workers learned to operate the CNC machine and robotic arm, and the R&D team learned to use the latest CAD software. Despite these changes, there was no material shift in the formal educational mix of the workforce.

III Winners–Losers Design

We estimate how EU technology subsidies affect firm employment and skill mix. The subsidies temporarily lower the cost of investment for recipient firms. Our event study tracks close winners and losers before and after the award. Its identification logic parallels that in Angrist (1998), Greenstone et al. (2010), and Kline et al. (2019).

A novel aspect is the use of text data to compare close winners and losers. To do so, we use evaluation reports written by the program officers. We map these report texts into propensity scores that reflect the likelihood of receiving a subsidy and control for the scores to compare similar winners and losers. We also compare subsidy winners with matched non-applicant firms.

The Online Appendix presents two complementary designs: (1) an RD design based on a change in the threshold that determines a priority for small firms (to address internal validity), and (2) a spikes design based on the precise timing of technology adoption events outside the program (to explore external validity).

III.A ELY Program

We analyze the effects of technology subsidies in manufacturing firms in Finland, 1994–2018.

The technology subsidy program is administered in Finland by the Centers for Economic Development, Transport and the Environment (the ELY Centers). These centers promote regional

business policy through various activities, including advisory, financing, and development services. Technology subsidies are part of a service called Business Development Aid. The service provides funding for technology adoption, export promotion, R&D, and several smaller categories, such as starting a new company. Based on the subsidy records, the service granted EUR 2 billion over 1994–2018 and directed EUR 758 million toward technology subsidies. Technology subsidies were, on average, 0.7% of machinery and equipment investment in Finland. This paper is the first quantitative evaluation of the program.

The program is part of the European Structural and Investment Funds (ESIF), one of the world's largest industrial policy programs. ESIF aims to promote economic development across all EU countries, especially in remote regions. The 2014–2020 program granted EUR 535 billion from the EU budget (European Commission, 2023). In the Finnish context, the national government and the EU fund technology subsidies together, typically 50/50. Decisions are made locally by the ELY Centers. This study speaks to the firm-level effects of the broader EU program.

The technology subsidies aim to promote the adoption of new technologies. The broader agenda behind this objective is to improve firms' competitiveness. Technology subsidies in Finland have a long tradition based on the idea that the government can foster growth and structural change through industrial and regional policy. The program follows the EU's technology neutrality principle from the bottom up—firms can choose their technology as long as it is new—and is not primarily about the direction of technology, for example, automation vs. non-automation.

A typical technology subsidy is a EUR 80K cash grant, reimbursing 15–35% of technology costs. Subsidized investments are primarily new production technologies for manufacturing: new CNC machines, robots, laser cutters, and enterprise-resource-planning (ERP) and computer-aided-design (CAD) software. The most common industries are fabricated metal products and machinery. The firms are mostly small and medium-sized enterprises (SMEs), often contract manufacturers producing specialized intermediate goods in small batches, such as pistons for engines, sold to large exporters. Workers are primarily production employees (median share: 70%), for example, machinists, welders, and machine operators, who are typically vocationally trained. Grants are paid by ELY Centers as reimbursements for verified costs, and approximately 30% of recipients are audited. Figure G2 illustrates typical technologies, firms, and workers in our sample.

The selection process typically works in three stages, outlined in Figure A1.

1. Application. From the universe of firms, some apply for technology subsidies. This means we compare firms that all planned a technology investment. Firm that do not apply typically (a) do not plan to invest, (b) do not know about the program, (c) anticipate they are not eligible, or (d) consider the opportunity cost higher than benefits.

2. Pre-screening. In the pre-screening stage, firms contact ELY Centers before submitting formal applications. This stage is helpful for our design: after pre-screening, the centers' goal is that all firms have a realistic chance of winning the subsidy. The coarse evaluation criteria are size, industry, and general economic position. The program targets manufacturing SMEs not owned by large firms, not in financial difficulties, and able to carry out their technology plan.

Firms may skip this stage, but skipping does not improve their chances of winning and can produce rejections of otherwise high-performing firms—for example, firms that are not SMEs.

3. Decision. Firms submit a formal application explaining the investment and timeline. Funding is discretionary. Subsidy winners are selected based on the program rules and local and temporal budget priorities and constraints, and an identical firm could receive a subsidy in one year but not another. ELY Centers do not score the applications on a formal scale, but we use the evaluation reports to match applicants. In the decision stage, ELY Centers re-evaluate the coarse criteria from pre-screening. The centers make an impact assessment to evaluate the effectiveness of the subsidy. Cases where the subsidy is more likely to have an impact are more likely to receive it. In addition, firms satisfying the criteria for small firms and firms in remote regions are prioritized. ELY Centers evaluate potential market distortions and sometimes reject applications if they suspect the subsidy negatively interferes with local competition. About 15% of applications are rejected.

We use text from application records to analyze the selection process. Records for approved applications typically explain how the project meets the program criteria, while records for rejected applications generally indicate the reasons for denial. Typical reasons for rejection are (1) effectiveness: the subsidy is not expected to change the project, the project is already underway or too small to meaningfully affect the firm, or a similar project was previously subsidized, (2) industry, size, and investment-type restrictions: the firm is not an SME, the industry is not supported, or the machinery is second-hand, (3) budget constraints, (4) technical issues: missing information or missed deadlines, (5) the firm's financial position, and (6) interference with local competition. Employment-related reasons do not appear as typical primary reasons for rejection; we discuss this aspect in Section VII.

Table A1 compares the main sample to all Finnish manufacturing firms. Technology subsidy applicants differ from non-applicants. The median subsidy-sample firm is larger (still an SME), more profitable, slightly more educated, and has a lower labor share than the median manufacturer. These differences suggest that non-winning applicants make a more credible comparison group than average manufacturers, since every applicant had already revealed a concrete intent to adopt new technology. Our estimates approximate the local average treatment effect (LATE) for firms close to investing in technologies; we return to this interpretation in Section VII.

We conceptualize the technology subsidy as a temporary price reduction. If a firm is close to the margin on whether to invest, a temporary price reduction might push it to invest. Firms and subsidy officers reported in our interviews that subsidies affect investment because they lower the price of technology, including the associated costs, future risk of debt, mental investment, and required courage. At the same time, they said that the moderately sized subsidies are unlikely to affect firms far from investing; instead, the grants are more likely to affect marginal investments. Appendix J clarifies the source of variation using a dynamic model adapted from Cooper et al. (1999).

III.B Baseline Design

Our main empirical strategy is an event-study design that contrasts similar firms, one of which was approved for technology subsidies while the other was not. The identification strategy is based on the idea that subsidy decisions are quasi-randomly assigned with respect to the counterfactual changes in firm outcomes after conditioning on the information used in the screening process. We assess the comparability of winners and losers and provide several alternative estimation strategies, including a matched non-applicant control group and matching with text data.

We estimate two types of equations. Our main specification is a stacked event study:

$$Y_{jt} = \alpha_j + \kappa_t + \sum_{\tau \in \mathcal{T}} \left[I_{jt}^{\tau} \cdot (\gamma_\tau + \beta_\tau \cdot D_j) \right] + X_{jt}^{\tau} + \varepsilon_{jt} \tag{1}$$

where Y_{jt} is an outcome for firm j in year t, D_j is the treatment indicator, I_{jt}^{τ} is the event-time indicator for firm j's decision having occurred τ years ago, and the set $\mathcal{T} = \{-5, -4, \ldots, 4, 5\}$ defines the five-year horizon over which we study dynamics. Our parameters of interest are the coefficients β_τ. They summarize the differential trajectory of mean outcomes for winning and losing firms by the time relative to their application. Event-time is defined for both winners and losers using their application year, and each firm remains in the treatment or control group for the entire panel.[3] Estimates before the event serve as a test of differential pre-trends between the treatment and the control group. The coefficients γ_τ capture the common event-time τ effects. The term α_j denotes the set of firm indicators, κ_t is the set of calendar-time t indicators, and X_{jt}^{τ} contains potential pre-period controls interacted with both time indicators (the main figures are reported without). We designate $\tau = -3$ as our base event period and omit it. We set the base clearly before the event to avoid contrasting the post-period with any anticipation effects. For clarity, we present all main estimates in reduced form (i.e., intention to treat, ITT).

To summarize the dynamic estimates into a single number, we estimate the stacked first-difference specifications:

$$\Delta Y_j = \beta \cdot D_j + X_j + \varepsilon_j \tag{2}$$

where ΔY_j is the change in the outcome from the base year $\tau = -3$ to the post period that we define in each context. The main regressor is D_j, an indicator for whether the firm won the subsidy. We also estimate continuous versions where D_j refers to the amount of subsidies. The control term X_j controls for potential differential trends across firm and application characteristics. We report heteroskedasticity-robust standard errors clustered by firm.

We report the event studies without additional controls. In the first-difference specifications, we control for 2-digit industry, firm size at $\tau = -3$, and calendar-time fixed effects. We later show that the results are robust to different controls.

We construct the analysis sample as follows. We first restrict to technology applications based on the text data (explained in Section IV). We then restrict to manufacturing and construction

[3] We focus on a never-treated control group to avoid biases from staggered adoption. Robustness to alternative DiD estimators is shown in Section V.

industries for three reasons: the program targets these industries, they produce physical outputs, and we have a concrete understanding of what their new technologies are based on our fieldwork. We exclude the largest 5% of applications because they tend to have poor control units. Event-time indicator $\tau = 0$ refers to the year the subsidy application was submitted. The treatment group is defined by selecting the largest approved subsidy application for each firm, and the control group is defined by the largest rejected application. Repeated applications for the same project are generally not allowed and rare in practice. Finally, we restrict to a balanced sample over the five-year horizon.[4]

The ideal experiment that could capture the causal effects of technology subsidies on employment, skill mix, and firm performance would randomly assign the subsidies to firms. Our identification strategy is based on the quasi-random assignment of technology subsidies, D_j. The identifying assumption is that treatment assignment is conditionally independent of the potential outcomes:

Assumption 1 (Rosenbaum and Rubin, 1983): $(\Delta Y_{1j}, \Delta Y_{0j}) \perp\!\!\!\perp D_j \mid X_j$,

where ΔY_{1j} and ΔY_{0j} are potential outcomes for the firm if it wins or loses a subsidy.

Our research design compares subsidy-receiving firms with firms that applied for subsidies but did not receive them. These comparisons are limited to applicants, so the contrast holds fixed selection into applying. But this restriction does not control for the program's own selection among applicants. We address this remaining selection by conditioning on observed inputs to the grant decision, including the evaluation report itself.

Table 1 reports summary statistics for the treatment and control groups. The groups are reasonably similar in terms of revenue, employment, and worker composition. The main differences are that the losing firms are smaller and applied for smaller subsidies. While our design is based on comparing short-term trends between the treatment and control, the pre-period level differences motivate our matching strategy in the next subsection.

An alternative counterfactual is similar firms that did not apply for subsidies. We use coarsened exact matching (CEM; Iacus et al., 2012) to define these similar firms. This matching strategy addresses the concern that the losing firms are not a reasonable counterfactual for what would have happened if the approved firms had not received the subsidy. We match by revenue, employment, and wages at $\tau = -3$; revenue and employment changes in percentages from $\tau = -3$ to $\tau = -1$; and industries' main sectors (letter classes). The CEM percentiles are 10, 25, 50, 75, 90, and 99. The match is 1:1 with replacement. We define matched control samples for both winning and losing firms; the latter is a placebo test. Tables B1 and B2 show the covariate balance for the matched samples. The matched control groups also serve to assess whether the patterns in the losing firms are typical or specific to losing applicants.

III.C Text Matching

We demonstrate a new method that uses text data to control for differences among applicants. The subsidy records contain a report written by the officer who evaluated the application. Given

[4] With ±5 years of follow-up, event years span 1999–2013 in data from 1994–2018.

similar reports, treatment assignment is more likely to reflect quasi-random variation than systematic differences. Text matching allows us to control for underlying characteristics captured in the reports (see also Roberts et al. 2020; Mozer et al. 2020).

Our main approach controls for propensity scores computed from the evaluation reports. The score is the predicted probability that the firm will win a subsidy, conditional on a text (W_j):

$$p(W_j) \equiv P\left[D_j = 1 | W_j\right]. \tag{3}$$

The propensity score theorem (Rosenbaum and Rubin, 1983) provides the formal basis for reducing the report to this scalar. Propensity scores are valuable here as a dimension-reduction tool, since directly controlling for texts is not feasible.

The subsidy records contain three types of texts that track the decision process: (1) description, (2) evaluation, and (3) decision. The descriptions and evaluations are written by a middle-rank officer responsible for administering the subsidy and presenting the case to a manager for a decision. We use the evaluations to compute the propensity scores, since they are most likely to capture potential differences between firms. Based on our interviews, the subsidy officers' goal is to present an unbiased evaluation. The evaluation text is available for 89% of the main sample.

The text propensity score method works in three steps.

1. We represent the text as data. We use a vector representation based on word embeddings. In particular, we employ the FastText library (Bojanowski et al., 2016) for the Finnish language. The vector representation aims to capture the semantic meaning of the text. This is helpful as we want to extract information beyond clear markers of success or failure.

2. We estimate the propensity scores. We use a stochastic gradient descent classifier and Platt scaling, fit by five-fold cross-validation. Figure 1 shows the calibration plot for our analysis sample: the text-based predicted probabilities closely match the empirical probabilities.

3. We control for confounders using the propensity score. Regression adjustment is our preferred approach. We compare the estimates to CEM and inverse probability weighting (IPW).

As an alternative, we use cosine similarity, which measures the overall textual similarity between evaluation reports without projecting them to a propensity score. Table A2 reports the summary statistics for the cosine-similarity matched sample.

We examine the text data in detail in Appendix H and present anonymized examples of the applications. Our main observation is that despite some differences, the evaluation texts appear relatively similar across propensity score levels above 0.60. One interpretation would be that the propensity scores fail to capture the key elements predicting acceptance. The evidence from the texts and our interviews with the program officials offers a more likely interpretation: many applications are relatively similar to one another, and the decisions are based more on idiosyncratic factors than on differences in evaluations. While this makes text matching less powerful in our context, it lends credibility to our research design more generally.

IV Data on Workers, Firms, and Subsidies

We link administrative records on workers, firms, and subsidies. Appendix F provides details.

IV.A Workers

Our data track all individuals in Finland over time, independent of their labor market status. We measure employment and wages from Statistics Finland's registers. We link these data to various skill measures: education (level and type), 9th-grade GPA, high-school exit exams, and cognitive and personality assessments from universal male conscription. Occupations are classified at the 3-digit ISCO level based on employment records. We construct occupation-level task measures from the European Working Conditions Survey (EWCS).

IV.B Firms

Our primary source for technology investments is the Finnish Financial Statement Register, which provides data on firms' machinery investments and IT expenditure. Machinery includes hardware such as CNC machines and robots. IT expenditure covers software and consulting services, excluding internal staffing costs. Statistics Finland compiles these figures from tax registers and through direct data collection. The dataset covers all major Finnish industries from 1994 to 2018.

We use four additional types of data on technologies. (1) Text data from the subsidy program enable us to investigate firms' motivations for seeking grants. We code each application into categories that describe firms' technologies (e.g., robots, CNC machines) and intentions (e.g., expansion, new products, quality). (2) Survey data from the EU Community Innovation Survey (CIS), the ICT survey, and the Etla survey give us a view of firms' technology plans and their environment. (3) Literature-based data from the SFINNO project, drawing on articles from 15 trade journals and linked to our sample firms, provide qualitative information on technology investments. (4) Customs data record 621 types of imported machinery, enabling a comparison between automated and non-automated technologies, building on Acemoglu and Restrepo (2022).

We measure firm performance—revenue, productivity, and profits—from the Financial Statement Register. We gauge productivity by revenue per worker and total factor productivity (TFP), calculated via a Cobb-Douglas production function and, for robustness, using the methods of Olley and Pakes (1996) and Levinsohn and Petrin (2003). We winsorize firms' monetary values at the 5% level and deflate them to 2017 euros using the Statistics Finland CPI. Export data, including destinations and the number of products, come from the Finnish Customs Register; we classify products at the 6-digit CN level and track new and discontinued items. We derive product prices from the revenue per unit sold, using information from the Customs Register and Industrial Production Statistics. R&D and marketing expenditure data come from the Financial Statement Register. Patent data come from the Finnish Patent Database.

IV.C Subsidies

The ELY subsidy dataset tracks the application process from submission to decision for both successful and unsuccessful applications. This confidential dataset offers direct firm-level measures that researchers have not systematically assembled before. The dataset includes unstructured text data, generated as part of the application process. We use these text data for three purposes: text matching (Section III), identifying technology subsidies (below), and measuring firms' intentions (Section VI). We track firms' other subsidies using Statistics on Business Subsidies.

This subsection details our method for identifying technology-related subsidies within the dataset, which includes 42,909 applications across categories such as technology, export, R&D, and startup grants. While these categories are routinely recognized by ELY Centers, they are not explicitly coded in the dataset and instead appear in the application texts. Our method follows a two-step process: We first manually code half of the applications to establish a benchmark, then use support vector machines (SVMs) to extend the coding to the remaining applications.

We code 21,210 randomly selected texts as technology-related or not. Our decision rule is simple: the text must explicitly mention a technology or technological advancement. We interpret technology in its everyday sense. Robots, CNC machines, laser cutters, CAD software, and upgrading production methods or modernizing production count as technology-related. In most cases, technology applications involve acquiring new machinery and equipment. While this could be a nuanced issue, in our context the decision rule is fairly straightforward; the answer is often clear from the application title. Table A3 provides anonymized examples.

We then use machine learning to scale the manual classification to the remaining 21,699 texts. We pre-process texts into a clean format, use the bag-of-words representation with TF-IDF weights, and employ SVMs for prediction. Our method achieves 95% accuracy in predicting technology applications across the full pool (Table H3). We manually re-check the applications in our main design. Figure H2 presents features that best predict being in the technology category. The top positive predictors include machine, line, device, cutter, CNC, and robot.

V Main Estimates on Employment and Skills

Core estimates show how technology subsidies affect employment and skill mix. The main result is clear: We find that technology subsidies raised employment with no detectable changes in the skill mix, wages, or the labor share. After winning a technology grant, firms invested sharply more in machinery and hired more workers, but did not alter their skill composition. Before receiving a technology subsidy, winners and losers had similar trends in investment, employment, and skill mix. Results are robust to conditioning on text propensity scores and additional covariates. The spikes and RD designs in Appendices C and D give similar results.

The First Stage Figure 2 shows the first-stage event-study estimates β_τ from Equation 1, with technology investment as the outcome, measured by machinery and equipment investment. Winning

a subsidy is associated with a sharp increase in technology investment. Before the subsidy application, the groups follow parallel trends. Figure A2 shows alternative first-stage estimates for all subsidies granted and received. The results indicate that winners and losers were granted different amounts of subsidies exactly in the event year $\tau = 0$, not before or after. The pattern for received subsidies matches technology investment. Table 2 reports first-stage estimates for the main winners–losers design, both with and without text matching. Outcomes include technology subsidies, technology investment, and capital. The first stage remains robust when controlling for the text propensity score. The dynamics suggest these investments were additive rather than shifted from later periods. We return to analyzing the specific types of investments in Section VI.

Employment, Wages, and the Labor Share Figure 3 displays the event-study estimates β_τ from Equation 1. The outcome is employment relative to the base period $\tau = -3$. The estimates indicate that technology subsidies led to approximately 20% higher employment in the five years after receiving the subsidy. As the figure shows, the employment pre-trends were similar between the treatment and control groups. Table 3 presents the first-difference estimates from Equation 2, with and without the text propensity control, and with the matched non-applicant control group. The specification with the propensity control indicates a statistically precise 23% increase in employment.

Another way to assess the potential replacement effects of advanced technologies is through the labor share—the share of revenue a firm pays workers. A declining labor share is a telltale sign of task automation (Acemoglu and Restrepo, 2018b). We find a precise zero estimate, reported in Table 3. We also generally find zero effects on wages; in some specifications, there is a small, statistically insignificant negative effect. We return to wage setting and local labor markets in Section VIII.

Figure A4 presents the effects on incumbent workers, those employed at the firm before the subsidy event, tracked regardless of later status. We generally find small effects: a 0.5% positive effect on employment anywhere but approximately the same-sized negative effect on the likelihood of staying in the treated firm. We find an income bump of EUR 500 at the time of application, potentially reflecting additional hours during technology adoption or rent-sharing from the grant.

The employment estimates are similar when using the matched non-applicant control group (Table 3 and Figures B1, B3), regression discontinuity design (Figure D4 and Table D4), and spikes design without subsidies (Figures C3, C5). The employment results are also robust to different text matching versions (Table A4) and different controls (Tables A5, A7), and the effects are clearly visible in the mean graphs comparing treatment and control groups over time (Figure A3).

Skill Composition Figure 4 displays the event-study estimates for the main firm-level skill measures: the average years of education, college-educated workers' share, and production workers' share. We find no change in these measures, either before or after the technology subsidy. Table 3 summarizes the estimates. Our 95% confidence interval excludes effects above 0.15 years of education. The results stand in contrast with the view that technology subsidies would necessarily increase the share of more educated workers and decrease the share of production workers in manufacturing firms. We observe that trained machinists and skilled welders were needed before and

after the subsidy receipt. The main skill-mix estimates hold in all of our research designs and are robust to a variety of controls referenced in the employment results, including text matching.

We zoom in on more detailed skill outcomes: education groups (Figure A6), occupation groups (Figure A7), cognitive performance (Figure A8), school performance (Figure A9), personality (Figure A10), demographics (Figure A11), and task composition (Figure A12). The big picture is that the effects are primarily skill neutral in the sense that the skill composition does not change. Another central observation is that the baseline skill levels of workers in the sample firms are well below the average. For example, average cognitive performance is 0.3 standard deviations below the population mean, and 9th-grade GPA 0.56 standard deviations below. The sample workers also score lower in tests designed to measure personality traits valued by the Finnish Defence Forces, such as achievement aim and dutifulness. We return to the role of baseline skills in Section VIII. Finally, some patterns in skill composition are consistent with our fieldwork, though not statistically significant and subject to multiple testing. The treatment effect on the 9th-grade GPA is +0.1 standard deviations (Figure A9), and the treatment effects on activity-energy, achievement aim, and sociability are approximately +0.05 standard deviations (Figure A10). The managers and workers we interviewed emphasized non-cognitive skills—initiative, cooperation, and adaptability—rather than higher education or non-production occupations, and noted that workers perform multiple tasks. One CEO said: "A company does not just pay a welder to weld."

Selection Bias A natural concern when estimating the impact of industrial subsidies is bias from a correlation between subsidy receipt and firms' potential outcomes. Because our comparisons are limited to applicants, the contrast holds fixed selection into applying: every firm in both groups had proposed a technology investment. We observe several inputs to the award decision, including the evaluation reports themselves. Our results are robust to conditioning on pre-application characteristics and evaluation-report text (Tables 3, A4, A5, A6, and A7).

To investigate whether the rejected applications are a reasonable counterfactual for the approved applications, we examined the approved and rejected applications in the analysis sample. We found ten rejected applications that seemed unlikely to receive subsidies in any situation: either the entrepreneur had a concerning history or the firm's financial position was unstable. Our results are robust to excluding these applications.

Text propensity scores enable us to narrow the sample to specific types of applications. A potential concern is that particularly good or bad applications, or undesirable comparisons across them, might drive our results. Table A8 addresses this concern by excluding applications with the highest and lowest 5%, 10%, and 20% propensity scores. The results are robust and consistently point in the same direction.

We also find similar effects when using a matched non-applicant control group (Table 3). As a placebo test, we contrast the losing firms to matched non-applicants (Figure B1). We find no first stage on investment and a small positive transitory effect on employment, indicating that the subsidy losers grew somewhat faster than similar non-applicant firms.

We employ three different research designs: (1) the winners–losers design, (2) a regression dis-

continuity design using unanticipated changes in the subsidy program rules (Appendix D), and (3) an event-study design focusing on technology adoption events (Appendix C). These designs generate similar results. This suggests that selection bias in any single design is unlikely to drive our findings.

The remaining concern is selection bias common to all our research designs. The concern would be that none of the control groups we analyze here represents a reasonable counterfactual for subsidy recipients. To address this concern, we can analyze trends in treatment firms without any control group. Figure A3 shows the evolution of treatment group means for machinery investment, employment, and years of education. Machinery investment increased sharply after the technology subsidy application; winners increased their employment but did not change their skill composition disproportionately.

Statistical Power A concern particularly relevant to presenting a null result is statistical power. Are our results precise and technology-adoption events large enough to justify our conclusion about no significant changes in skill composition measured by education and occupation? The estimates with the text propensity score indicate a -0.004 change in the average years of education at the firm level, with a standard error of 0.075 years, meaning that we can exclude increases above 0.15 years in average education. In comparison, the treatment and control firms increase their education on average over the 11-year event window by 0.4 years.

The null effects are unlikely to be driven by small events: (1) The average effect on machinery investment is EUR 100K, which doubles the firm's annual investment. This is a lower bound, as the purchase price covers only part of the total cost. In US manufacturing, Berger (2020) documents that machinery costs account for just 25% of the total investment, with the remainder covering installation, the machine bed, and integration work. (2) The subsidy program requires technology investments to represent significant advances for the firm. (3) We also examine large technology investment events in the spikes design (Appendix C) and find no effect on skill mix, measured by education and occupations.

Estimation Methods We probe our results' robustness to different estimation methods. Recent research proposes estimators that address issues that arise when estimating dynamic treatment effects (Borusyak et al., 2024; de Chaisemartin and D'Haultfoeuille, 2020; Callaway and Sant'Anna, 2021; Sun and Abraham, 2021). Figure A13 shows consistent results across these estimators.

To define our treatment group, we selected the largest subsidy event for firms with multiple applications to focus on meaningful events. The first stage (Figures 2, A2) shows that this choice isolates distinct technology-subsidy events. Table A9 shows our results are robust to this choice.

We focus on a balanced panel of firms because we are interested in within-firm changes in our key variables—employment and skill mix. Table A10 reports results from an unbalanced panel, in which effects are larger but consistent. Figure A14 directly examines firm survival, indicating incremental impacts: the effects are 10% or less within the first five periods.

VI Understanding Firm Responses

This second part examines the firm-level mechanisms behind these effects. Based on our fieldwork, we first outline two potential responses to technology subsidies. One is that technology subsidies lead firms to automate human tasks. The other is that subsidies enable firms to expand or shift into new activities, with potentially limited effects on productivity. These channels are not exhaustive and may be complementary.

We then present evidence on the mechanisms at play. We find that technology subsidies led to higher revenues without changing firms' productivity. Firms entered new markets, changed product mix, and increased marketing. In the subsidy application texts, firms wrote about their plans to expand rather than automate. While our data have limitations, and firms often implemented a combination of changes, our observations are consistent with the view that the subsidies primarily encouraged expansion.

VI.A Conceptual Motivation

The idea is simple: Subsidies lower the effective price of technology, but firms choose which technology to adopt and how to use it. Consider the composite function:

$$F\left(T_E; f\left(T_I; \mathbf{X}\right)\right). \tag{4}$$

The inner function $f(\cdot)$ captures how firms combine inputs $\mathbf{X}$ with technology T_I. The outer function $F(\cdot)$ maps this into market outcomes with T_E.

In one view, subsidies accelerate automation. Firms adopt technologies that replace human tasks, cut costs, and raise productivity. The intensive margin, $f\left(T_I; \mathbf{X}\right)$, captures this idea: technology changes how existing inputs are used. A welding robot that substitutes for a human welder is a textbook case.

Another view is that subsidies support expansion—entering new markets, developing new products, or simply scaling up operations—without fundamentally altering how production is done. This is the extensive margin, $F\left(T_E; \cdot\right)$. In the *Moore's Law for Pistons* example, the firm used new machines to make new kinds of pistons, not to automate away labor.

The point is that technology subsidies have uncertain effects. Subsidies may support automation, but not all subsidized machinery investments necessarily automate tasks. Some could instead help firms scale up or improve quality, with limited impact on workforce composition or productivity.[5]

VI.B Evidence on Firm Performance

We now turn to firm outcomes. Tables 3 and 4 summarize the results. We find increases in revenue, exports, product turnover, and marketing. In contrast, we observe no increases in productivity, profit margins, or capital intensity. We conclude by discussing magnitudes and dynamics.

[5]Appendix I presents these ideas in a simple model, building on Melitz and Redding (2014).

Revenue, Productivity, and Profits Table 3 reports the first-difference estimates from Equation 2 for revenue, productivity, and profit margin. Figure 5 summarizes the overall pattern. We find that technology subsidies led to approximately 30% higher revenues five years later. In contrast, we find no evidence of increases in productivity or the profit margin. The results are robust to controlling for text propensity scores and to comparison with the matched non-applicant control group.

We measure productivity as revenue per worker and TFP. Revenue per worker, our preferred measure, does not require specifying a production function. Because both measures are revenue-based, we interpret them cautiously when firms change their product mix. Alternative TFP estimates using the Olley and Pakes (1996) and Levinsohn and Petrin (2003) methods also show null effects (Figure A17).

Firm survival, often cited as a measure of firm performance in our interviews, improves with subsidies, as shown earlier in Figure A14. This is relevant to interpreting our productivity results: We see no productivity effects, but productivity is observed only for surviving firms.

The zero productivity effects are consistent with the existing evidence: Criscuolo et al. (2019), Curtis et al. (2022), and Cerqua and Pellegrini (2014) also find no productivity effects from capital-investment incentives in the UK, US, and Italy.

Table 4, Panel E, reports more detailed estimates on profits. The average profit margin is 5.2%. The subsidies do not appear to increase it, as noted above. But as the firms expand, winning a subsidy increases gross profits by EUR 143.5K and net profits by EUR 123.6K over the post-period, without discounting. Discounting the future profits at a 5% rate yields present-value net profits of EUR 95.8K, and at a 10% rate, EUR 73.7K. The average received subsidy (EUR 81.8K) falls within the 95% confidence intervals of both present-value estimates, suggesting that profits increased roughly one-for-one with the subsidies.

Exports and Products Figure 6 visualizes the event-study estimates for firms' export status. Subsidy winners are more likely to become exporters. Table 4, Panel A, reports an effect of four percentage points from a baseline of 28%. The effect on the export revenue share is 0.9 percentage points (baseline: 5.2%). The winners exported to an additional 0.2 regions (baseline: 1.5).

Export data allow us to examine changes in exported products and product mix. The number of exported products rose by 0.15 from a baseline of 1.55 per firm. We also observe an increase in product turnover: treatment firms both introduce and discontinue products at higher rates. However, subsidy-winning firms switch to products and regions with similar skill intensity (Table A11). A potential reason is that these changes are a normal part of how these firms operate (Section VIII).

Inputs and Imports Table 4, Panel B, examines firms' inputs and imports. While subsidies increased machinery investment, this did not lead to higher capital per worker. We also find no change in input intensity, measured by the share of input costs to revenue. These null results align with our finding of no effect on labor productivity: Firms did not make production more capital- or input-intensive but scaled operations proportionally.

Patents, R&D, and Marketing We next analyze three outcomes tied to different stages of innovation: patents, R&D, and marketing, reported in Table 4, Panel C. A relatively clear picture emerges: We observe no effect on patenting, a small EUR 564 increase in R&D expenses, and a larger EUR 13,890 increase in marketing costs.[6]

Prices Table 4, Panel D, reports price effects, measured from the Customs Register and Industrial Production Statistics (a survey of manufacturing firms). We compute prices as product-level revenue divided by quantity and focus on the firm-level unweighted average. We find a 29.1% increase in customs prices and a 30.8% increase in the manufacturing survey. The price data are noisy and cover only a subset of the sample, making these estimates more sensitive than our main findings. At a minimum, we do not observe price declines.

Magnitudes Table A6 reports the first-difference estimates from Equation 2 with a continuous treatment variable, the subsidy granted in EUR. The estimates indicate that one euro in subsidies is associated with a one euro increase in machinery investment. Revenue increased by five euros per euro of subsidy. We interpret these results cautiously because the continuous treatment is subject to additional selection concerns.

What is the implied cost per job from this industrial policy, based on our estimates? The employment increase is 0.25 jobs per EUR 10K in subsidies, indicating a back-of-the-envelope estimate of EUR 40K (≈USD 45K) per job. The binary specification implies EUR 20K per job, based on a 23% employment effect, average baseline employment of 18, and average subsidy of EUR 80K. Accounting for positive firm survival effects raises the employment effect to 27% and lowers the implied cost per job to EUR 17K. These magnitudes are broadly consistent with what managers reported for machinery per worker in our interviews, often about EUR 35K.

Our estimated range falls in the middle of the cost-per-job estimates reviewed by Criscuolo et al. (2019). For reference, Cerqua and Pellegrini (2014) estimate EUR 46K–78K per job in the context of capital subsidies in Italy's least developed regions. Criscuolo et al. (2019) report USD 27K at the firm level in the UK, and Garrett et al. (2020) find that the bonus capital depreciation policy in the US had a cost per job between USD 20K and 50K.

Dynamics The effects unfold at different speeds. Investment and exporting react rapidly, largely within the first two years. Employment and firm survival increase more gradually, with most gains in the first four years (Figure A15). Skill mix remains flat throughout.

A key question is whether the grants produced temporary or persistent improvements. The gradually rising effects on employment and firm survival favor persistent gains. For context, firms explained in our interviews that, when applying for subsidies, they already had an idea for how to expand with the funds. For example, in application texts, several firms cite responding to foreign demand as a reason to acquire new technology. This would likely show up in the data as an

[6] These results are less stable than our core estimates on employment, revenue, and skills. One reason may be that the underlying measures include many zeros and some large values.

immediate increase in export status measured as an indicator, and later expansion in employment if the exporting were successful (consistent with the scaling up from a blueprint discussed by Berger, 2013, and Gruber and Johnson, 2019). By contrast, if cost-reducing task automation were the main goal, we would expect to see higher productivity and lower output prices simultaneously with potential expansion (Acemoglu et al., 2020b). We do not detect these dynamics, or these effects more broadly.

VI.C Text and Survey Evidence

We use three complementary sources to explore firms' intentions: (1) subsidy application texts describing how firms planned to use the grants, (2) survey responses on technological goals, and (3) trade journal articles detailing firms' projects, combined with targeted surveys. Each source has limitations, but together they paint a consistent picture.

Text Evidence from Subsidy Applications The application texts offer a window into firms' motivations. Consider a firm buying a robot—a multipurpose technology. Some firms may use it to automate tasks previously done by workers; others may use it to expand capacity, introduce new products, improve quality, or shorten delivery times.

Our approach is straightforward. We use the application records to answer the question: What did the firm plan to do with the subsidy? Previously, we examined how firms changed their operations after receiving the grant. Here, we focus on firms' plans prior to treatment. The analysis is descriptive rather than causal.

We hand-code each application into detailed categories that reflect firms' stated intentions. Table 5 defines these categories—such as expansion, new products, quality, productivity, and work—and gives examples. The categories are grounded in our fieldwork and reflect goals that firms described in interviews and the texts themselves. The texts provide sufficient information for coding in 88.9% of cases (1,805 firms).

We use narrow, concrete categories to ensure that they capture clear and interpretable objectives. We are conservative: A category is assigned if the text is explicit about it. For example, "new products" requires a direct mention of a new product, service, or business area. "Productivity" applies only if the firm refers to efficiency, cost savings, or a similar goal. A firm that writes, "The project contributes to improving the company's competitiveness as it expands the product range," is coded as "new products." Another that says, "The new production line increases the company's capacity by approximately 30–40%," is classified as "expansion." Appendix F contains full descriptions, additional examples, and details on the coding process.

Figure 7, Panel A, presents the main findings. About 74% of firms reported expansion motives, such as growth, scaling up, or increased capacity. When we combine objectives directly tied to expansion—expansion, new customers/markets, and exports—80.1% of firms referenced at least one. Another 8.8% aimed to expand through vertical or horizontal integration, such as mergers and acquisitions.

New products (31.1%), quality (29.9%), and new capabilities or flexibility (29.4%) were common objectives. New capabilities often meant offering customized solutions. Other goals included improved delivery (9.5%), precision (6.1%), and brand (4.2%). Overall, 64.5% of firms reported plans to modify their output—via new products, quality, delivery, branding, precision, or capabilities and flexibility.

Fewer firms mentioned automation of work: only 13.9% referenced work-related motives. We group all mentions of reducing labor costs or improving labor productivity into a single category. This choice helps avoid understating work-related motives. The firms do not always distinguish between automation and other work-related objectives, so we cannot reliably separate these categories. To isolate narrowly defined labor-saving uses, we focus on firms that reported work-related motives but did not mention changes in output—excluding new products, quality, precision, and capabilities/flexibility. Under this criterion, 4.7% of our sample could be classified as "work only." In either case, the findings suggest that automation was not a primary stated objective for many firms—at a minimum, these firms pursued additional objectives. This differs from settings where task-replacing automation is the explicit focus and may partly explain differences in findings (Bessen et al., 2025; Feigenbaum and Gross, 2024).[7]

At the same time, productivity was a common stated goal: 51.6% of firms mentioned it. This contrasts with our earlier null results on measured productivity. Several explanations are possible. TFP might take time to materialize (though we do not observe trends within 5 years). Firms may have aimed to improve productivity but failed. In interviews, firms often used "productivity" to mean overall success—higher revenue or firm survival. They may have referred to productivity improvements in parts of their operations—"solving bottlenecks" as they sometimes described it—which eventually led them to hire more workers, and hence, there was no firm-level improvement in labor productivity (echoing standard scale effects). Finally, our productivity results condition on firm survival.

We examine heterogeneity in effects by stated intentions (Table A12). The few firms that focused narrowly on labor-saving objectives show some skill bias and smaller employment effects. But overall, heterogeneity across stated objectives is limited. Most firms appear to have pursued expansion, though their strategies varied.

This text analysis has several limitations. (1) Classification is qualitative and requires judgment. (2) Coding is based on what firms stated; lack of mention does not imply lack of intent. (3) Intentions are self-reported, and firms may have framed their intentions strategically, for example, by emphasizing expansion. Our main check is whether the patterns in the texts are consistent with patterns in the survey and trade journal data.

[7]The term *automation* in the application texts does not necessarily refer to the automation of work. It also describes automation in a narrow part of the production process—for example, automatic defect sensing in piston quality—or refers to the technology itself operating autonomously.

Survey Evidence on Firms' Intentions Survey data provide a complementary perspective. The EU's Community Innovation Survey (CIS) asks firms about their objectives for product and process innovation, such as expanding product lines, enhancing quality, and reducing labor costs. We matched these survey responses to 708 firms in our sample. Firms are likely to respond truthfully because the survey is confidential and unrelated to the subsidy program.

Figure 7, Panel B, shows that firms rated customer-specific solutions (43.4%), replacing outdated products (30.0%), better quality (26.6%), accessing new markets (26.0%), and larger product selection (25.7%) as highly important. Lower labor costs, while relevant, were not a top priority; only 20.6% of firms reported them as highly important. Lower material costs were a high priority for 9.2% of firms.

Evidence from Trade Journals Technical and trade journals report industry news and cover firms' technology investments. The SFINNO database, which tracks innovations in Finland, is based on a systematic review of 15 such journals from 1985 to 2020. We matched 213 articles from this database to firms in our sample.

These data have two advantages: the articles were written by independent journalists rather than the firms themselves, and the coding was conducted by independent researchers, not by us. SFINNO surveyed the firms to supplement the text-based information.

Figure 7, Panel C, shows that common motivations included filling a market niche (93.1%) and meeting customer needs (91.0%). Figure A19 breaks down innovation types: 89.7% were product innovations, 8.9% process innovations, and 1.4% services. Most innovations were low- to medium-complexity, such as redesigned pistons. Overall, the evidence from trade journals, combined with targeted surveys, aligns with our earlier findings.

VI.D Which Technologies Did Firms Adopt?

Next, we measure specific technologies and categorize them into groups. We define these technologies ex ante, rather than based on observed effects ex post. The evidence shows that firms primarily adopted machinery, with far less investment in IT. This distinction sets our study apart from research focused on IT and digitalization (Akerman et al., 2015; Gaggl and Wright, 2017).

The Financial Statement Register separates machinery investment from IT expenditure (Section IV). While both categories can have significant effects, they represent distinct types of technology. Tables 2 and A13 show the first-stage results for each. At baseline, firms invest EUR 108.0K in machinery and EUR 46.2K in IT. Subsidies increased machinery investment by EUR 103.8K but raised IT expenditure by only EUR 4.1K.

Text data reveal a similar pattern. We manually code each application into machinery and IT. As before, machinery refers to physical investments like CNC machines, robots, and laser cutters, while IT includes software and digital tools such as ERP and CAD software. This division is informed by our fieldwork and uses terms familiar to most manufacturers. The coding shows that only 5% (107 firms) of our sample mentioned IT, software, or digital technologies.

We also examine specific types of machinery. Table A16 reports results for firms mentioning CNC, robots, or lasers in their applications. We find no significant differences in effects across machinery types.

We analyze specific technology-occupation pairs, such as machining and machinists, welding and welders, and painting and painters, identified by keywords and occupational registers (Table A17). We find no evidence that these technologies displaced related workers. The share of machinists rises by 4.5 percentage points in firms adopting machining technologies; for the other technology-occupation pairs, we detect no changes in the shares, wages, or education levels of related workers. These results align with our overall findings.

Finally, we group technologies into two categories—automated and non-automated—building on Acemoglu and Restrepo (2022). We manually code all sample application texts and 621 technologies from customs records. While not always clear-cut, the main criterion is whether the technology requires active human operation. Automated technologies include robots, CNC machining centers, conveyor belts, and other automatic non-hand-operated machinery that can run independently for at least some time. Non-automated technologies, such as manual welding tools, vehicles, and hydraulic presses, require human operators. This coding reflects the type of machinery, not whether firms used it for automation. Figures A20 and A21 show that the effects are consistent across both categories, with increases in employment and no shifts in skill composition.

VII What Is Our LATE?

We proposed that technology subsidies may produce a range of effects, depending on how firms respond. What causal impacts does the quasi-experiment identify, and whose effects do we estimate—in other words, what is our LATE?

The EU subsidy program functions as an intent-to-treat policy. Not all firms take up the policy; only a subset chooses to make qualifying technology investments. The treatment effects approximate local effects for firms induced to invest by the program, capturing a non-random, non-representative subset of firms that respond to the subsidy.

This section notes that the observed effects likely reflect incremental investments induced by the subsidies. We consider alternative explanations, such as credit constraints, signaling effects, and employment bias. While these factors may matter, we find limited evidence that they primarily drive the results. Our firm-level estimates do not rule out spillovers.

Incremental Investments One hypothesis is that financial frictions limit technology adoption, and EU subsidies lift these constraints, enabling firms to make large investments. Alternatively, firms may already be capable of investing, and the subsidies lead to incremental investments with limited productivity effects. Our results support the latter view.

We focus on firms close to the investment margin. Both winners and losers intended to invest, as shown by their applications. Winners received an average EUR 80K grant, reimbursed after the investment. This reduced their effective investment cost. These firms were plausibly near the margin

of indifference, where the subsidy pushed them to invest moderately more.

Five observations support this interpretation: (1) Firms already used some technology and did not shift from no technology to full automation. (2) The average subsidy (EUR 80K) was moderate—about a typical year of investment—suggesting incremental changes unless it caused substantial additional investment. (3) Machinery investment rose roughly in proportion to subsidies (between one-for-one and 1.2-for-one across specifications), indicating that firms typically invested the grant amount but not substantially more. (4) Profits increased approximately one-for-one with the average subsidy, consistent with subsidies financing marginally profitable projects. (5) Losing firms also invested (Figure A3), suggesting we estimate a relative effect among firms that all invested.

This may help explain why subsidies did not lead to substantial productivity gains. If large productivity gains from automation were feasible and profitable, firms would likely have pursued them without the subsidy. Firms that responded to the program appear to have expanded incrementally with discrete machinery investments.

Fieldwork and interviews reinforce this interpretation. On the one hand, program officers would like to see larger-than-marginal changes and sometimes reject applications deemed too small. On the other hand, several CEOs explained that the program most likely affects firms on the verge of making incremental investments, and is less effective for firms already planning significant investments or those too constrained to benefit.

Credit Constraints A related alternative explanation is that the effects are primarily due to credit access rather than technology—an exclusion restriction concern for the interpretation. While credit constraints may play a role in allowing the subsidies to induce firms to invest more, several observations argue against them as the primary driver of the employment and skill results: (1) We observe a strong first stage on technology investment; at a minimum, the firms acquired new machinery. (2) We do not observe consistently larger effects for firms that are more likely to be credit-constrained, such as small firms (Table A18) or those with higher financial costs or relative debt (Table A19). Controlling for these measures, we observe similar effects. (3) To measure credit constraints more directly, we link our sample firms to CIS data on self-reported credit constraints. When asked about obstacles to innovation, only 6.3% cited lack of credit as highly relevant. (4) We observe similar patterns around investment spikes outside the program (Appendix C).

Signaling Value A reasonable concern is that winning the subsidy might not only reduce the cost of technology adoption but also serve as a positive signal about the firm's prospects. If these signaling effects were significant, we would expect to see positive employment effects even when the subsidy is small relative to the firm's size. To probe this, we narrow the sample to cases where the subsidy is a small share of the firm's total costs in the base year ($\tau = -3$), progressively reducing the sample from the bottom 50% to the bottom 10% (Figure A16). Reassuringly, the relative employment effects of a subsidy shrink to zero when the subsidy constitutes a negligible share of firms' total costs.

Employment Bias Another possible explanation is that the program itself is biased toward employment. This hypothesis has a factual basis: One objective of the ELY program is to stimulate employment by supporting the adoption of advanced technologies in manufacturing firms. However, several factors suggest that program biases are not the main driver of our findings: (1) We find similar results when we analyze technology adoption events outside the program in Appendix C. (2) The text propensity scores address this concern: If employment-related phrases predicted acceptance, the scores would capture them, and controlling for the scores would lower the employment effects, which we do not find. Controlling for text categories directly gives consistent results (Table A5). (3) Interviews with firm managers suggest that technology adoption events supported by subsidies tend to be similar to typical technology adoption events in this context. Administrators told us that major technology projects are rarely rejected for low employment effects. The program does not enforce employment outcomes; firms make their own hiring decisions after receiving subsidies. Additionally, the program does not specifically target low-skill jobs. (4) We reviewed rejected applications and found that none appeared to be denied primarily for employment-related reasons. Five reports mentioned employment concerns, but these were secondary to low projected technological advancement. (5) The matched non-applicant control version, where the comparison group is different, produces similar estimates.

Spillover Effects These firm-level results do not rule out spillovers. A firm's subsidized machinery adoption can affect others, so total employment and skill impacts may differ from our estimates. Estimating these effects is challenging because (1) these firms are small, and (2) they trade globally (our sample is primarily in tradable industries, following Mian and Sufi, 2014), making spillovers likely dispersed. Empirically, we find similar results using the matched non-applicant control group, suggesting that spillovers from winning to losing firms are unlikely to drive our findings. Furthermore, new hires at winning firms rarely come from losing firms (Table A20).

Whether these externalities are positive, negative, or neutral depends on the types of investments. We found that firms sometimes expanded by introducing new products, typically intermediate goods used by final goods producers. In Romer (1990), this type of variety expansion promotes growth, raising the possibility that some externalities may be positive. But new intermediate goods could also replace older versions, creating business-stealing effects (Aghion and Howitt, 1992).

VIII Manufacturing Context

We study small and medium-sized firms that tend to produce non-standard products in short, flexible runs and adapt to client demand. Our estimates may not extend to mass production.

VIII.A Flexible Manufacturing $\neq$ Mass Production

Our conceptual framework outlines two potential responses to technology subsidies: automation versus expansion. A central question raised by our empirical analysis is: When and why is one more likely to occur than the other? We mostly observe the latter in our context. But both responses occur

in practice, and studies document capital substituting for workers' tasks elsewhere (Feigenbaum and Gross, 2024; Acemoglu and Restrepo, 2020). We explain next why our findings are distinctive but logical, and potentially applicable to other settings with similar incentives for technology adoption.

Consider two contrasting views of manufacturing: mass production (Taylor, 1911; Ford, 1922) and flexible manufacturing (Piore and Sabel, 1984; Milgrom and Roberts, 1990). Mass production is characterized by standardized products, high-volume output, and stable environments. The production process is divided into tasks that can be automated. Flexible manufacturing is characterized by specialized products and low volumes. Firms in this setting rely on multi-skilled workers to adapt. Examples include the production of specialized machinery, defense equipment, and high-end textiles.

Understanding the manufacturing context helps interpret our findings. While automation is well understood, not all investments in manufacturing technologies aim to automate worker tasks (Acemoglu and Restrepo, 2018b; Restrepo, 2024). Most importantly, Piore and Sabel (1984) argue that in flexible manufacturing, different technology–labor relations emerge: firms often use new technologies to expand, replace outdated products, and provide customer-specific solutions, rather than primarily to automate work.

Automation often entails significant fixed costs, justifiable only in long, uninterrupted production runs typical of mass production. Flexible manufacturers, whose competitive advantage lies in creating specialized products and differentiating their offerings, typically do not find this model attractive. For example, one manufacturer we interviewed could automate its assembly but would need to commit to specific models and parts. This commitment was unattractive due to the frequent need to update its products to stay competitive. "I could see the business case for automation," the CEO said, "in a standardized environment, with many workers performing routine work, and with unions and strikes." But that was not their context.

A body of literature documents the shift from mass production to more flexible, specialized production since the 1980s (Dertouzos et al., 1989; Berger, 2013). Piore and Sabel (1984) call this the second industrial divide, Kenney and Florida (1993) describe it as moving beyond mass production, and Milgrom and Roberts (1990) call it modern manufacturing. Despite different approaches, the common observation is that "the business environment is no longer conducive to producing standardized products for a stable market" (Piore, 1994). A manager in Berger (2020) explained: "American manufacturing has been transformed. It's become highly engineered, highly specialized, and highly customized. I see this across all manufacturing. This is a different country. It's no longer the mass production of the past."

VIII.B Further Evidence About the Context

Contextual evidence shows that typical firms produce differentiated products in low volumes, adapt often, and employ skilled manufacturing workers.

Most firms we interviewed told us they produce specialized products for markets with limited demand. In the Etla survey of Finnish SMEs, 88.3% of our sample firms (N = 202) reported producing at least somewhat differentiated products, though only 2.1% considered their products

strongly differentiated (Figure A18). Looking ahead three years, 45.3% planned to introduce new products or services. Using the Rauch (1999) classification, 91% produce differentiated goods—for example, fabricated metal products, machinery and equipment, and wood products. Few, if any, produce commodity goods or are engaged in high-volume assembly.

There is often no production line; instead, firms produce in small batches or even one unit at a time. Several firms explained that the possibilities for automation are limited because they would need higher production volumes to cover the upfront investment needed to automate workers' tasks. Subsidy application texts frequently describe small-batch production.

Firms described adapting to changing client needs and technological obsolescence. One firm reported using robots to handle demand volatility: it could not hire skilled workers for short periods during demand peaks, so it used robots as temporary support.

One interpretation of our evidence is that because the workers are already skilled and continue to learn new skills, we do not observe changes in skill composition even if the technologies are skill-biased. To some degree, this is likely true.

The workers are already skilled: The typical workers are skilled welders and trained machinists with vocational degrees, rather than unskilled workers. They apply foundational skills to non-standard manufacturing tasks—closer to modern artisanal work than routinized mass production. In the CIS survey, only 7.5% of our sample firms cited a lack of skilled employees as a highly relevant obstacle to innovation. But these workers are not highly educated: only 15% have a college degree, and their 9th-grade GPA is 0.56 SD below the population average.

The workers also learn new skills: Managers reported that technology adoption was combined with worker training. New technologies required new skills, but current workers were typically best suited to learn and apply them. Importantly, the skills managers discussed were not about formal education or replacing production workers with college graduates, but about upskilling production workers (echoing Bartel et al., 2007). Nearly half of manufacturing workers participated in continuing vocational training in 2015 (CVTS, 2015).

Several aspects of the Finnish labor market are relevant. Firms have autonomy in hiring, firing, and technology decisions. Unions do not directly negotiate technology adoption or new hires. Short-term contracts are common. Terminating permanent staff is somewhat restricted, but Finland's employment protection is in line with the OECD average (OECD, 2020). Firms can terminate contracts for "production-related grounds," including changes in production technology.

Wage setting in Finland is moderately inflexible: unions and employer associations negotiate contracts for industries and occupations, and collective agreements cover close to 90% of employees (OECD and AIAS, 2025). This relative rigidity may contribute to the absence of observed wage effects. Our wage estimates could have been different in a context with a more flexible wage setting.

IX Zooming Out

Finally, we step outside the subsidy program. Using subsidy records and a broader sample of Finnish manufacturers, we examine firm- and industry-level patterns in technology adoption and workforce composition. This descriptive evidence shows that IT expenditure is more strongly linked to skill upgrading than machinery investment. Because the subsidy program primarily supported machinery rather than IT, this contrast may reconcile our findings with earlier studies reporting skill-biased effects of IT adoption (Autor et al., 1998; Akerman et al., 2015).

IX.A Big Trends

Are Finland's broad skill trends comparable to those in other countries? The backdrop of our study is the general shift in Finnish manufacturing toward greater skill demand. This trend is visible in the rising share of educated labor, declining share of production workers, and increasing college wage premium from 1994 to 2018 (Figure E1). These trends align with global patterns (Acemoglu and Autor, 2011) and are also reflected within our treatment and control firms, where skill levels increase steadily over time (Figure A3).

IX.B Compositional Effects

Our firm-level estimates may differ from macro-level impacts. At the macro level, technology subsidies could increase skill demand in two ways: (1) within-firm effects, where subsidized firms raise their demand for skilled workers, and (2) compositional changes, where firms with more educated workers receive grants, grow faster as a result, and this combination leads to skill bias at the market level. We examine whether subsidies primarily flow to high-skill, low-labor-share firms and, if so, whether these grants raise aggregate skill demand and reduce labor shares by favoring such firms.

Table A1 compares subsidy applicants to non-applicants in Finnish manufacturing. Applicants have lower labor shares and are slightly more educated. These differences are consistent with compositional effects that could lower aggregate labor shares and modestly raise skill levels. Table A20 shows that new hires joining winning firms come from larger, more productive firms, with higher labor shares and more skilled workforces. This movement aligns with labor-share-reducing compositional effects. But the flow of workers from more educated firms to less educated ones contrasts with skill-biased compositional effects. Nearly all new hires come from outside the program.

In sum, the evidence provides mixed support for skill-biased compositional effects. Pre-adoption skill differences point in this direction but are modest—less than 0.1 years—creating a natural upper bound on these effects, even if the program were scaled up. Worker flows point in the opposite direction. Evidence on labor shares is more clear-cut, with respect to both pre-adoption differences and worker movement.

IX.C Cross-Sectional Correlations

Outside the subsidy program, how does skill mix predict technology adoption? We use register data on machinery investment and IT expenditure for all Finnish manufacturing firms from 1999 to 2018. We find that the relationship between skill mix and technology adoption varies by type: Firms adopting machinery have skill levels similar to non-adopters, while IT adopters are distinctly more skilled. Both technologies are associated with lower labor shares.

Table 6 shows the results from predicting firms' machinery investment per worker and IT expenditure per worker with college share, production workers' share, labor share, and productivity. We observe that the skill mix has some but limited predictive power for machinery investment. Without controls, a one percentage point higher college share predicts 36.5 euros more machinery investment per worker, with a low R^2 of 0.01. After controlling for the 2-digit industry and firm size, the coefficient becomes an insignificant 9.2, which is close to zero compared to the mean of 5,321.7 euros per worker.

The results for IT are strikingly different: Skill mix strongly predicts IT expenditure. A one percentage point higher college share predicts 77.8 euros more IT expenditure per worker, with an R^2 of 0.26. After controlling for industry and firm size, the estimate remains significant at 55.7 euros. Compared to the mean of 1,016.1 euros of IT expenditure per worker, this corresponds to a 5.5% increase. A commonality between machinery and IT adoption is that both are robustly predicted by lower labor shares and higher productivity.

Survey data (Table E1) complement the register evidence with smaller samples but greater specificity. We use three surveys—CIS, ICT, and Etla—which include measures of robots and IT. As in the register data, robot users are about as educated as firms that responded to the survey but did not use robots. In contrast, IT users are consistently more educated than non-users.

IX.D Industry Trends

At the industry level, how do technology investments relate to shifts in skill composition? We find that industries investing more heavily in machinery have not experienced faster skill upgrading, whereas industries adopting more IT have significantly increased their skill shares.

Figure 8 shows the relationship between our technology measures—machinery and IT—and changes in the employment shares of college graduates from 1999 to 2018. We find no consistent relationship between industries' machinery investment and college shares. However, there is a strong positive relationship between IT expenditure and college shares. Machinery investment per worker was among the highest in the oil products and basic chemicals industries and among the lowest in textile fiber and footwear. Despite this, all these industries saw roughly average increases in their college shares. The highest IT expenditure per worker was in the electronics industry, which also saw the largest increase in college shares. Industries with lower IT expenditures, such as metal treatment and coating (the most common industry in our design), experienced less educational upgrading.

Table 7 reports estimates for the employment shares of college graduates, high school graduates, and workers with less than a high school education. Machinery investment per worker predicts

limited changes in educational composition, as shown in Panel A. In contrast, IT expenditure predicts an increase in the college-graduate share and a decline in the high-school graduate share in Panel B. Concretely, 1,000 euros in total IT expenditure per worker-year from 1999 to 2018 predicts a one percentage point higher college share. For reference, the average industry IT expenditure is 2,306 euros per worker-year, and the average machinery investment is 9,236 euros. When predicting college shares, the R^2 is below 0.001 for machinery but 0.59 for IT. Excluding electronics leaves the machinery results and the IT–college-share estimate similar, though the high-school graduate share no longer shows a consistent decline. The unweighted estimates are less precise but overall consistent.

Looking at other outcomes, Table E2 shows the estimates for labor share, productivity, and production workers' share. Both machinery and IT predict lower labor shares. IT predicts higher productivity, while the machinery coefficient is not significant. Machinery investments predict higher production workers' shares, whereas IT expenditures predict lower shares.

X Conclusion

We examine the micro-level effects of EU technology subsidies on the quantity and quality of jobs. The main finding is that firms receiving subsidies invested in advanced machinery and increased employment, with no detectable change in their skill mix or labor share. The technology grants primarily led firms to expand rather than automate work.

A relevant question is whether similar effects would emerge in other contexts. The program focuses on machinery rather than IT. Our descriptive evidence shows that IT is strongly associated with skill upgrading, while machinery is not.

We develop a new method for using text data in program evaluation. Many policy programs generate text records that capture details otherwise difficult to measure. We demonstrate how to use these text data for matching—in this case to identify "close winners" and "close losers" of the grant. This approach could prove helpful in other settings where decisions are based on textual case records, such as estimating the effects of judge decisions.

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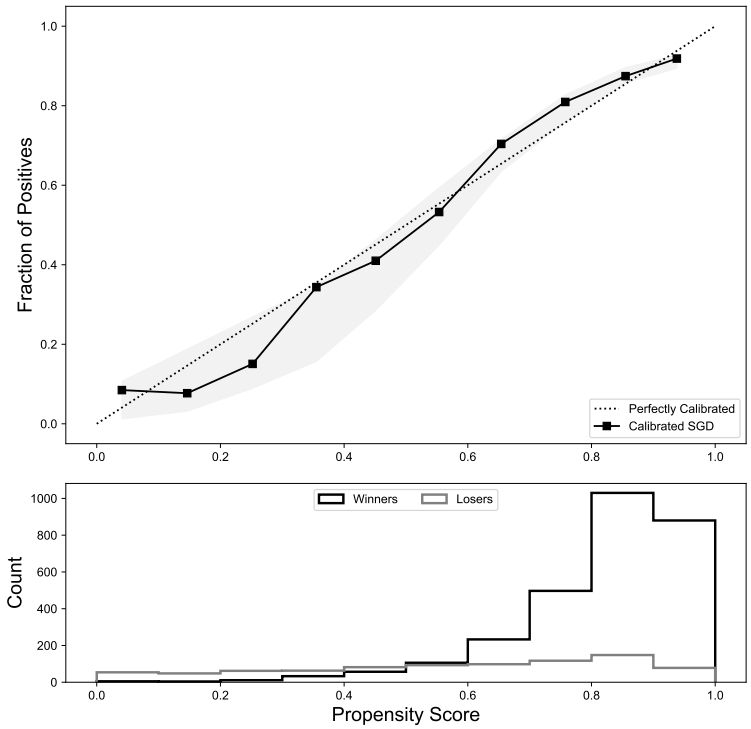


Figure 1: Text Propensity Score Calibration Plot.

Notes: **Upper panel:** The predicted probabilities of winning the subsidy based on text data are on the x-axis, and the observed probabilities on the y-axis. The text data are evaluation reports of the applications written by the subsidy program officers. The predicted probabilities are obtained from vector representations of the texts using a stochastic gradient descent (SGD) classifier with modified Huber loss and calibrated using Platt scaling. Standard errors are estimated by non-parametric bootstrap. The calibration is performed using all possible subsidy applications and plotted for a 10% sample. The predicted probabilities across the bins closely match the empirical probabilities. **Lower panel:** Distribution of the predicted values. Most of the applications have high predicted values reflecting the overall high acceptance rate. Back to Section III.

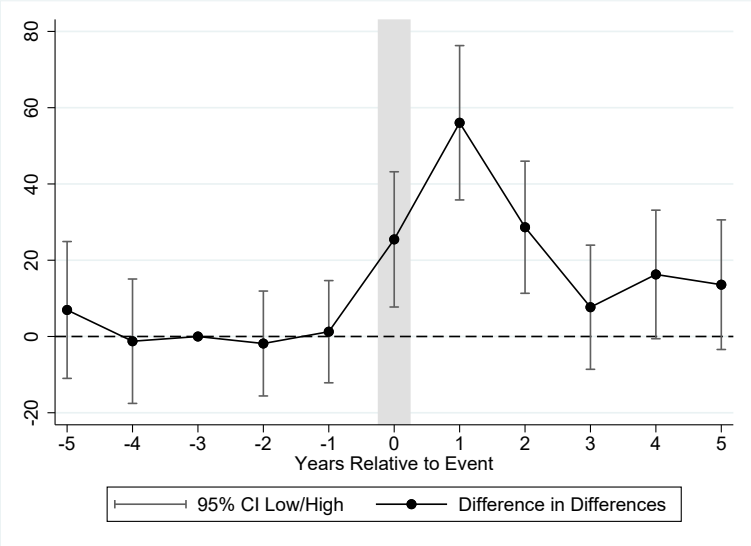


Figure 2: The First Stage: Effect of Technology Subsidies on Machinery Investment.

Notes: Event-study estimates from Equation 1. The outcome is investment in machinery and equipment (in EUR K) measured from the Financial Statement Register. The grey bar at event time $\tau = 0$ refers to the application year when the treatment begins. The estimate for $\tau = 1$ indicates that the treatment group invested EUR 60K more than the control group in the year after subsidy application. The estimates indicate a cumulative EUR 100K effect on machinery investment over $\tau \in [0, 2]$. This event-study specification contains no controls in the term X_{jt}^{τ} of Equation 1. Back to Sections V and VI.

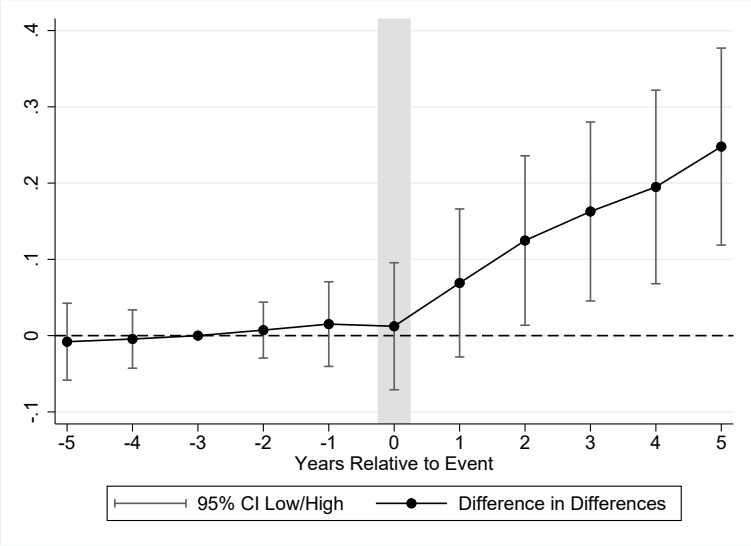
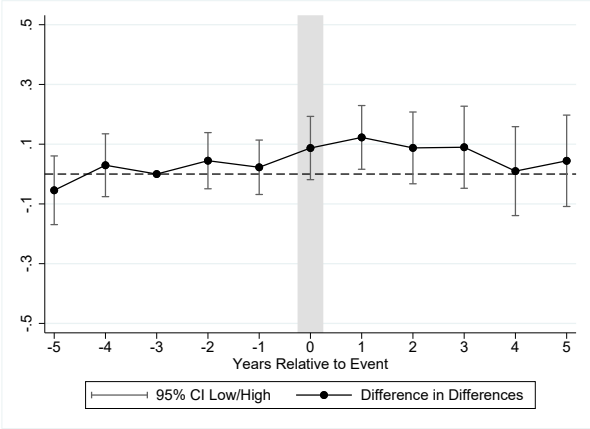
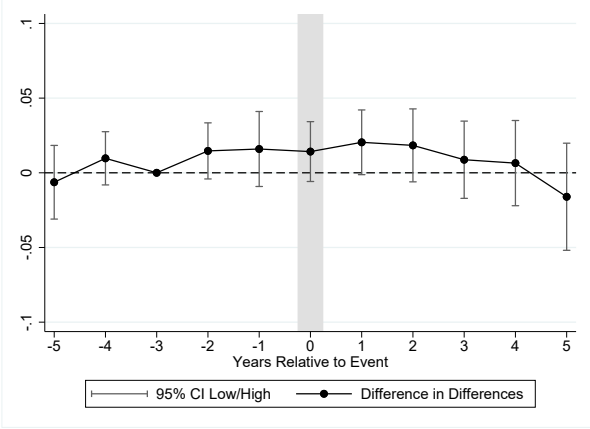


Figure 3: Employment Effects: Effect of Technology Subsidies on Employment (%).

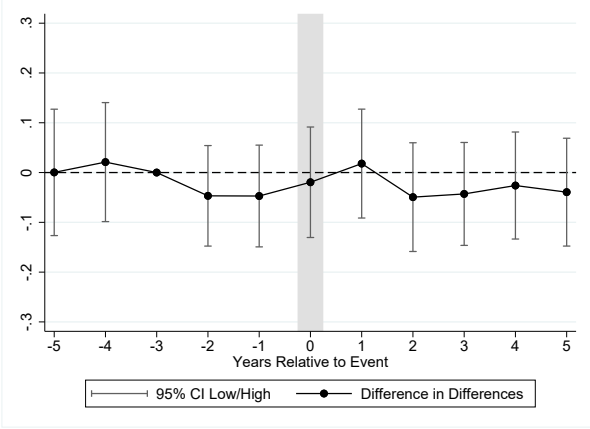
Notes: Event-study estimates from Equation 1. The outcome is employment relative to the base year $\tau = -3$. The grey bar at event time $\tau = 0$ refers to the application year when the treatment begins. The estimates indicate approximately a 20% increase in employment after five years. This event-study specification contains no controls in the term X_{jt}^{τ} of Equation 1. Back to Sections V and VI.



(A) Education Years.



(B) College-Educated Workers' Share.



(C) Production Workers' Share.

Figure 4: Skill Effects: Effect of Technology Subsidies on Main Skill Outcomes.

Notes: Event-study estimates from Equation 1. The outcomes are relative to the base year $\tau = -3$. The grey bar at event time $\tau = 0$ refers to the application year when the treatment begins. The estimates indicate approximately zero changes in the main skill measures. Education years are defined as the average years of education among the workers in the firm (measured in years); college-educated workers' and production workers' shares are the shares of employment of that group (measured in percentage points; 0.01 corresponds to 1 percentage point). These event-study specifications contain no controls in the term X_{jt}^{τ} of Equation 1. Back to Sections V and VI.

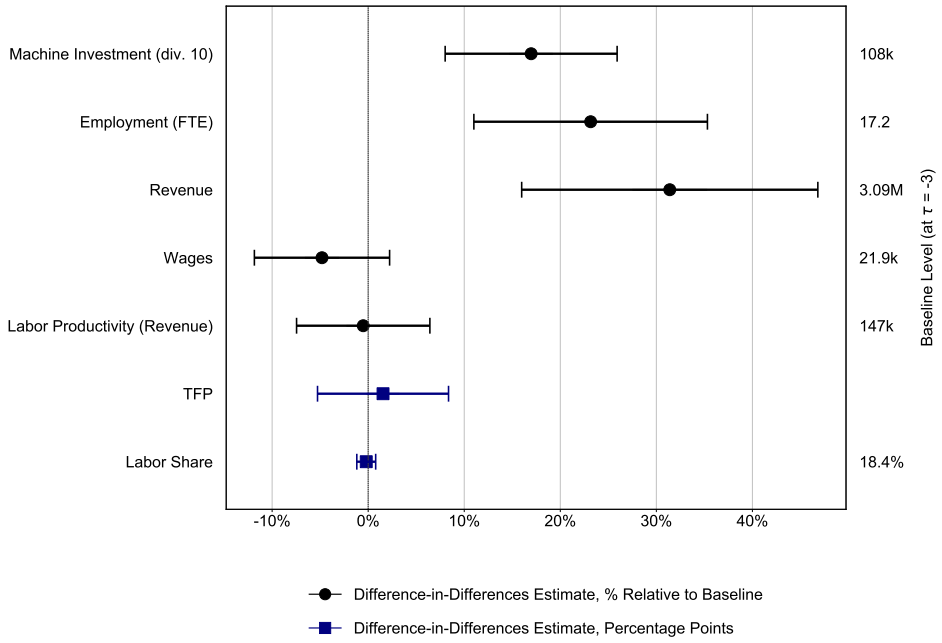


Figure 5: Firm-Level Effects: First-Difference Estimates on Main Firm-Level Outcomes.

Notes: Difference-in-differences estimates from Equation 2. The right-hand side reports means at $\tau = -3$ (omitted for TFP). Machine Investment, Employment, Revenue, Wages, and Productivity are measured by relative changes to baseline at $\tau = -3$. For Machine Investment, the post-period outcome is the average of investment over $\tau \in [0, 2]$ and for other outcomes, the average over $\tau \in [2, 5]$. The specifications include two-digit industry and firm size as controls. Back to Sections V and VI.

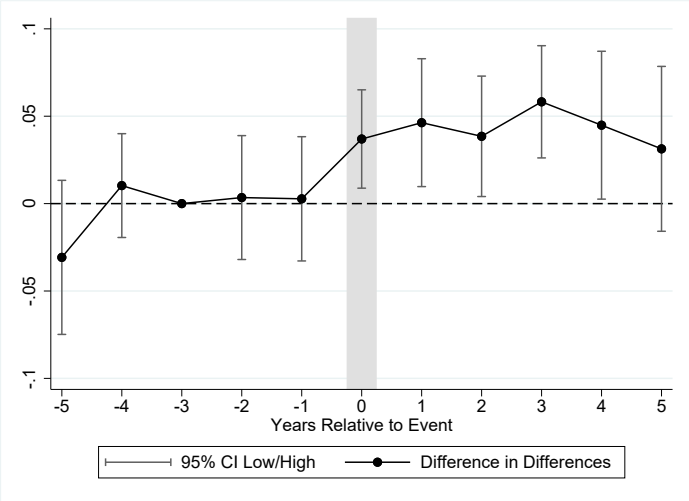
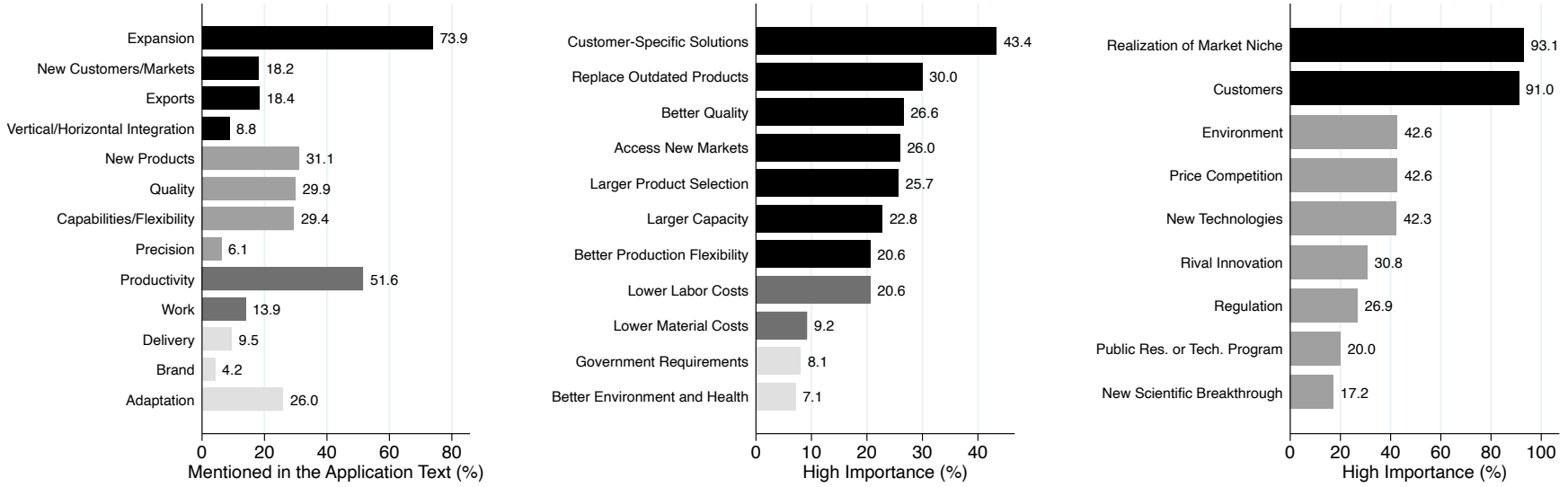


Figure 6: Export Effects: Effect of Technology Subsidies on Export Status.

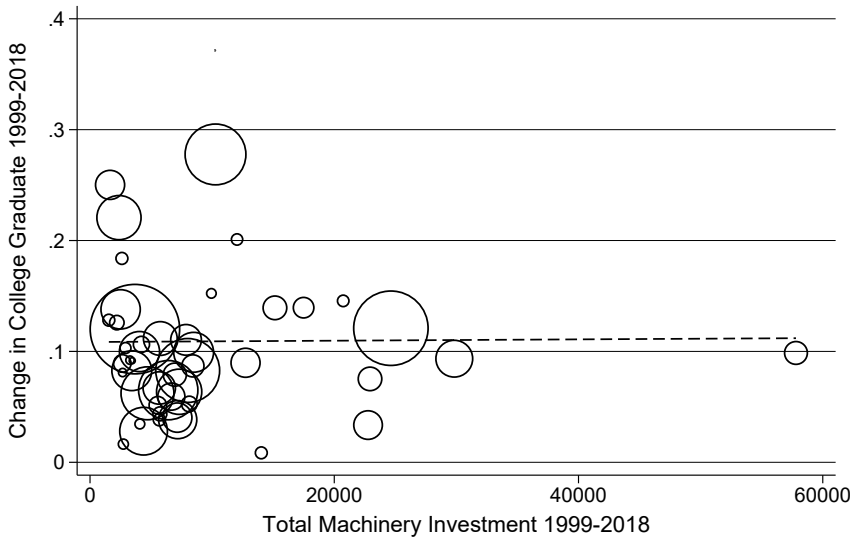
Notes: Event-study estimates from Equation 1. The grey bar at event time $\tau = 0$ refers to the application year when the treatment begins. The outcome is the firm's export status indicator (exporter vs. non-exporter). The estimates indicate an approximately 4 percentage point increase. The baseline value is 0.28 (28%). Exports are measured from the Finnish Customs' Foreign Trade Statistics. The definition from Statistics Finland identifies a firm as an exporter in a given year if its annual exports exceed EUR 12K across at least two months, or if it has a single export transaction exceeding EUR 120K. This event-study specification contains no controls in the term X_{jt}^{τ} of Equation 1. Back to Section VI.



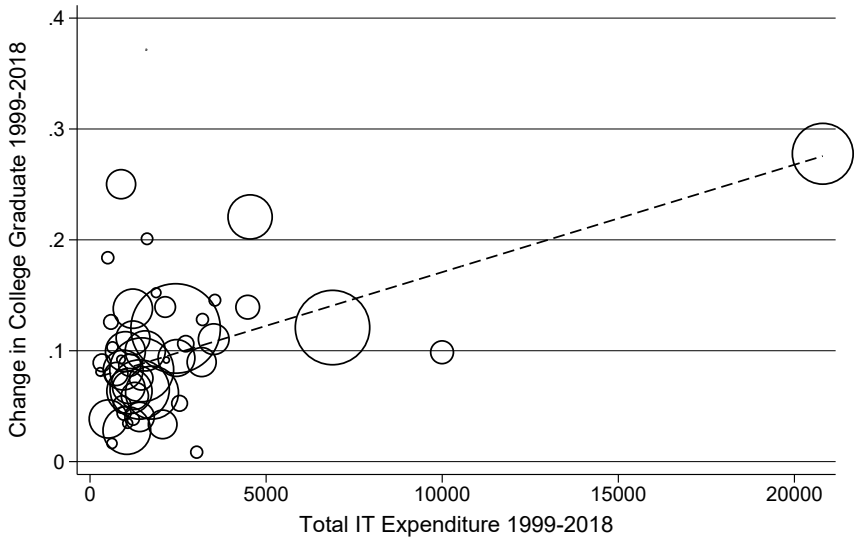
(A) Text Evidence. (B) Survey Evidence. (C) Literature-Based Evidence.

Figure 7: Evidence on Firms' Intentions From Application Texts, Surveys, and Trade Journals.

Notes: **Panel A:** The application texts contain information on the firms' intentions for the grant. This graph shows the prevalence of each objective. Colors reflect thematic grouping of the intentions: expansion, product changes, productivity and work, and other intentions. **Panel B:** The Community Innovation Survey (CIS) asks firms about the importance of different objectives for process and product innovations, including technology adoption. This figure shows the share of firms in our main sample that said a certain objective was highly important to them. The data come from survey rounds 1996, 2000, -04, -06, -08, -10, -12, -14, -16, -18. If the firm has responded to multiple rounds of CIS, we consider the closest survey to its subsidy application. N = 708 (the number of main-sample firms covered in these CIS rounds). **Panel C:** Evidence from the SFINNO database that collected information from 15 technical and trade journals. The figure reports answers to a targeted survey question: "How significant have the following factors been for the commencement of innovation's development?" The numbers refer to the share of firms in our main sample that said the factor was important to them. The options are not mutually exclusive. N = 73 (the firms in our main sample matched to the survey). Details in the main text and Appendix F. Back to Sections VI and VIII.



(A) Machinery Investment.



(B) IT Expenditure.

Figure 8: Industry-Level Evidence on Machinery and IT.

Notes: Industry-level graphs on predicting skill mix changes with total machinery investment (Panel A) and IT expenditure (Panel B) between 1999–2018. The technology variables are measured in EUR per worker-years (FTE). The outcome measure is the change in share of college-educated workforce in the industry in percentage points (i.e., .1 refers to 10 p.p.). Each bubble indicates an industry. The observations are weighted by industry employment, reflected in the size of the bubble. The related estimates are in Table 7. Back to Section IX.

Table 1: Summary Statistics: Winners–Losers Design.

	Treatment Group		**Control Group**		**Both**			**Tests**
Variable	Mean	Std. Dev.	Mean	Std. Dev.	p10	Median	p90	p-value
Machine Investment (EUR K)	109.93	369.14	82.60	233.11	0.00	27.24	233.80	0.378
Revenue (EUR M)	3.20	25.39	1.64	5.29	0.16	0.96	5.67	0.458
Employment	17.81	47.16	9.67	21.29	1.40	7.90	37.00	0.039
Wages (EUR K)	22.23	9.08	18.40	10.22	11.26	22.30	31.61	0.000
Labor Share (%)	18.68	9.31	15.47	10.28	6.49	18.03	30.70	0.000
Labor Productivity (EUR K)	146.82	163.84	145.61	87.86	72.35	118.60	228.49	0.930
Subsidy Applied (EUR K)	112.05	129.25	47.01	81.30	8.89	58.13	290.06	0.000
Subsidy Granted (EUR K)	81.77	103.02	0.00	0.00	3.24	35.64	200.23	0.000
Educ. Years	11.71	0.99	11.45	1.12	10.50	11.73	12.67	0.004
College Share (%)	15.51	16.80	11.63	18.42	0.00	12.50	33.33	0.012
Production Worker Share (%)	70.53	21.53	70.37	28.61	42.86	72.73	100.00	0.966
F-test								0.000
N	1885		146		2031			2031

Notes: All variables measured at $\tau = -3$ except for subsidies applied and received which are sums over $\tau \in [0, 2]$. Machinery investment and revenue are measured from the Financial Statement Register. Data on employment and wages come from the firm- and worker-level registers. Subsidies applied and granted are from the subsidy application data. Education years, college share, and production worker share are measured based on the worker composition within the firm. The variables are not winsorized prior to calculating the summary statistics. Back to Sections III and VIII.

Table 2: The First Stage: First-Difference Estimates.

	(1)		(2)		(3)	
	Subsidies Granted		Machine Inv.		Capital Stock	
Treatment	66.06***	70.22***	103.8***	100.3***	49.78**	41.60
	(3.119)	(4.907)	(22.56)	(29.29)	(18.26)	(23.60)
Propensity Score		✓		✓		✓
Observations	2031	1812	2031	1812	1560	1540

Standard errors in parentheses.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Notes: Difference-in-differences estimates from Equation 2 with and without the text propensity score control. The outcomes are in EUR K. To measure capital stock, we use the official records on firms' balance sheets. The post-period outcomes are means over $\tau \in [0, 2]$ multiplied by three (the number of periods). The specifications include two-digit industry and firm size as controls. Back to Section V.

Table 3: Main Firm-Level Estimates: Different Specifications.

Panel A: Investment, Employment, and Revenue.

	Machine Investment (EUR K)			Employment			Revenue		
	Baseline	Prop. Score	Match	Baseline	Prop. Score	Match	Baseline	Prop. Score	Match
Treatment	103.8***	100.3***	124.5***	0.232***	0.234**	0.217***	0.314***	0.333***	0.261***
	(22.56)	(29.29)	(9.358)	(0.0614)	(0.0746)	(0.0183)	(0.0779)	(0.0958)	(0.0232)
Observations	2031	1812	3200	2031	1812	3200	2031	1812	3200

Panel B: Wages, Profit Margin, and Productivity.

	Wages			Profit Margin			Productivity		
	Baseline	Prop. Score	Match	Baseline	Prop. Score	Match	Baseline	Prop. Score	Match
Treatment	-0.0481	-0.0285	0.00306	0.00121	-0.00791	-0.00685*	-0.00516	-0.00622	0.0117
	(0.0355)	(0.0407)	(0.00290)	(0.00772)	(0.00978)	(0.00290)	(0.0350)	(0.0427)	(0.0120)
Observations	1952	1738	3080	2031	1812	3200	2031	1812	3200

Panel C: Labor Share and Skill Composition.

	Labor Share			Education Years			College Share			Production Worker Share		
	Baseline	Prop. Score	Match	Baseline	Prop. Score	Match	Baseline	Prop. Score	Match	Baseline	Prop. Score	Match
Treatment	-0.00202	-0.000700	-0.00293	0.0246	-0.00385	0.0303	0.00557	0.00592	0.00542	0.000723	-0.0213	-0.00464
	(0.00496)	(0.00601)	(0.00203)	(0.0611)	(0.0752)	(0.0207)	(0.00935)	(0.0116)	(0.00330)	(0.0181)	(0.0212)	(0.00605)
Observations	2031	1812	3200	1884	1676	2999	1884	1676	2999	1891	1692	3011

Standard errors in parentheses.
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Notes: Difference-in-differences estimates from Equation 2. The table reports the treatment effects on selected outcomes for the main sample with and without the text propensity-score control and the matched control sample. "Baseline" refers to a baseline specification with calendar-year indicators, two-digit industry, and firm size as controls. "Prop. Score" refers to estimation with the text propensity score included as an additional control. "Match" refers to estimation in the matched sample, where the control group is formed from matched non-applicant firms. **Panel A:** Machinery investment is in EUR K. Employment and revenue are in relative changes, e.g., 0.20 would refer to a 20% increase. **Panel B:** Wages and productivity are relative changes; the profit margin is in percentage points; 0.01 corresponds to 1 percentage point. **Panel C:** Education years is in years (absolute change). The labor, college, and production worker shares are in percentage points. For machinery investment, the post-period outcome is the average of investment between $\tau \in [0, 2]$ multiplied by three (the number of periods) and for other outcomes, the average of $\tau \in [2, 5]$. The pre-period for the estimations is $\tau = -3$. Back to Sections V and VI.

Table 4: Detailed Firm-Level Estimates.

Panel A: Exports and Products.

	(1) Exporter Status	(2) Export Share	(3) Export Regions	(4) Products Number	(5) Products Introduced	(6) Products Discontinued
Treatment	0.0404**	0.00935*	0.219***	0.155**	0.0880**	0.0664**
	(0.0134)	(0.00451)	(0.0568)	(0.0599)	(0.0282)	(0.0223)
Mean	0.284	0.0523	1.498	1.546	0.498	0.539
N	2031	2031	2031	2031	2031	2031

Panel B: Inputs and Imports.

	(1) Capital per Worker	(2) Input Value	(3) Input Share	(4) Import Value	(5) Import Share	(6) Machine Imp. Value	(7) Machine Imp. Share
Treatment	-0.085	-115.6	-0.0521	20.60***	0.0029*	3.437***	0.0005
	(0.087)	(663.6)	(0.0438)	(5.989)	(0.0014)	(1.031)	(0.0002)
Mean	23.85	3457.9	0.292	152.9	0.0203	27.80	0.0037
N	1550	321	321	2031	2031	2031	2031

Panel C: Patents, R&D, and Marketing.

	(1) Patents	(2) R&D Costs	(3) Marketing Costs
Treatment	0.002	0.564*	13.89***
	(0.008)	(0.239)	(4.162)
Mean	0.056	7.101	31.42
N	1535	1842	2028

Panel D: Prices.

	(1) Prices, Exports	(2) Prices, Manuf. Survey
Treatment	0.291	0.308**
	(0.328)	(0.102)
N	400	217

Panel E: Profits.

	(1) Profit Margin	(2) Gross Profits	(3) Net Profits
Treatment	0.00121	143.5***	123.6**
	(0.00772)	(51.15)	(51.61)
Mean	0.052	274.8	-16.07
Median	0.050	52.85	37.56
N	2031	2031	2031

Standard errors in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Notes: Difference-in-differences estimates from Equation 2. The specifications include two-digit industry and firm size as controls. The baseline means are measured at $\tau = -3$. All level estimates are in EUR K and shares in percentage points; 0.01 corresponds to 1 percentage point. The export, input, and import shares are defined over revenue. Patents are measured as the average annual patenting rates. Prices are computed as product-level revenue divided by quantity from the Customs Register and the Industrial Production Statistics (a survey of manufacturing firms). Post-period values calculated as the average over $\tau \in [2, 5]$, except for the profit levels that are over $\tau \in [0, 5]$ multiplied by the number of periods, 6, to scale them to the overall effect without discounting. The pre-period for the estimations is $\tau = -3$. More details are in the main text and Appendix F. Back to Sections VI and VIII.

Table 5: Text Categories of Firms' Intentions.

Category	Description	Examples
Expansion	Expand production, operations, or firm scale.	*Increase output, build production capacity, pursue growth.*
New Customers/Markets	Target new customers or markets.	*Win new customers, enter a new market.*
Exports	Expand or improve exporting.	*Start exporting, appeal to foreign markets.*
Vertical/Horizontal Integration	Shift firm's position in the value chain.	*Bring tasks in-house, investments related to mergers and acquisitions.*
New Products	Introduce or develop new products or services.	*New product, expand product range.*
Quality	Raise product or production quality.	*Improve product quality, meet stringent quality requirements.*
Capabilities/Flexibility	Enhance production capabilities or flexibility.	*Offer customized and diverse solutions.*
Precision	Improve precision or accuracy.	*Increase precision, enable more accurate measurement.*
Productivity	Raise productivity or reduce costs.	*Cut costs, increase productivity, enhance efficiency.*
Work	Affect labor costs, tasks, or productivity.	*Automate tasks, reduce labor costs, increase labor productivity.*
Delivery	Improve delivery speed, reliability, or logistics.	*Accelerate deliveries, enhance delivery reliability.*
Brand	Improve branding or public image.	*Enhance image, improve reputation.*
Adaptation	Respond to new challenges or demands.	*Adapt to changing customer needs.*

Notes: This table lists categories of firms' intentions that we first defined and then hand-coded from the subsidy application texts. Each application may belong to one or more of these categories (not mutually exclusive). Bold text indicates the category name, regular text provides a description, and italicized text gives examples of common phrases appearing in the applications. Full details and more detailed examples are available in Appendix F. Back to Section VI.

Table 6: Predicting Machinery Investment and IT Expenditure in Firm Register Data.

	(1)	(2)	(3)	(4)
A: Machinery Investment				Mean
College Share	3651.1***	1646.3**	916.7	0.217
	(727.8)	(523.8)	(574.9)	
Production Workers' Share	1481.9*	99.26	888.9	0.612
	(611.1)	(476.8)	(512.6)	
Labor Share	-38251.2***	-16093.1***	-15733.8***	0.236
	(1966.0)	(937.5)	(972.2)	
Productivity	19460.9***	9069.6***	8958.6***	0.173
	(1217.1)	(821.8)	(858.5)	
B: IT Expenditure				
College Share	7779.8***	6607.8***	5569.2***	0.217
	(426.2)	(283.1)	(286.6)	
Production Workers' Share	-5646.4***	-4577.6***	-3579.1***	0.612
	(394.6)	(237.7)	(208.5)	
Labor Share	-10719.0***	-7339.8***	-6200.8***	0.236
	(581.2)	(314.4)	(325.8)	
Productivity	5850.5***	4418.4***	3961.3***	0.173
	(288.5)	(250.0)	(266.6)	
Year Controls	Yes	Yes	Yes	
Industry Controls	No	Yes	Yes	
Firm Size Control	No	No	Yes	
Machinery	R^2	R^2	R^2	N
College Share	0.013	0.298	0.299	159,583
PW Share	0.009	0.311	0.313	125,817
Labor Share	0.103	0.310	0.310	161,336
Productivity	0.179	0.318	0.318	161,338
IT				
College Share	0.256	0.475	0.508	159,583
PW Share	0.177	0.453	0.510	125,817
Labor Share	0.182	0.428	0.478	161,336
Productivity	0.332	0.472	0.517	161,338
	Mean	10p	90p	Obs. $\neq 0$
Machinery	5321.7	0	14054.0	116,032
IT	1016.1	106.0	2343.0	149,988

Standard errors in parentheses.
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Notes: Pooled cross-sectional estimates predicting machinery investment (Panel A) and IT expenditure (Panel B) with college share, production workers' share, labor share, and productivity. Each estimate is from a separate firm-level regression. Shares are on a scale of 0 to 1 (estimates reflect a change from 0 to 100%), productivity in EUR M (revenue per worker), and machinery and IT in euros per worker. N refers to firm × year observations. The data contain all Finnish manufacturing firms with > 2 full-time equivalent employees (FTE) from 1999–2018. Estimates are weighted by firm employment. Back to Section IX.

Table 7: Industry-Level Evidence on Machinery and IT.

	(1)			(2)		
	College	HS grad	Less than HS	College	HS grad	Less than HS
A: Machinery Investment	0.0001	0.0017	-0.0018	-0.0006	0.0027***	-0.0022**
	(0.0007)	(0.0009)	(0.0009)	(0.0008)	(0.0005)	(0.0007)
Pre-Period College Share				0.396*	-0.657***	0.261***
				(0.165)	(0.167)	(0.0586)
B: IT Expenditure	0.0097***	-0.0114***	0.0017	0.0082***	-0.0065*	-0.0017
	(0.0006)	(0.0018)	(0.0015)	(0.0010)	(0.0025)	(0.0020)
Pre-Period College Share				0.123	-0.411***	0.288**
				(0.0862)	(0.104)	(0.0833)
N	49	49	49	49	49	49
Mean Pre-Period	0.245	0.471	0.284	0.245	0.471	0.284
Mean Change	0.104	0.0494	-0.153	0.104	0.0494	-0.153
R^2, Machinery	0.000	0.035	0.118	0.356	0.646	0.421
R^2, IT	0.592	0.509	0.036	0.614	0.660	0.268
	Mean Total	10p	90p			
Machinery	9.236	2.560	21.75			
IT	2.306	0.610	4.011			

Standard errors in parentheses.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Notes: Industry-level long-difference estimates on predicting skill mix changes with total machinery investment (Panel A) and IT expenditure (Panel B) between 1999–2018 in Finnish manufacturing industries. The technology variables are measured in EUR K per worker-year (FTE). The educational variables are in long-changes over the timeline. The educational groups are mutually exclusive. "College" refers to the employment share of college-educated employees in the industry, "HS grad" to the share of exactly high-school educated workers including vocational degrees, and "Less than HS" to the share of those without a high-school or vocational degree. The estimates and means are weighted by pre-period industry employment sum between 1999–2002. The pre-period is elsewhere defined as the year 1999. N denotes the number of 2-digit manufacturing industries. Figure 8 illustrates the estimates for the college share. Back to Section IX.

What Do Technology Grants Do?

Online Appendix

Johannes Hirvonen
Aapo Stenhammar
Joonas Tuhkuri

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A Winners–Losers Design

B Matched Control Group

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D Regression Discontinuity Design

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H Text Processing

I Conceptual Framework

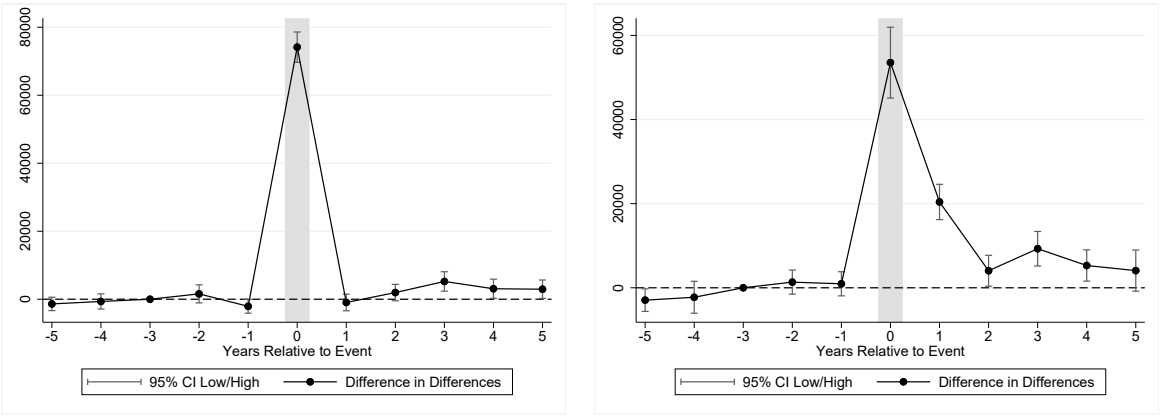
J Dynamic Model

A Winners–Losers Design



Figure A1: ELY Program Application Process.

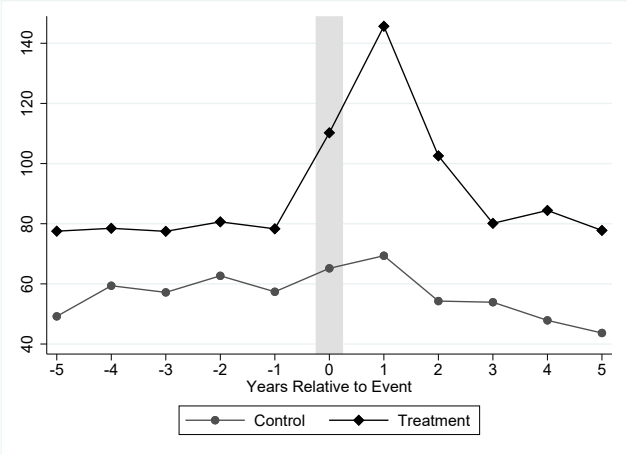
Notes: Details in the main text. Back to Section III.



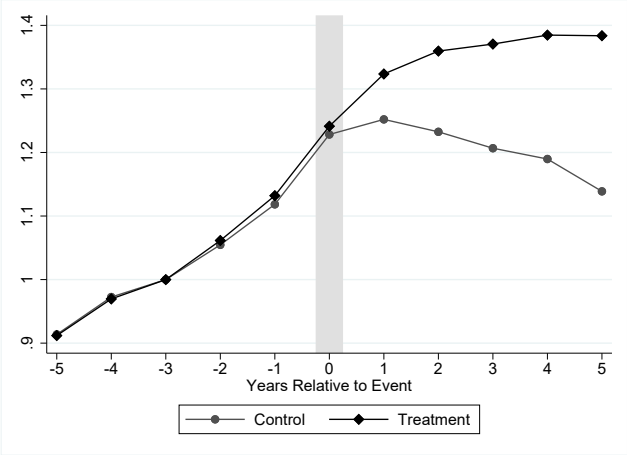
(A) Granted. (B) Received.

Figure A2: The First Stage: Effect of Winning a Subsidy on Granted and Received Subsidies.

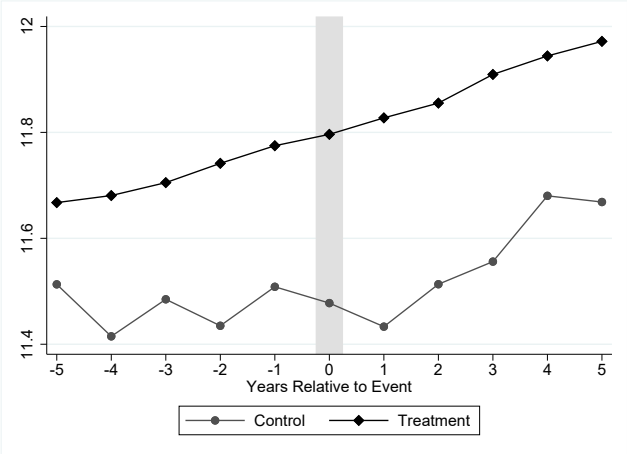
Notes: Event-study estimates from Equation 1. The outcomes are (A) any subsidy granted and (B) received, measured from the Finnish Statistics on Business Subsidies. Event time $\tau = 0$ refers to the application year. This event-study specification contains no controls in the term X_{jt}^{τ} of Equation 1. Back to Section V.



(A) Machinery Investment.



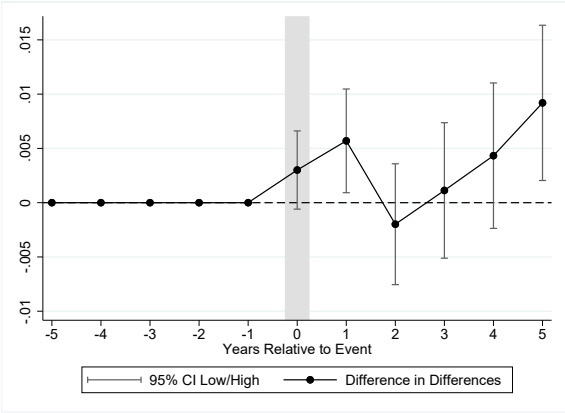
(B) Employment.



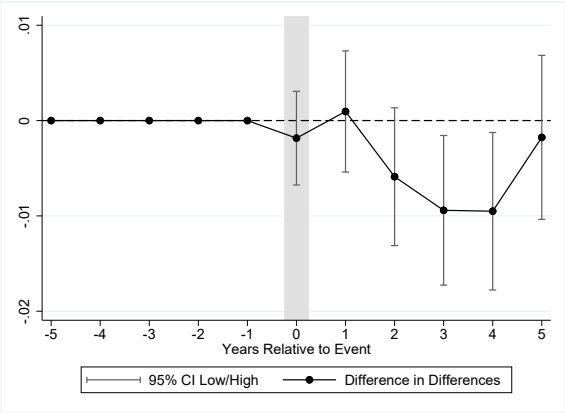
(C) Education Years.

Figure A3: Raw Means of Machinery Investment, Employment, and Education.

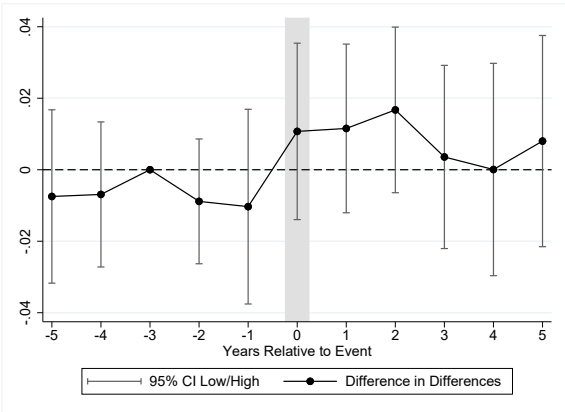
Notes: Means over time for the main treatment and control groups (winners vs. losers). The grey bar at event time $\tau = 0$ refers to the application year when the treatment begins. Machinery investment in EUR K, employment in % relative to $\tau = -3$, and education in years. Back to Sections V, VII, and IX.



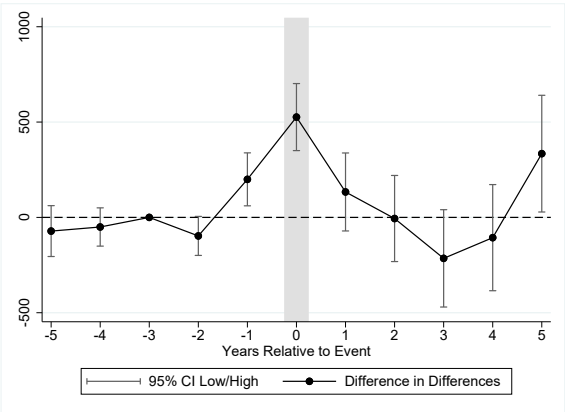
(A) Employed.



(B) Employed at the Baseline Firm.



(C) Income in % Relative to Baseline.



(D) Income in Euros.

Figure A4: Effects of Technology Subsidies on Incumbent Workers.

Notes: The sample is the baseline workers (employed at the firm from $\tau = -5$ to $\tau = -1$) in the main analysis sample (subsidies design). The first two outcomes are in percentage points, the third in percentages, and the fourth in euros. The baseline workers in treatment group firms are slightly more likely to be employed in general, but less likely to be employed in the baseline firm after the event. The same workers also receive extra income of about 500 euros in the year of the application. This corresponds to a salary of about one week. We present here results for the high-tenure workers. The modest effects are robust to relaxing that assumption. Back to Section V.

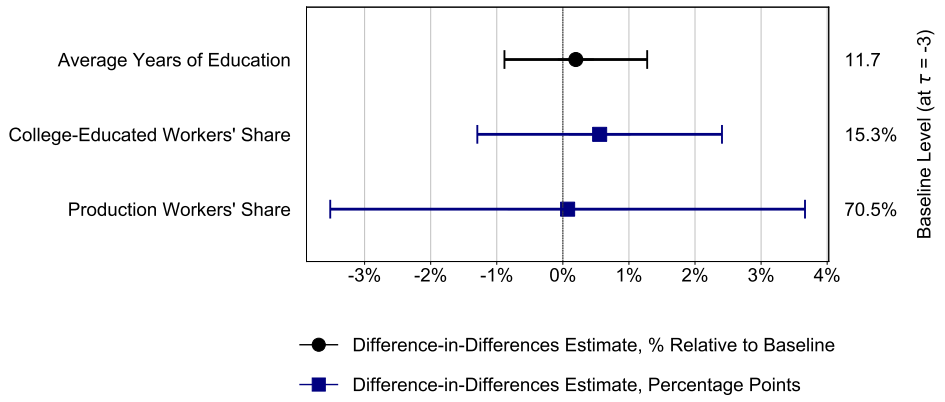


Figure A5: Skill Effects: First-Difference Estimates on Main Skill Outcomes.

Notes: Difference-in-differences estimates from Equation 2. The right-hand side reports means at $\tau = -3$. Education is measured as a relative change (%) in the average years of education in the firm between $\tau = -3$ and the average over $\tau \in [2, 5]$. The shares are measured in percentage-point changes. The estimates indicate no detectable changes in the skill composition. The specifications include two-digit industry and firm size as controls. Back to Section V.

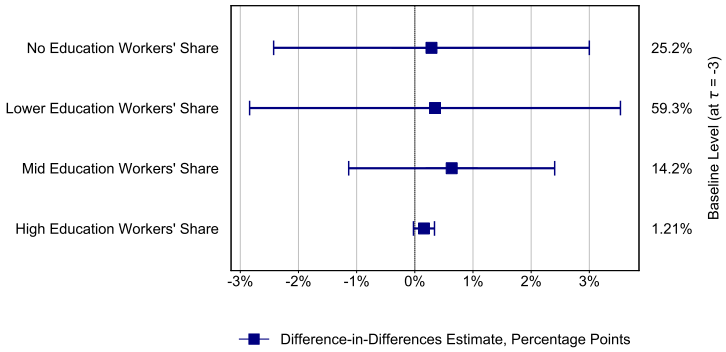


Figure A6: Skill Effects: Effect of Technology Subsidies on Education Group Shares.

Notes: Difference-in-differences estimates from Equation 2. The right-hand side reports means at $\tau = -3$. The data are from Finnish educational registers. Education groups are defined in Appendix F. Back to Section V.

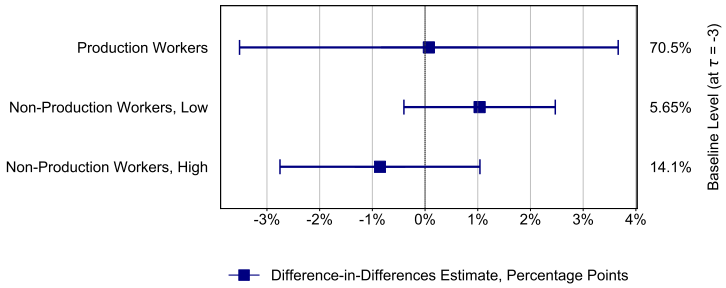


Figure A7: Skill Effects: Effect of Technology Subsidies on Occupation Group Shares.

Notes: Difference-in-differences estimates from Equation 2. The right-hand side reports means at $\tau = -3$. The data are from the Finnish occupation registers. Occupation groups are defined in Appendix F. The shares do not sum to 100% because some workers do not have occupational info. Back to Section V.

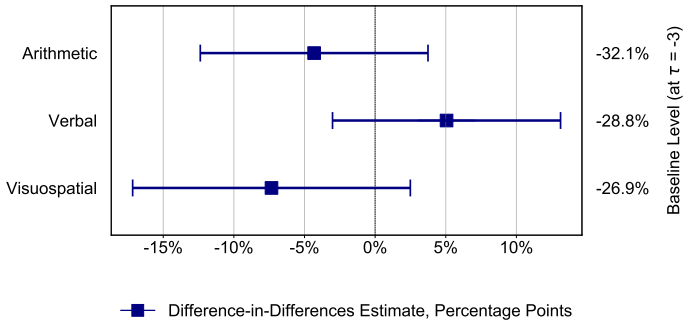


Figure A8: Skill Effects: Effect of Technology Subsidies on Average Worker Cognitive Performance Measures.

Notes: Difference-in-differences estimates from Equation 2. The right-hand side reports means at $\tau = -3$. The estimates are in percentages of standard deviations. The data are from the Finnish Defence Forces. Appendix F provides more information. Back to Section V.

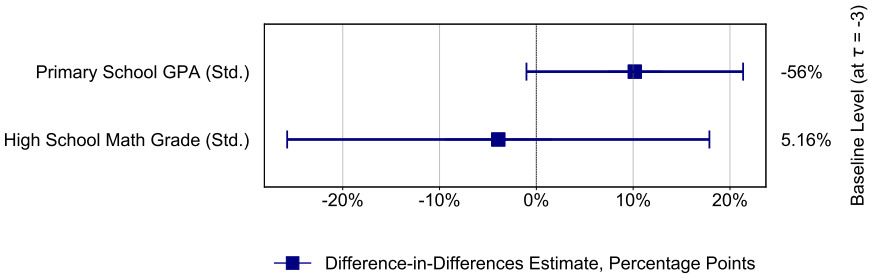


Figure A9: Skill Effects: Effect of Technology Subsidies on Average Worker School Performance.

Notes: Difference-in-differences estimates from Equation 2. The right-hand side reports means at $\tau = -3$. The estimates are in percentages of standard deviations. The data are from the Secondary Education Application Register and the Finnish Matriculation Examination Board Register. School variables are defined in Appendix F. Back to Section V.

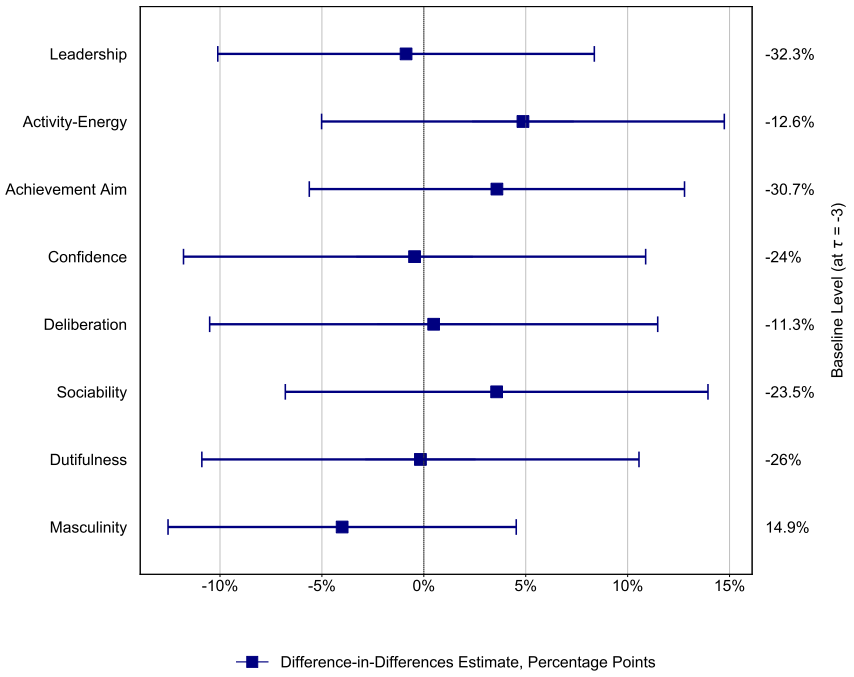


Figure A10: Skill Effects: Effect of Technology Subsidies on Average Worker Personality Measures.

Notes: Difference-in-differences estimates from Equation 2. The right-hand side reports means at $\tau = -3$. The estimates are in percentages of standard deviations. The data are from the Finnish Defence Forces. Personality variables are defined in Izadi and Tuhkuri (2024). Back to Section V.

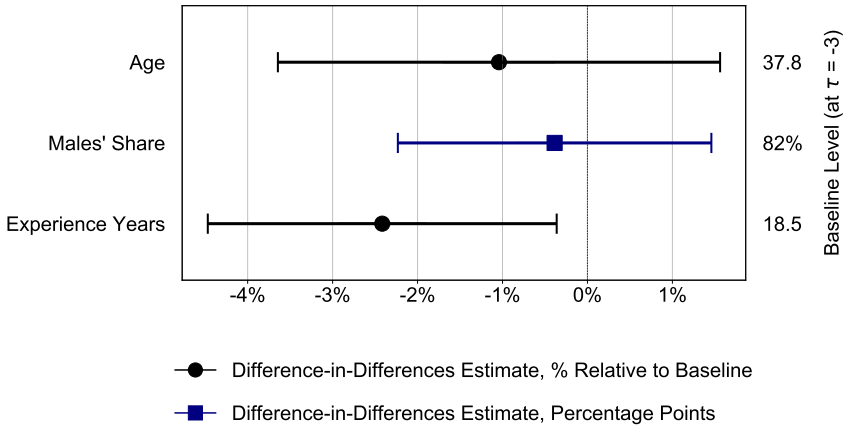


Figure A11: Skill Effects: Effect of Technology Subsidies on Average Worker Demographics.

Notes: Difference-in-differences estimates from Equation 2. The right-hand side reports means at $\tau = -3$. The data are from the Finnish worker and population registers. Back to Sections V and VIII.

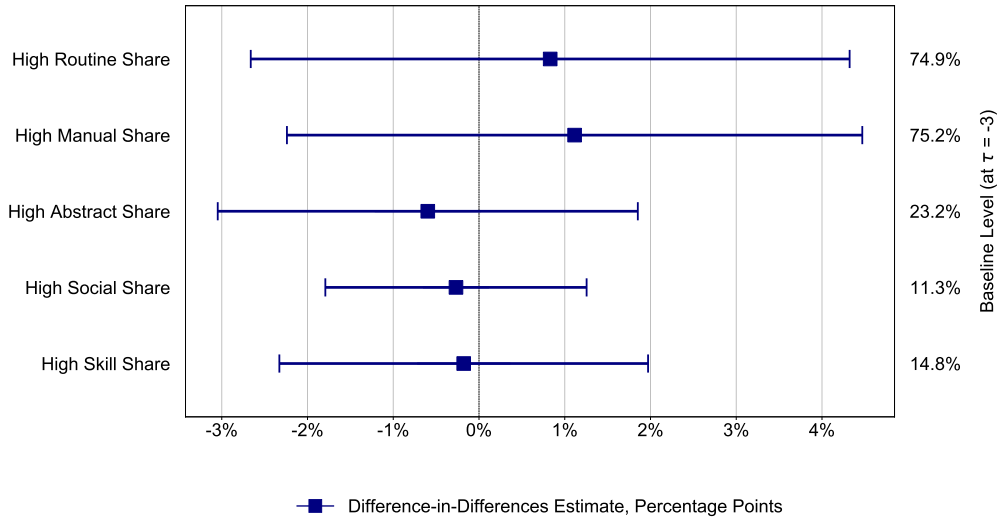
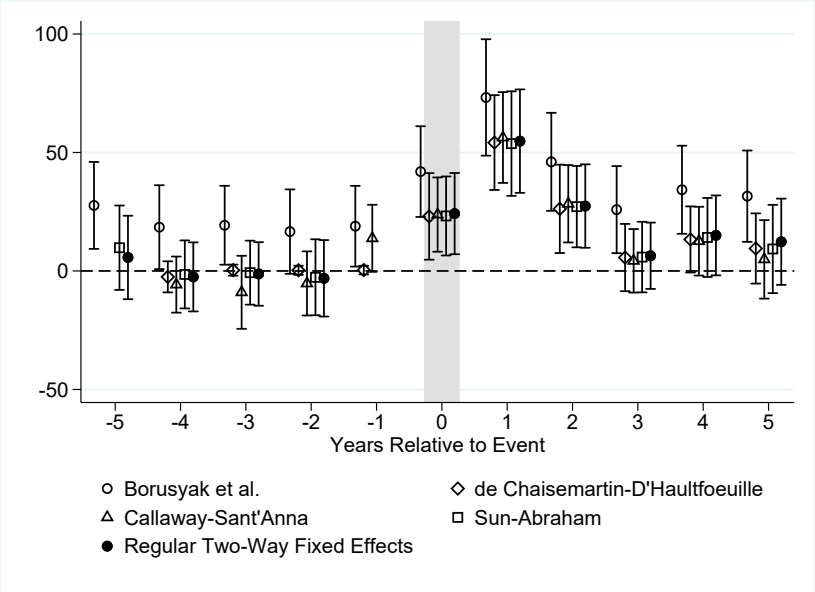
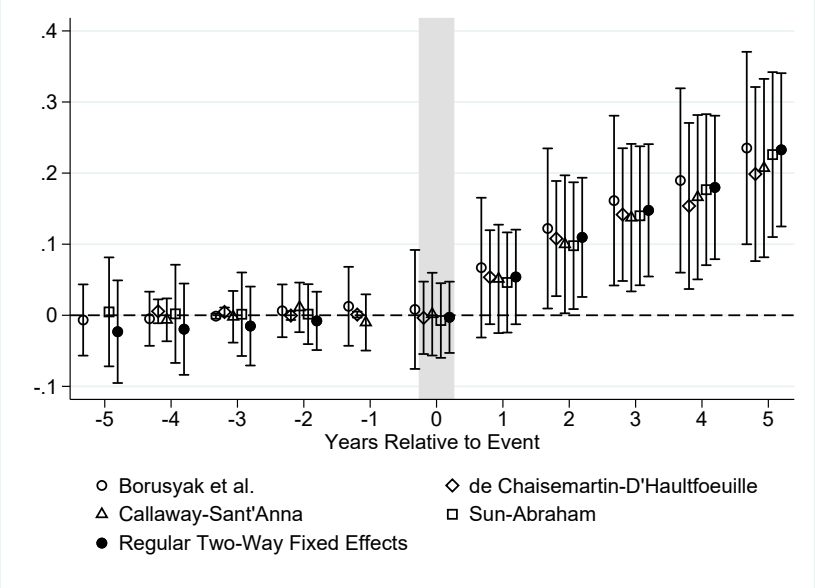


Figure A12: Skill Effects: Effect of Technology Subsidies on Task Content.

Notes: Difference-in-differences estimates from Equation 2. The right-hand side reports means at $\tau = -3$. We discretize the data for each worker into above vs. below median (high/low) because it has no natural scale. Median refers to the median task intensity in the Finnish labor force. For example, the first row indicates that 74.9% of workers in our sample firms are in an occupation times industry cell that is above the median in routine task content. The treatment group increases the share of these workers by a statistically insignificant 1% compared to the control group. The data are from the Finnish occupation registers and the European Working Conditions Survey (EWCS). More details in F. Back to Section V.



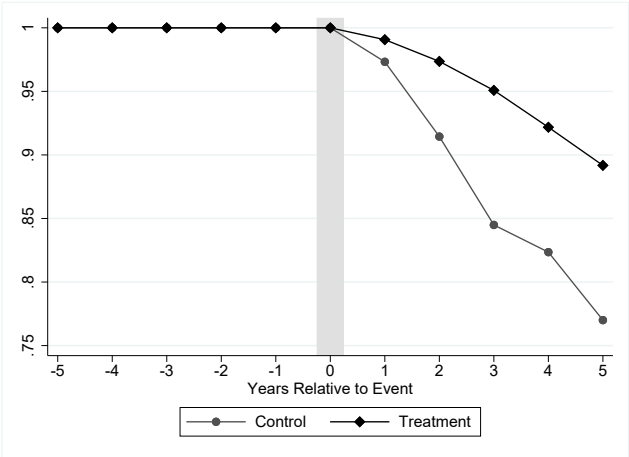
(A) Effect of Technology Subsidies on Machinery Investment.



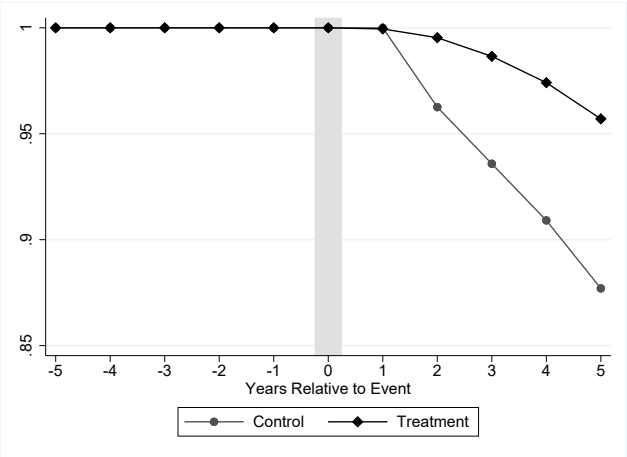
(B) Effect of Technology Subsidies on Employment (%).

Figure A13: Robustness to Different Event-Study Estimators.

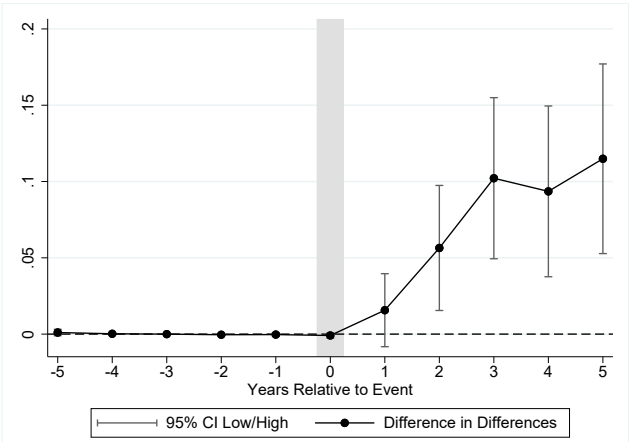
Notes: Results from different recent event-study estimators: Borusyak et al. 2024; de Chaisemartin and D'Haultfoeuille 2020; Callaway and Sant'Anna 2021; Sun and Abraham 2021. The regular two-way fixed effects refer to event-study estimates from Equation 1. The grey bar at event time $\tau = 0$ refers to the application year when the treatment begins. **Panel A:** The outcome is investment in machinery and equipment (in EUR K) measured from the Financial Statement Register. **Panel B:** The outcome is employment relative to the base year $\tau = -1$. These event-study specifications contain no controls in the term X_{jt}^{τ} of Equation 1. Back to Section V.



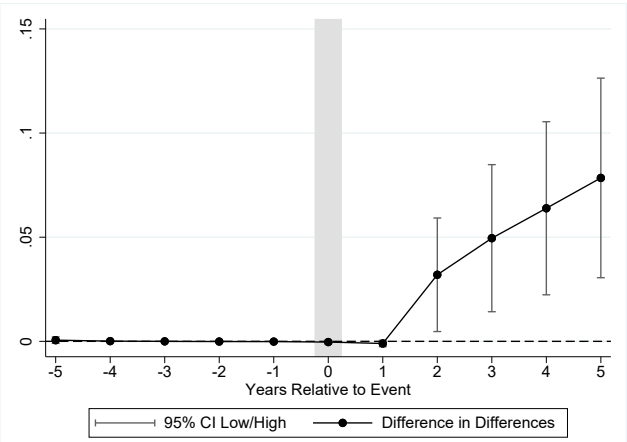
(A) Firm Survival Based on the Firm Register.



(B) Firm Survival Based on Worker Flows.



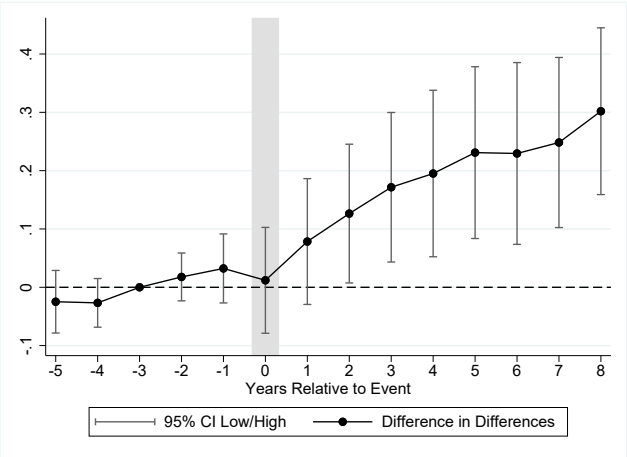
(C) Firm Survival Based on the Firm Register.



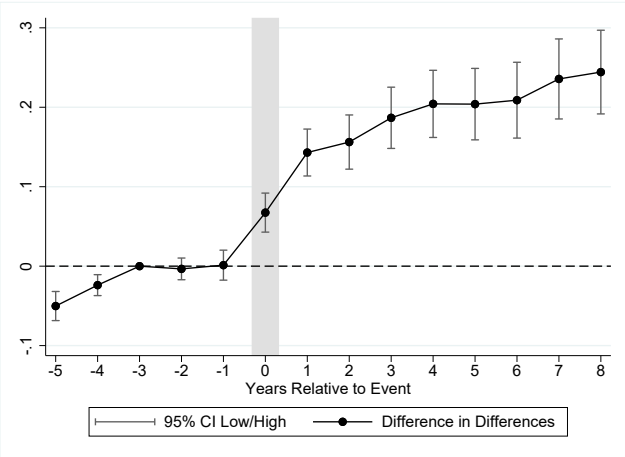
(D) Firm Survival Based on Worker Flows.

Figure A14: Effect of Technology Subsidies on Firm Survival.

Notes: Group means and event-study estimates from Equation 1. **Panels (A, C):** Survival is measured from whether the firm ID exists in the firm register. **Panels (B, D):** Survival is extended to include mergers and acquisitions (and other cases the firm ID changes), where at least 50% of workers continue under the same new firm ID. The main estimates of the paper are robust to a non-balanced sample, shown in Table A10. Back to Sections V, VI, and VIII.



(A) Winners vs. Losers.



(B) Matched Control Group.

Figure A15: Longer-Term Effects: Effect of Technology Subsidies on Employment until $\tau = 8$.

Notes: Event-study estimates from Equation 1 with an extended post-event horizon. The outcome is employment relative to the base year $\tau = -3$. The grey bar at event time $\tau = 0$ refers to the application year when the treatment begins. The balanced sample requirement is extended to $\tau = 8$. This event-study specification contains no controls in the term X_{jt}^{τ} of Equation 1. Back to Section VI.

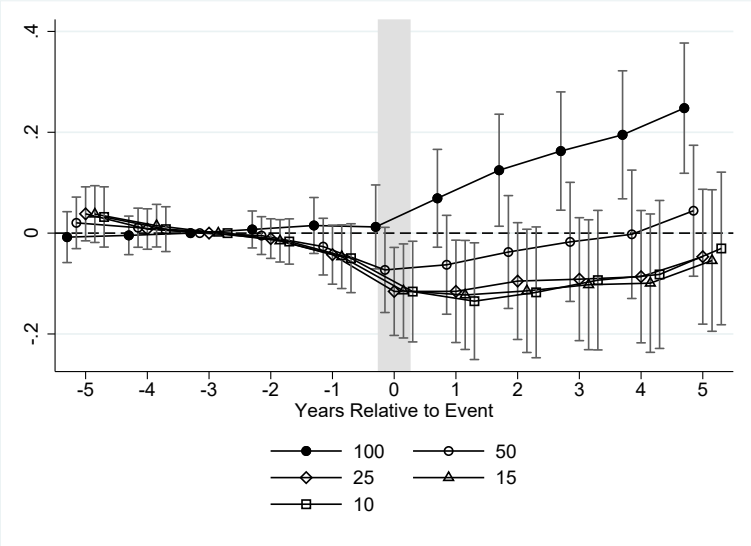
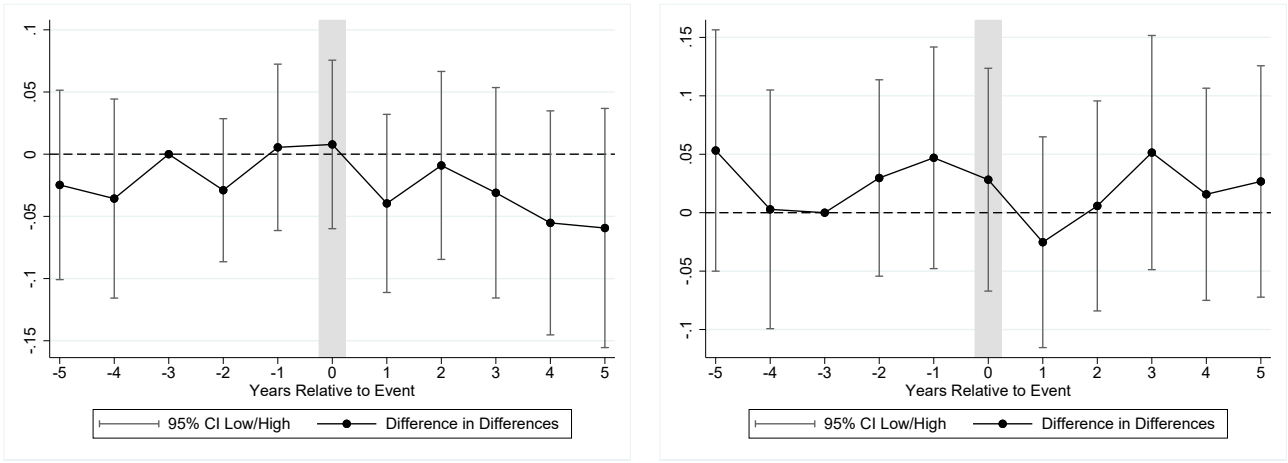


Figure A16: Placebo Test: The Employment Effects of Insignificant Subsidies (%).

Notes: Event-study estimates from Equation 1. The graphs restrict the sample to cases where the applied subsidy is a progressively smaller share of the firm's total costs in the base year $\tau = -3$. 100% contains full sample, 50% the subsidies that are below median share of total costs, 25% below the first quartile of total costs, etc. The control group here contains all losing firms, as the control sample is small. This event-study specification contains no controls in the term X_{jt}^{τ} of Equation 1. Back to Section VII.



(A) Olley-Pakes.

(B) Levinsohn-Petrin.

Figure A17: TFP: Alternative Versions.

Notes: The sample is the main analysis sample (subsidies design). Event study graphs of log total factor productivity, estimated as in (a) Olley and Pakes (1996) and (b) Levinsohn and Petrin (2003). The results are in line with the Cobb-Douglas version, showing no effect. Back to Section VI.

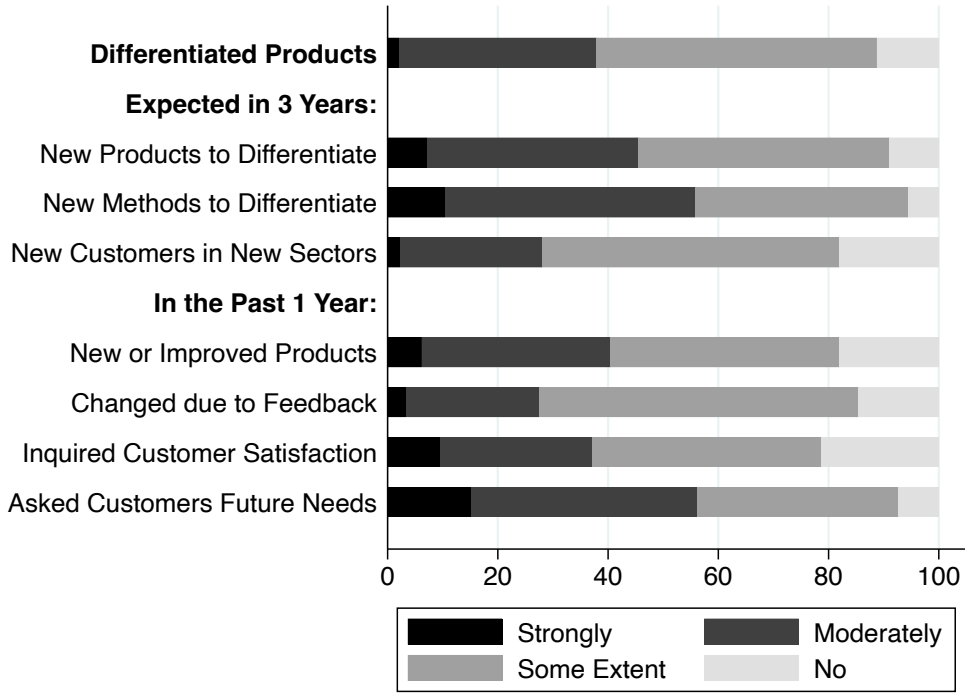
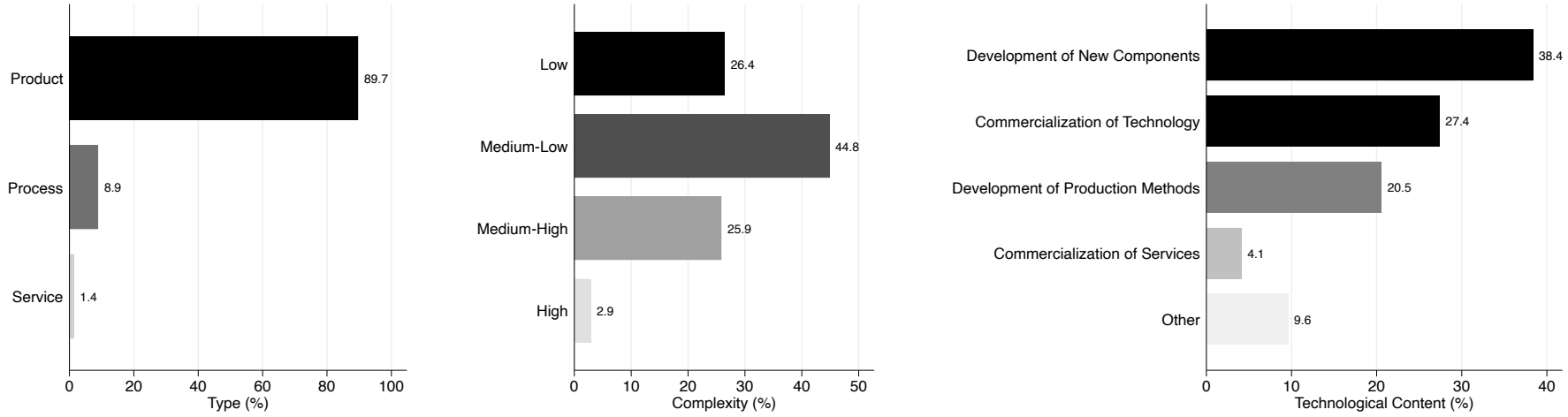


Figure A18: Etla Survey: Intentions.

Notes: The Etla survey provides information on firms' objectives and strategy. The full questions are "To what extent do your own products/services differ from other options on the market?", "To what extent do you think your company will change in the next 3 years?", where the subtopics are "We will bring products/services to the market that differentiate us from our competitors.", "We will develop production methods for our products/services that will differentiate us from our competitors.", and "We will acquire customers from new sectors.", and whether "During the last 12 months, we have..." "introduced new or substantially improved products to the market;" "changed our operations based on feedback received from customers"; "surveyed customer satisfaction;" and "asked customers about their future needs." N = 202 (firms matched to our winners–losers design). Back to Sections VI and VIII.



(A) Type. (B) Complexity. (C) Technological Content.

Figure A19: Literature-Based Evidence on Technologies from Trade Journals.

Notes: Evidence from 15 technical and trade journals that featured our sample firms, supplemented with targeted surveys and collected by the SFINNO project. **Panel A:** Shares of cases coded as product, process, or service using the definitions of the Oslo Manual (OECD, 2018) based on the journal article. N = 213 (the number of firms in our main sample matched to the journal data). **Panel B:** Shares of cases coded by their complexity based on the journal article. N = 174. Low complexity: *Innovation is a single coherent unit. Examples: glue-laminated timber, mobile phone cover.* Medium artefactual complexity / low developmental complexity: *Innovation is a unit, development is based on knowledge base from one discipline. Examples: electronic wheel chair, drill.* Medium artefactual complexity / high developmental complexity: *Innovation is a unit, development is based on knowledge bases from several disciplines. Examples: pharmaceuticals, software, generator.* High complexity: *Innovation is a system consisting of several functional parts, development is based on several disciplines. Examples: paper machine, mobile phone network, cruise ship.* **Panel C:** The survey question asked to choose the primary technological content of the innovation. The options are mutually exclusive. N = 73. Details in Appendix F. Back to Section VI.

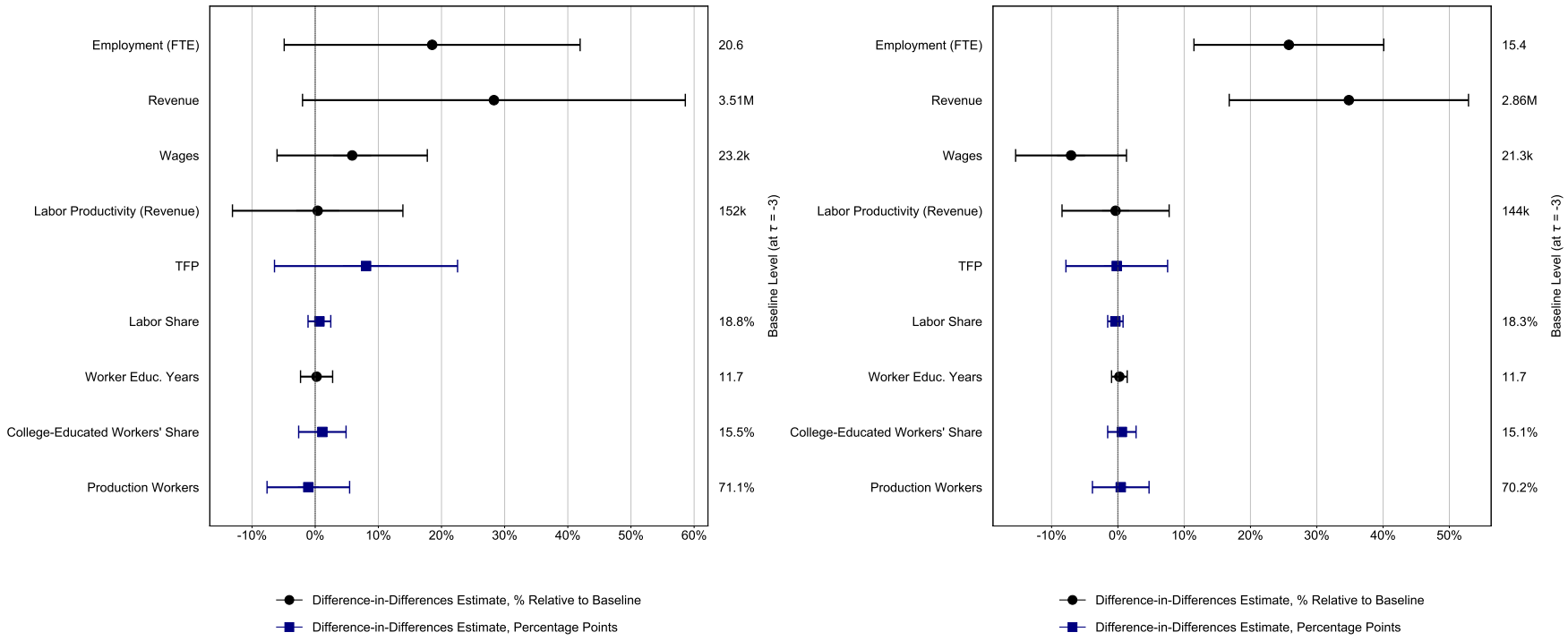


Figure A20: Firm-Level Effects: Automated (left) vs. Non-Automated (right) Technologies from Text Data.

Notes: Difference-in-differences estimates from Equation 2. Automated vs. non-automated technologies are measured from text data as described in Section IV and Appendix F. Automated (N): Treatment 678, Control 30. Non-Automated (N): Treatment 1207, Control 116. Back to Section VI.

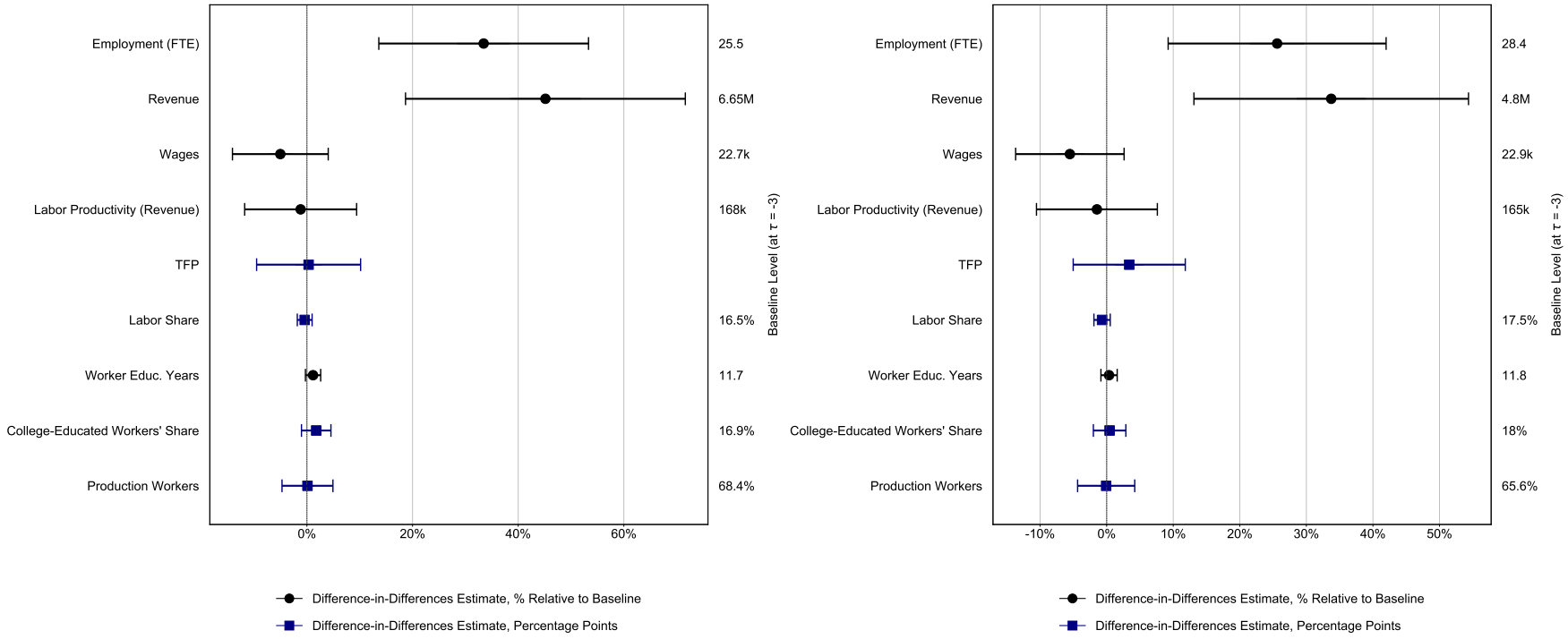
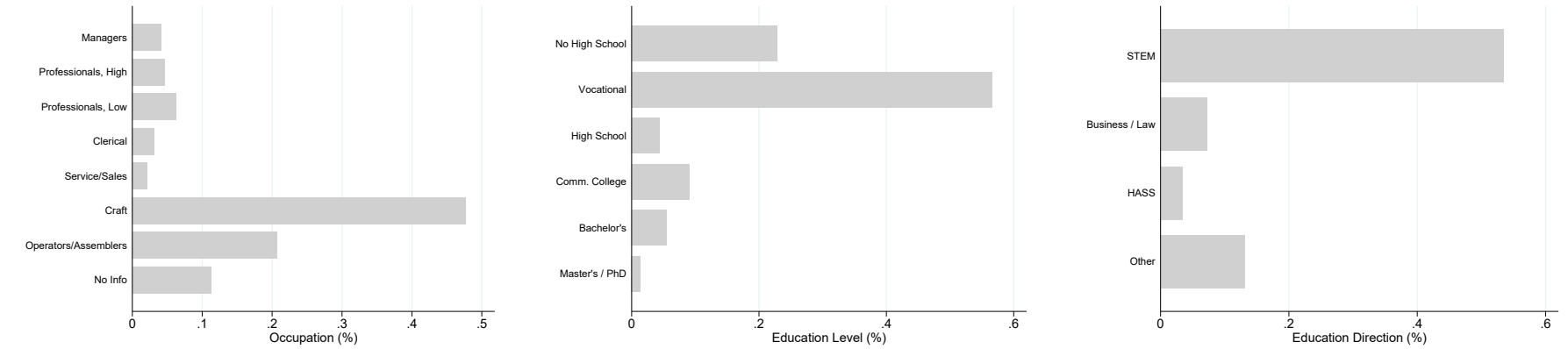


Figure A21: Firm-Level Effects: Automated (left) vs. Non-Automated (right) Technologies from Customs Data.

Notes: Difference-in-differences estimates from Equation 2. Automated vs. non-automated technologies are measured from customs data as described in Section IV and Appendix F. A project is classified as automated if over 50% of the imported machinery are automated technologies. A project is classified as non-automated if over 50% of the imported machinery are non-automated technologies. Automated (N): Treatment 220, Control 146. Non-Automated (N): Treatment 319, Control 146. Back to Section VI.



(A) Occupations. (B) Education Levels. (C) Education Types.

Figure A22: Sample Worker's Occupations and Education.

Notes: The figures show the distribution of sample workers' 1-digit occupations, and education levels and types. The sample is the main analysis sample (winners–losers design). The shares are unweighted means of the sample firms at $\tau = -3$. We study production work: a vast majority of the workers in the sample firms are craftworkers, operators, and assemblers. The mean share of production workers in the sample firms is approximately 70% of all workers. Notably, the share of clerks and other operation support workers and workers in sales is low. Most workers in the sample hold a vocational school degree or only a primary school degree. The share of workers with a bachelor's degree or higher is low, accounting for less than 20%. A majority, over 50%, of the degrees the workers in the sample firms hold are in STEM fields. Note that the shares in each subfigure do not add up to hundred percent because not all workers have data on occupation or education. Back to Section VIII.

Table A1: Summary Statistics: Benchmarking Subsidy Sample to All Manufacturing.

	Subsidy Sample				**Finnish Manufacturing**			
Variable	Mean	p10	Median	p90	Mean	p10	Median	p90
Revenue (EUR M)	3.09	0.17	0.96	5.67	7.26	0.20	0.63	5.46
Employment	17.22	2.20	7.90	37.00	25.69	2.50	5.50	32.80
Wages (EUR K)	21.95	13.72	22.30	31.40	24.84	14.98	24.17	34.59
Labor Productivity (EUR K)	146.73	72.35	118.60	228.49	150.63	62.51	114.95	246.23
Labor Share (%)	18.45	6.53	18.03	30.70	25.86	7.98	20.09	36.74
Profit Margin (%)	5.22	-3.44	4.97	18.51	-15.10	-6.75	4.12	18.03
Employment Change (%, Five Year)	88.74	-26.61	20.67	165.83	90.27	-31.67	16.67	180.95
Revenue Change (%, Five Year)	100.69	-20.08	37.75	218.08	211.46	-33.51	27.08	220.78
Subsidy Applied (EUR K)	107.38	8.89	58.13	290.06	4.74	0.00	0.00	0.00
Subsidy Granted (EUR K)	75.89	3.24	35.64	200.23	2.61	0.00	0.00	0.00
Educ. Years	11.69	10.50	11.74	12.67	11.61	10.29	11.67	12.81
College Share (%)	15.28	0.00	12.50	33.33	14.76	0.00	9.09	38.63
Production Worker Share (%)	70.74	44.08	72.73	100.00	69.98	33.33	74.19	100.00
Number of Observations	2031				254,008			
Number of Unique Firms	2031				19,661			

Notes: Manufacturing means are measured for each firm-year and pooled together over 1994–2018. Manufacturing firms include all firms that satisfy the subsidy sample's balance-sheet-based restrictions and have over two full-time employees. The subsidy sample is measured at event-time $\tau = -3$, except for subsidies applied and received which are sums over $\tau \in [0, 2]$. The subsidy variables for the manufacturing sample are sums for the firm for its overall life span. The summary statistics are not winsorized. Back to Sections III and IX.

Table A2: Summary Statistics: Text Matching Using Cosine Similarity.

	Treatment Group		**Control Group**		**Both**			**Tests**
Variable	Mean	Std. Dev.	Mean	Std. Dev.	p10	Median	p90	p-value
Machine Investment (EUR K)	96.86	240.06	91.43	229.23	0.00	20.03	201.79	0.525
Revenue (EUR M)	2.26	4.44	1.68	3.85	0.13	0.72	4.68	0.000
Employment	15.77	26.04	11.15	24.65	1.10	5.90	27.40	0.000
Wages (EUR K)	21.24	8.15	19.28	10.29	6.73	21.27	29.23	0.000
Labor Share (%)	18.49	9.31	15.76	10.59	3.54	16.62	30.65	0.000
Labor Productivity (EUR K)	138.81	100.83	151.79	93.55	73.51	121.83	231.61	0.000
Subsidy Applied (EUR K)	110.02	128.33	64.64	105.44	4.60	38.35	241.32	0.000
Subsidy Granted (EUR K)	78.31	99.14	0.00	0.00	0.00	0.34	124.65	0.000
Educ. Years	11.67	0.98	11.42	1.04	10.50	11.63	12.50	0.000
College Share (%)	15.18	16.75	11.05	16.30	0.00	10.30	33.33	0.000
Production Worker Share (%)	70.62	22.17	72.65	27.18	40.00	75.00	100.00	0.215
F-test								0.000
Observations	1508		1508		3016			3016

Notes: All variables measured at $\tau = -3$, expect for subsidies applied and received which are sums over $\tau \in [0, 2]$. The summary statistics are not winsorized. We construct a matched sample for the winners by selecting the nearest neighbor from the losing firms with replacement. Back to Section III.

Table A3: Examples of Application Texts Classified as Technology.

Technology:

- Machinery and equipment.
- Vertical machining center.
- Laser welding technology investment.
- Acquisition of Mazak Multiplex 6200 machine tool.
- Acquisition of laser cutter, measuring equipment, and edger steel sets, obtaining the CE SFS-EN 1090-2 certificate for the manufacturing of steel structures, and improving the utilization rate of production facilities.
- Streamlining the production of demonstration samples. The goal of the project is to invest in and improve the manufacturing of demonstration samples to meet the demand from the market. The objective is to have the enhanced equipment for the production of demonstration samples operational by the end of August.

Not Technology:

- Internationalization project in Germany.
- Market research in the USA. The goal of the project is to assess market opportunities in the USA, including market size, pricing and distribution structures, product adaptation needs, consumer preferences, and requirements for market entry.
- Product development project.
- Development of a fuel-efficient snow blower.
- Start-up support.
- Launch of a short-term care and rehabilitation facility.

Notes: Examples of what is classified as a technology and what is not from the application data. The first four of technology examples are one-sentence versions of the description while the latter two are the full description texts available in the data. The second example of not technology is also the full description text. The Appendix F presents additional examples. Back to Section IV.

Table A4: Firm-Level Effects: Different Text Matching Versions.

Panel A: Coarsened Exact Matching (CEM).

	(1) Machine Inv. (EUR K)	(2) Employment	(3) Revenue	(4) Educ. Years	(5) College Share	(6) Production Worker Share
Treatment	95.82***	0.260***	0.332***	-0.0501	-0.000688	-0.0105
	(27.41)	(0.0684)	(0.0944)	(0.0664)	(0.0106)	(0.0206)
Observations	1254	1254	1254	1161	1161	1162

Panel B: Inverse Probability Weighting (IPW).

	(1) Machine Inv. (EUR K)	(2) Employment	(3) Revenue	(4) Educ. Years	(5) College Share	(6) Production Worker Share
Treatment	146.0***	0.358***	0.455***	-0.0430	0.00569	-0.0274
	(31.50)	(0.0904)	(0.116)	(0.0846)	(0.0161)	(0.0301)
Observations	1812	1812	1812	1676	1676	1692

Panel C: Cosine Similarity.

	(1) Machine Inv. (EUR K)	(2) Employment	(3) Revenue	(4) Educ. Years	(5) College Share	(6) Production Worker Share
Treatment	112.7***	0.169***	0.195***	0.0133	-0.00224	0.00278
	(10.50)	(0.0249)	(0.0335)	(0.0219)	(0.00542)	(0.0103)
Observations	3016	3016	3016	2678	2678	2715

Standard errors in parentheses.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Notes: Difference-in-differences estimates from Equation 2 with different text matching versions. Details in the main text. Back to Section V.

Table A5: Firm-Level Effects: Robustness to Different Controls.

	(1) Machine Inv. (EUR K)	(2) Employment	(3) Revenue	(4) Wages	(5) Productivity	(6) Labor Share	(7) Educ. Years	(8) College Share	(9) Prod. Work. Share
No Controls	114.3***	0.185**	0.261***	-0.0553	-0.00559	-0.00242	0.0225	0.00480	-0.0235
	(22.25)	(0.0606)	(0.0770)	(0.0353)	(0.0341)	(0.00492)	(0.0599)	(0.00927)	(0.0359)
Controls 1	110.7***	0.219***	0.302***	-0.0499	-0.00379	-0.00247	0.0252	0.00587	-0.0263
	(22.53)	(0.0615)	(0.0779)	(0.0356)	(0.0351)	(0.00496)	(0.0611)	(0.00936)	(0.0357)
Controls 2	103.8***	0.232***	0.314***	-0.0481	-0.00516	-0.00202	0.0246	0.00557	-0.0256
	(22.56)	(0.0614)	(0.0779)	(0.0355)	(0.0350)	(0.00496)	(0.0611)	(0.00935)	(0.0357)
Controls 3	100.4***	0.249***	0.327***	-0.0385	-0.00670	-0.000862	0.0252	0.00572	-0.0255
	(22.43)	(0.0609)	(0.0773)	(0.0350)	(0.0349)	(0.00490)	(0.0612)	(0.00942)	(0.0363)
Controls 4	67.69 **	0.210***	0.284***	-0.0344	-0.00658	-0.000101	0.0247	0.00509	-0.0268
	(23.64)	(0.0607)	(0.0770)	(0.0351)	(0.0350)	(0.00493)	(0.0614)	(0.00946)	(0.0363)
Controls 5	56.89*	0.191**	0.259**	-0.0360	-0.0141	0.00186	0.0181	0.00374	-0.00103
	(24.05)	(0.0623)	(0.0796)	(0.0354)	(0.0356)	(0.00505)	(0.0629)	(0.00976)	(0.0185)
Observations	2031	2031	2031	1952	2031	2031	1884	1884	1891

Standard errors in parentheses.
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Notes: Difference-in-differences estimates from Equation 2 with different controls.

Controls 1: industry (2-digit).

Controls 2: industry (2-digit), employment (at the base year).

Controls 3: industry (2-digit), employment (at the base year), ELY Center indicators.

Controls 4: industry (2-digit), employment (at the base year), ELY Center indicators, applied subsidy amount.

Controls 5: industry (2-digit), employment (at the base year), ELY Center indicators, applied subsidy amount, text category indicators.

Back to Section V.

Table A6: Continuous Treatment Estimates.

	(1) Machine Inv.		(2) Employment		(3) Revenue	
Granted Amount	1.011***	0.973***	0.249***	0.230***	5.292***	4.973***
	(0.115)	(0.118)	(0.0213)	(0.0220)	(0.468)	(0.478)
Propensity Score		✓		✓		✓
Observations	2031	1812	2031	1812	2031	1812

Standard errors in parentheses.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Notes: Difference-in-differences estimates from Equation 2. Treatment is the subsidy amount in EUR, scaled to EUR 10K for employment. For machinery investment, the post-period outcome is the sum of investment between $\tau \in [0, 2]$ and for other outcomes, the average over $\tau \in [2, 5]$. The pre-period for the estimations is $\tau = -3$. The specifications include two-digit industry and firm size as controls. Back to Section VI.

Table A7: Continuous Treatment Estimates: Controlling for the Subsidies Applied.

	(1) Machine Inv. (EUR K)		(2) Employment		(3) Revenue	
Granted Subsidy	0.915***	0.960***	0.129**	0.140**	1.546	2.074*
	(0.228)	(0.239)	(0.0464)	(0.0500)	(0.960)	(1.038)
Applied Subsidy	✓	✓	✓	✓	✓	✓
Propensity Score		✓		✓		✓
Observations	2031	1812	2031	1812	2031	1812

Standard errors in parentheses.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Notes: Difference-in-differences estimates from Equation 2. Treatment is the received subsidy amount in EUR. Treatment is scaled to EUR 10K for employment. Applied subsidy is the applied subsidy amount in EUR. Machinery investment is the sum between $\tau \in [0, 2]$. Other outcomes are averages over $\tau \in [2, 5]$. Back to Section V.

Table A8: Firm-Level Effects: Propensity-Score Trimmed Samples.

Panel A: Investment, Employment, Wages, and Firm Performance.

	(1)		(2)		(3)		(4)		(5)	
	Machine Inv. (EUR K)		Employment		Revenue		Wages		Profit Margin	
5%	108.0**	105.5**	0.245**	0.248**	0.301**	0.327**	-0.0124	-0.00497	-0.00420	-0.00628
	(34.22)	(34.76)	(0.0815)	(0.0828)	(0.116)	(0.118)	(0.0439)	(0.0441)	(0.0118)	(0.0119)
10%	115.8**	113.2**	0.251**	0.254**	0.313*	0.324*	-0.00737	-0.00578	-0.00304	-0.00472
	(40.07)	(40.16)	(0.0953)	(0.0956)	(0.132)	(0.133)	(0.0472)	(0.0473)	(0.0134)	(0.0134)
20%	76.47	76.89	0.188	0.188	0.242	0.242	0.00738	0.00652	0.000227	0.000249
	(51.98)	(51.88)	(0.124)	(0.125)	(0.174)	(0.174)	(0.0561)	(0.0559)	(0.0151)	(0.0152)
Propensity Score		✓		✓		✓		✓		✓
N, 5%	1631	1631	1631	1631	1631	1631	1570	1570	1631	1631
N, 10%	1449	1449	1449	1449	1449	1449	1395	1395	1449	1449
N, 20%	1088	1088	1088	1088	1088	1088	1049	1049	1088	1088

Panel B: Skill Composition, Productivity, and the Labor Share.

	(1)		(2)		(3)		(4)		(5)	
	Productivity		Labor Share		Educ. Years		College Share		Production Worker Share	
5%	-0.0311	-0.0203	0.00105	0.000261	-0.0636	-0.0499	0.00144	0.00226	-0.0316	-0.0309
	(0.0489)	(0.0494)	(0.00677)	(0.00686)	(0.0900)	(0.0911)	(0.0136)	(0.0138)	(0.0227)	(0.0229)
10%	-0.0314	-0.0294	0.000653	0.000813	-0.0273	-0.0251	0.00519	0.00449	-0.0496*	-0.0489
	(0.0553)	(0.0554)	(0.00759)	(0.00765)	(0.102)	(0.103)	(0.0154)	(0.0154)	(0.0252)	(0.0253)
20%	-0.0193	-0.0193	0.00113	0.00106	-0.0281	-0.0286	0.00570	0.00562	-0.0128	-0.0125
	(0.0681)	(0.0682)	(0.00924)	(0.00927)	(0.128)	(0.128)	(0.0180)	(0.0181)	(0.0296)	(0.0296)
Propensity Score		✓		✓		✓		✓		✓
N, 5%	1631	1631	1631	1631	1519	1519	1519	1519	1533	1533
N, 10%	1449	1449	1449	1449	1352	1352	1352	1352	1366	1366
N, 20%	1088	1088	1088	1088	1018	1018	1018	1018	1030	1030

Standard errors in parentheses.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Notes: The estimated effects for the main sample where firms with top and bottom 5%, 10%, and 20% of propensity score values are dropped. Our main results are robust to excluding firms with small and large values of the propensity score. The samples are subsets of the main analysis sample (winners–losers design). Back to Sections III and V.

Table A9: Robustness to Different Sample Specifications: Single and First Events.

	(1)		(2)		(3)	
A. Single Event	Machine Inv. (EUR K)		Employment		Educ. Years	
Treatment	51.79**	56.67*	0.165*	0.188*	0.0178	-0.0319
	(18.24)	(24.15)	(0.0665)	(0.0805)	(0.0624)	(0.0782)
B. First Event						
Treatment	81.67***	87.79**	0.286***	0.312***	0.00550	-0.0350
	(20.03)	(26.96)	(0.0635)	(0.0776)	(0.0539)	(0.0659)
Propensity Score		✓		✓		✓
N, Single	1011	888	1011	888	937	819
N, First	1980	1668	1980	1668	1895	1589

Standard errors in parentheses.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Notes: Difference-in-differences estimates from Equation 2. **Panel A**: We restrict the sample to firms with only one (either successful or unsuccessful) subsidy application. **Panel B**: We select the first subsidy application in the data to serve as the treatment or control. The specifications include two-digit industry and firm size as controls, and are presented with and without the text propensity-score control. For machinery investment, the post-period outcome is the average of investment over $\tau \in [0, 2]$ multiplied by three (the number of periods) and for other outcomes, the average over $\tau \in [2, 5]$. Back to Section V.

Table A10: Robustness to an Unbalanced Sample: Firm-Level Effects Allowing for Firm Exit.

Panel A: Investment, Employment, Wages, and Firm Performance.

	(1) Machine Inv. (EUR K)		(2) Employment		(3) Revenue		(4) Wages		(5) Profit Margin	
Treatment	73.04**	93.17***	0.310***	0.268***	0.400***	0.364***	-0.0371	-0.0442	0.000834	-0.0125
	(22.40)	(26.43)	(0.0547)	(0.0694)	(0.0667)	(0.0849)	(0.0372)	(0.0445)	(0.00782)	(0.00996)
Propensity Score		✓		✓		✓		✓		✓
Observations	2118	1880	2118	1880	2118	1880	1977	1754	2060	1831

Panel B: Skill Composition, Labor Share, and Productivity.

	(1) Productivity		(2) Labor Share		(3) Educ. Years		(4) College Share		(5) Production Worker Share	
Treatment	-0.00742	-0.0148	0.000989	0.00210	-0.0338	-0.0610	-0.00531	-0.00562	0.00735	-0.00679
	(0.0345)	(0.0417)	(0.00500)	(0.00620)	(0.0513)	(0.0649)	(0.00836)	(0.0106)	(0.0181)	(0.0213)
Propensity Score		✓		✓		✓		✓		✓
Observations	2056	1828	2060	1831	1953	1733	1912	1697	1896	1708

Standard errors in parentheses.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Notes: The sample is the main analysis sample (winners–losers design) without the balanced panel requirement. For the firms that exited, the first three outcomes in Panel A are defined as zero, all others are defined as missing. Back to Sections III and V.

Table A11: Effects on Export Products' and Regions' Skill Intensity.

	(1) Product Skill Intensity	(2) Region Skill Intensity
Treatment	-0.0267	-0.00139
	(0.0599)	(0.0316)
Baseline	12.64	12.87
N	401	401

Standard errors in parentheses.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Notes: The effects on the skill intensity of exported products and export regions. We use data on all Finnish exporting firms to construct the outcomes. We first compute the average worker education years for each export product and region by taking the average over all years from firms that export the given product or export to the given region. Then for each exporting firm in our sample, we calculate the skill intensity each year by taking the unweighted average of the skill intensities of the products the firm exports that year or the regions it exports to. Export regions and products are measured from the Finnish Customs' Foreign Trade Statistics. A concern about the lack of skill-bias effects in our sample is that it exists, but is subtle and hard to find empirically. One way to explore this possibility is to estimate whether, after adopting new technologies, the firms export products which require more skills or export to regions that do. If this is true, the firms are likely also to exhibit an increased need for skills, even if we do not detect these effects in the short term. This table explores these effects on export products' and regions' skill intensity. The coefficients on both outcomes are fairly precise zeros, implying that the hypothesis of undetected skill bias through this channel does not receive support. Back to Section VI.

Table A12: Heterogeneous Effects by Firms' Intentions.

Subgroup	Employment	Educ. Years	Labor Share	Productivity	N
All	0.217***	0.030	-0.003	0.012	3,200
	(0.018)	(0.021)	(0.002)	(0.012)	
	17.7	11.6	0.185	144,204	
Work Only	0.165*	0.220*	-0.012	0.027	138
	(0.068)	(0.094)	(0.010)	(0.057)	
	21.0	11.5	0.188	132,724	
Productivity Only	0.143***	0.064	-0.004	0.045	542
	(0.043)	(0.050)	(0.005)	(0.029)	
	22.9	11.6	0.188	141,218	
Work or Productivity Only	0.143***	0.068	-0.005	0.047	562
	(0.042)	(0.048)	(0.005)	(0.029)	
	22.5	11.6	0.187	142,013	
Excluding Work or Productivity Only	0.234***	0.021	-0.003	0.008	2,638
	(0.020)	(0.023)	(0.002)	(0.013)	
	16.7	11.6	0.184	144,671	

Standard errors in parentheses and baseline means on every third row.
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Notes: The table presents estimates by subgroup using the matched control group. Each estimate is from a separate firm-level regression. The **All** category contains all applications as a reference. The **Work Only** category contains applications that mention a work-related motive but no product changes (new products, quality improvements, capabilities/flexibility, or precision). The **Productivity Only** category similarly contains applications that mention a productivity-related motive but no product changes. The **Work or Productivity Only** category is the union of these two. **Excluding Work or Productivity Only** is the complement: It excludes the Work or Productivity Only category from all applications. Descriptions of the underlying narrow text categories are provided in Table 5 and Appendix F. Outcomes: Employment and productivity are in relative changes, e.g., 0.20 would refer to a 20% increase. Education years is in years. Labor share is in percentage points. Post-period values are calculated as the average over $\tau \in [2, 5]$. The pre-period for the estimations is $\tau = -3$. Back to Section VI.

Table A13: The First Stage: IT Expenditure.

	(1)	(2)	(3)	(4)
	IT Expenditure			
Treatment (Indicator)	4.101**	2.395		
	(1.355)	(1.831)		
Granted Subsidy (Continuous)			0.0516***	0.0489***
			(0.0073)	(0.0073)
Propensity Score		✓		✓
Observations	2031	1812	2031	1812

Standard errors in parentheses.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

The baseline IT expenditure is 46.2K EUR at $\tau = -3$.

Notes: Difference-in-differences estimates from Equation 2 with and without the text propensity control. To measure IT, we use the official records on the Financial Statement Register. The post-period outcomes are means over $\tau \in [0, 2]$ multiplied by three (the number of periods). The units are EUR K (Columns 1 and 2) and EUR (Columns 3 and 4). The specifications include two-digit industry and firm size as controls. Back to Section VI.

Table A14: Summary Statistics by IT Status.

A. Register Data

	IT above median		IT below median		Tests
Variable	Mean	Std. Dev.	Mean	Std. Dev.	p-value
Machine Investment (EUR K)	105.15	447.60	110.78	246.22	0.725
Revenue (EUR M)	4.10	34.39	2.08	4.23	0.063
Employment	20.63	61.70	13.82	19.40	0.001
Wages (EUR K)	23.05	10.23	20.85	7.94	0.000
Labor Share (%)	18.77	9.71	18.13	9.11	0.127
Labor Productivity (EUR K)	150.43	189.93	143.03	121.92	0.296
Subsidy Applied (EUR K)	96.53	121.23	118.22	132.67	0.000
Subsidy Granted (EUR K)	66.63	95.67	85.14	106.20	0.000
Educ. Years	11.78	0.99	11.60	1.00	0.000
College Share (%)	16.67	17.31	13.86	16.45	0.000
Production Worker Share (%)	69.17	22.40	72.58	20.86	0.028
Observations	1015		1016		2031

B. Text Data

	Software mentioned		Machinery mentioned		Tests
Variable	Mean	Std. Dev.	Mean	Std. Dev.	p-value
Machine Investment (EUR K)	134.89	308.26	108.10	362.15	0.440
Revenue (EUR M)	4.15	6.27	3.06	24.86	0.653
Employment	27.71	32.99	17.03	46.18	0.016
Wages (EUR K)	25.50	8.93	21.81	9.19	0.000
Labor Share (%)	21.11	9.34	18.45	9.46	0.003
Labor Productivity (EUR K)	185.39	496.94	146.17	161.47	0.010
Subsidy Applied (EUR K)	95.88	105.48	108.67	128.66	0.316
Subsidy Granted (EUR K)	66.96	87.50	76.99	102.47	0.323
Educ. Years	11.85	1.03	11.69	0.99	0.092
College Share (%)	18.87	16.75	15.19	16.94	0.026
Production Worker Share (%)	64.30	22.14	70.49	21.93	0.027
Observations	107		1971		2007

Notes: Summary statistics by IT expenditure and mentions of machinery vs. software in the application texts. The sample is the main analysis sample (winners vs. losers design). **Panel A:** The groups are defined by IT expenditure over $\tau \in [0, 2]$ in the register data. **Panel B:** The groups are defined by whether the application text data mentions software or machinery. The summary statistics are not winsorized. Back to Section IX.

Table A15: Firm-Level Effects by IT Status.

	(1)		(2)		(3)		(4)		(5)	
A. Register Data	Machinery Inv. (EUR K)		Employment		Educ. Years		Labor Share		Productivity	
IT below median	93.99*	85.38	0.288**	0.298**	-0.0498	-0.0183	0.00273	0.00503	0.0169	0.0475
	(41.16)	(55.57)	(0.0895)	(0.111)	(0.0919)	(0.118)	(0.00735)	(0.00942)	(0.0557)	(0.0659)
IT above median	60.57*	59.07	0.164	0.151	0.0892	0.0236	-0.00477	-0.00432	-0.0137	-0.0396
	(23.84)	(30.69)	(0.0849)	(0.102)	(0.0817)	(0.0957)	(0.00683)	(0.00800)	(0.0472)	(0.0576)
B. Text Data										
Machinery mentioned	95.66***	92.40**	0.234***	0.238**	0.00755	0.00240	-0.00151	-0.000185	-0.0137	-0.0101
	(23.07)	(29.62)	(0.0634)	(0.0761)	(0.0611)	(0.0754)	(0.00506)	(0.00610)	(0.0360)	(0.0435)
Software mentioned	115.1	108.6	0.329	0.450	-0.0938	-0.0685	0.00929	-0.0168	-0.113	0.00135
	(229.2)	(269.9)	(0.253)	(0.371)	(0.343)	(0.441)	(0.0264)	(0.0308)	(0.237)	(0.277)
Propensity Score		✓		✓		✓		✓		✓
N, Below Median IT	1016	886	1016	886	947	822	1016	886	1016	886
N, Above Median IT	1015	926	1015	926	937	854	1015	926	1015	926
N, Software Not Mentioned	1971	1758	1971	1758	1827	1625	1971	1758	1971	1758
N, Software Mentioned	107	98	107	98	105	96	107	98	107	98

Standard errors in parentheses.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Notes: The effects by IT expenditure and mentions of machinery vs. software in the application texts. The sample is a subset of the main analysis sample (winners vs. losers design). **Panel A**: The groups are below and above median IT expenditure over $\tau \in [0, 2]$. The results are robust to defining the groups by IT expenditure at $\tau = 3$. **Panel B**: We divide the sample by the firms' stated the intention to purchase hardware or software technologies in the application texts. The text-based categories are not mutually exclusive, so that an application can include both intended hardware and software investments and thus appear in both categories. The results for the text-based software events are imprecise, likely due to the small sample size (n = 107). Back to Section VII.

Table A16: Firm-Level Effects: Specific Types and Uses of Technologies Measured from the Text Data.

	(1) Machine Inv.	(2) Employment	(3) Revenue	(4) Wages	(5) Productivity	(6) Labor Sh.	(7) Educ. Years	(8) College Sh.	(9) Prod. Sh.	(10) N
Types of Technology										
CNC	134.0***	0.168***	0.180***	0.0367	-0.0100	0.00299	-0.0331	0.00304	-0.0211	628
	(16.87)	(0.0398)	(0.0492)	(0.0212)	(0.0266)	(0.00498)	(0.0513)	(0.00771)	(0.0144)	
Robot	207.3*	0.233*	0.416**	-0.0380	0.0124	-0.0133	0.0311	0.00256	0.0198	232
	(79.94)	(0.0921)	(0.136)	(0.0291)	(0.0593)	(0.00787)	(0.0581)	(0.00961)	(0.0192)	
Laser	110.2*	0.322**	0.313*	-0.0272	-0.0831	0.000777	0.0578	0.0180	0.0139	224
	(46.13)	(0.0979)	(0.132)	(0.0415)	(0.0501)	(0.0101)	(0.0919)	(0.0157)	(0.0286)	
Uses of Technology										
Machining	165.7***	0.246***	0.276***	-0.00449	-0.0187	-0.00126	0.0136	0.00862	-0.0121	584
	(30.65)	(0.0471)	(0.0663)	(0.0227)	(0.0302)	(0.00514)	(0.0505)	(0.00899)	(0.0137)	
Welding	103.8***	0.352***	0.385***	-0.00611	-0.0146	-0.0100	0.0185	-0.00120	0.00966	312
	(28.05)	(0.0835)	(0.0821)	(0.0352)	(0.0431)	(0.00760)	(0.0700)	(0.0122)	(0.0218)	
Painting	173.6***	0.267***	0.318***	-0.0223	0.00608	-0.00591	-0.0112	-0.00147	-0.00148	312
	(38.18)	(0.0634)	(0.0836)	(0.0302)	(0.0396)	(0.00636)	(0.0627)	(0.00946)	(0.0193)	
Logistics	163.5***	0.304***	0.404***	0.00781	0.0348	-0.00659	0.0207	0.0148*	-0.0106	822
	(22.48)	(0.0404)	(0.0544)	(0.0171)	(0.0255)	(0.00388)	(0.0370)	(0.00611)	(0.0108)	
Automation	136.3***	0.178***	0.217***	0.00216	0.0249	-0.00237	0.0546	0.00942	0.000592	678
	(29.12)	(0.0350)	(0.0446)	(0.0189)	(0.0259)	(0.00422)	(0.0391)	(0.00650)	(0.0113)	

Standard errors in parentheses.
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Notes: The effects of specific types and uses of technology on selected outcomes for subsets of treatment firms matched to non-applicant control firms. The first three rows show the effects for treatment firms that intend to buy the given technology, and the latter five for the firms that specify the listed uses for the technologies. We define the technologies using keywords: "cnc" for CNC, "robo" for robots, and "laser" and "plasma" for laser. The remaining keywords (in Finnish) are provided in Appendix F. N (Column 10) refers to the treatment group sample size: the number of firms with subsidy application texts containing keywords associated with the given technology or its use. We find broadly similar effects for all the subgroups. Machinery investment is in EUR K. The sample is the winners matched to non-applicants as the sample would otherwise be small (the matching procedure is described in Section III). Back to Section VI.

Table A17: Technology × Occupation Pairs: Effects of Specific Technologies on Specific Workers.

	(1) Worker Share	(2) Wages	(3) Educ. Years	(4) Labor Share
Machining-Machinists	0.0454***	0.0753	-0.00340	0.0360*
	(0.0106)	(0.0816)	(0.0179)	(0.0158)
Mean	0.0227	25751.0	11.78	0.107
N	554	51	51	51
Welding-Welders	0.0135	0.0598	0.0295	0.00971
	(0.00995)	(0.0938)	(0.0214)	(0.00839)
Mean	0.0683	25831.3	11.48	0.0594
N	300	88	88	88
Painting-Painters	0.00518	0.0114	0.0258	0.00825
	(0.00358)	(0.0750)	(0.0279)	(0.00590)
Mean	0.0250	21076.8	10.58	0.0411
N	307	65	65	65
Logistics-Logistics (Non-Office)	-0.000775	-0.0936	0.0413	-0.00759
	(0.000488)	(0.144)	(0.0524)	(0.00699)
Mean	0.0120	25330.9	10.76	0.0191
N	799	58	58	58
Logistics-Logistics (Office)	0.0000555	0.298	-0.0884	-0.000707
	(0.000246)	(0.301)	(0.112)	(0.00189)
Mean	0.00399	29623.5	11.70	0.00696
N	799	40	40	40

Standard errors in parentheses.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Notes: The firm-level effects of specific technologies on specific workers. The sample is winners matched to non-applicants (the matching procedure is described in Section III). The technologies refer to the description of the technology in the subsidy application text (the technology keywords are listed in Appendix F). The occupations come from the workers' occupational records (exact occupational codes available in the replication package). The technology-to-worker pairs are: (1) machining-machinists, (2) welding-welders, (3) painting-painters, and (4-5) logistics words (e.g., "driving," "hoisting") to logistics occupations (non-office and office). N refers to the number of treatment firms where the given outcome is defined. Note that worker count (employment) shares are defined for all firms with the given technology, but the other three outcomes require at least one worker with the specific occupation and wages observed both before and after, and the smaller sample size comes from combining these restrictions. We find positive effects on the employment share and the wage-bill (labor) share of machinists when the firm has applied for a subsidy and specified the intended use to be an investment in technologies associated with machining. Back to Section VI.

Table A18: Firm-Level Effects: Firm Size.

Panel A: Large Firms.

	(1) Machine Inv. (EUR K)		(2) Employment		(3) Revenue		(4) Productivity		(5) Labor Share		(6) College Share	
Treatment	221.8*	158.9	0.305***	0.309**	0.264	0.494**	-0.136	0.0236	0.0133	-0.00853	-0.00893	-0.0159
	(94.69)	(129.0)	(0.0722)	(0.104)	(0.137)	(0.160)	(0.0834)	(0.0967)	(0.00981)	(0.0111)	(0.0167)	(0.0201)
Propensity Score		✓		✓		✓		✓		✓		✓
N	676	609	676	609	676	609	676	609	676	609	675	608

Panel B: Medium-Sized Firms.

	(1) Machine Inv. (EUR K)		(2) Employment		(3) Revenue		(4) Productivity		(5) Labor Share		(6) College Share	
Treatment	99.31	132.8*	0.296***	0.280*	0.467***	0.399**	0.0707	0.0193	-0.0104	-0.0124	0.0185	0.0200
	(52.09)	(64.17)	(0.0858)	(0.113)	(0.114)	(0.150)	(0.0551)	(0.0718)	(0.00856)	(0.00969)	(0.0162)	(0.0213)
Propensity Score		✓		✓		✓		✓		✓		✓
N	685	603	685	603	685	603	685	603	685	603	683	601

Panel C: Small Firms.

	(1) Machine Inv. (EUR K)		(2) Employment		(3) Revenue		(4) Productivity		(5) Labor Share		(6) College Share	
Treatment	22.73	25.83	0.330**	0.373**	0.355**	0.370*	-0.0410	-0.0956	0.00216	0.0162	0.00334	0.00373
	(14.06)	(19.78)	(0.103)	(0.121)	(0.125)	(0.148)	(0.0526)	(0.0615)	(0.00781)	(0.00927)	(0.0158)	(0.0192)
Propensity Score		✓		✓		✓		✓		✓		✓
N	670	600	670	600	670	600	670	600	670	600	526	467

Standard errors in parentheses.
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Notes: Difference-in-differences estimates from Equation 2. The firm size groups are defined by FTE at $\tau = 3$, with thresholds at the 33.3 and 66.6 percentiles. Large Firms (FTE > 13.3; Median 25.8, Mean 41.7), Medium-Sized Firms (FTE >= 4.6 & FTE <= 13.3; Median 7.9, Mean 8.2), Small Firms (FTE < 4.6; Median 2.3, Mean 2.3). Back to Section VII.

Table A19: Credit Constraints: Estimates by the Cost of Capital and Debt Level.

Panel A: Effects by Average Financial Costs.

	(1) Machine Inv. (EUR K)		(2) Employment		(3) Revenue	
	High Costs	Low Costs	High Costs	Low Costs	High Costs	Low Costs
Treatment	128.0***	75.07***	0.276***	0.194*	0.313**	0.326**
	(33.55)	(29.60)	(0.0793)	(0.0959)	(0.105)	(0.114)
N	1016	1015	1016	1015	1016	1015

Panel B: Effects by Relative Debt.

	(1) Machine Inv. (EUR K)		(2) Employment		(3) Revenue	
	High Debt	Low Debt	High Debt	Low Debt	High Debt	Low Debt
Treatment	140.3***	63.38*	0.0676	0.384***	0.151	0.486***
	(32.24)	(30.52)	(0.0965)	(0.0687)	(0.125)	(0.0820)
N	1016	1015	1016	1015	1016	1015

Panel C: Controlling for Credit Constraint Measures.

	(1) Machine Inv. (EUR K)			(2) Employment			(3) Revenue		
Treatment	103.8***	103.4***	104.7***	0.232***	0.242***	0.232***	0.314***	0.342***	0.314***
	(22.56)	(22.62)	(22.70)	(0.0614)	(0.0597)	(0.0614)	(0.0779)	(0.0721)	(0.0778)
Relative Debt		✓			✓			✓	
Average Financial Costs			✓			✓			✓
N	2031	2031	2031	2031	2031	2031	2031	2031	2031

Standard errors in parentheses.
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Notes: Difference-in-differences estimates from Equation 2. Estimated effects on selected outcomes by the cost of capital (Panel A), debt level (Panel B), and with these proxies for credit-constraints as controls (Panel C). We measure baseline levels at $\tau = -3$. Average financial costs are financial expenses divided by non-current liabilities. Relative debt is the sum of current liabilities, non-current liabilities, and obligatory reserves divided by revenue. We divide the sample into two groups by whether the firms' average financial costs (Panel A) or relative debt (Panel B) are below or above the median in the sample. Panel C controls directly for the baseline value. Back to Section VII.

Table A20: Describing the Treatment Firms' New Workers.

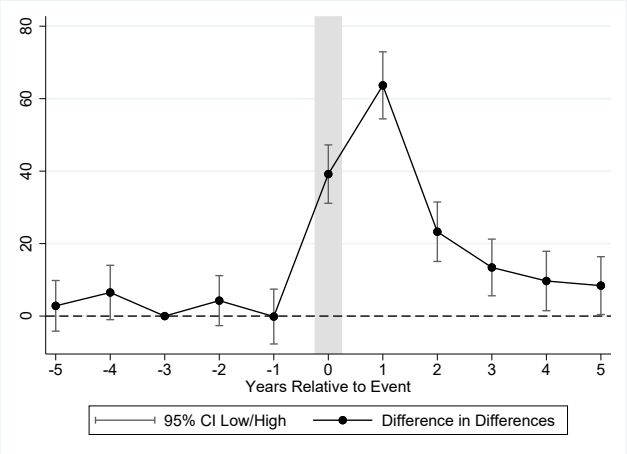
Variable	Mean
A. Origin Firm Characteristics:	
Employment	451.48
Labor Productivity (EUR K)	184.24
Average Education Years	12.04
Labor Share (%)	26.45
Average Wage (EUR K)	28.37
Survival (%)	93.00
B. Origin Firm Treatment Status:	
Other Winners (%)	0.65
Losers (%)	0.02
C. Prior Activity and Demographics:	
Employed (%)	53.63
Student (%)	21.28
Retraining (%)	3.54
Military service (%)	3.20
Retired (%)	1.64
Unemployed (%)	13.53
Other Non-Employed (%)	5.49
Male (%)	80.21
Age	33.77
Foreign-born (%)	4.13
D. Occupation, Industry, and Location:	
Same Occupation (%)	69.33
Same Industry (%)	50.97
Same County of Prior Firm (%)	41.14
Same County of Residence (%)	59.70
Change in Commute (km)	-2.39

N, Firms = 1,800.

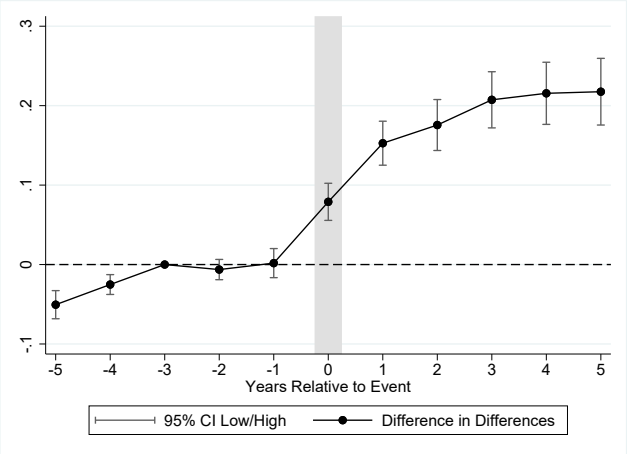
N, Workers = 40,768.

Notes: Characteristics of the new workers entering the winning firms at $\tau \in [0, 5]$. In alignment with our research design, we compute these statistics in two steps. First, we calculate the firm-level average for all treatment firms that hired new workers after winning the subsidy. Then, we average these figures across all such firms. Origin firm survival is a binary variable. It is assigned a value of one if the worker's previous firm remains operational five years following the event year of the new firm. Occupations are categorized at the 1-digit level, while industries are classified at the letter level, which represents the broadest categorization. For all variables derived from employment-related data, values are designated as missing for workers who were not employed prior to joining the new firm. Back to Sections VII, VIII, and IX.

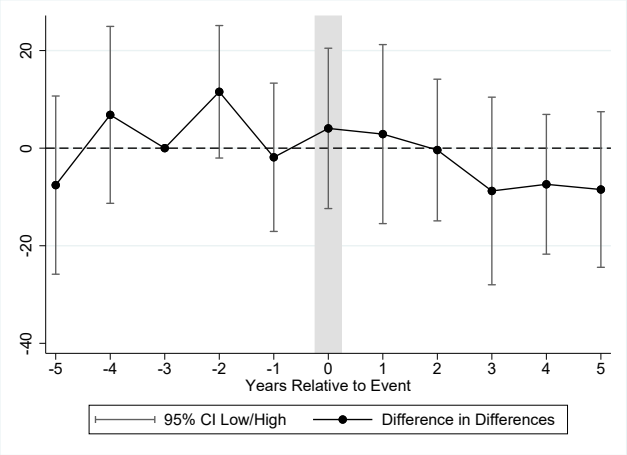
B Matched Control Group



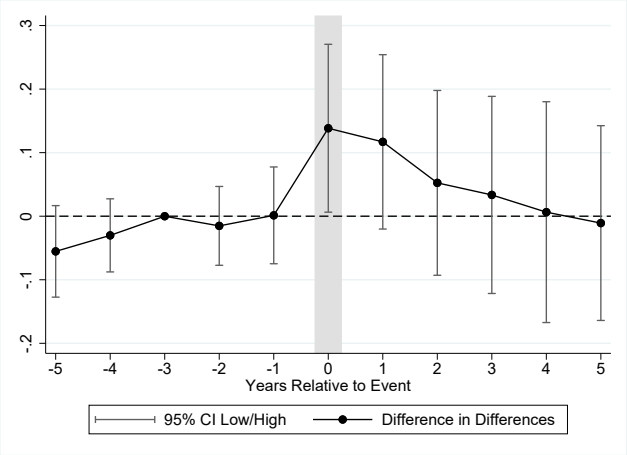
(A) Winners vs. Matched Control: Machinery Investment.



(B) Winners vs. Matched Control: Employment.



(C) Losers vs. Matched Control: Machinery Investment.



(D) Losers vs. Matched Control: Employment.

Figure B1: Matched Control Groups: The First Stage and Employment Effects.

Notes: Event-study estimates from Equation 1. **Panels (A, B):** Treatment group is the subsidy winners (the main treatment group), and control group is constructed via matching. **Panels (C, D):** Treatment group is the subsidy losers (the main control group), and the control group is constructed via matching, i.e., comparing two different control groups. We use coarsened exact matching (CEM). We match by revenue, employment, wages at $\tau = -3$ plus revenue and employment changes in percentages from $\tau = -3$ to $\tau = -1$ and industries' main sectors (letter classes). The CEM percentiles are 10, 25, 50, 75, 90, and 99. The match is 1:1 with replacement. The grey bar at event time $\tau = 0$ refers to the application year when the treatment begins. Back to Sections V and VI.

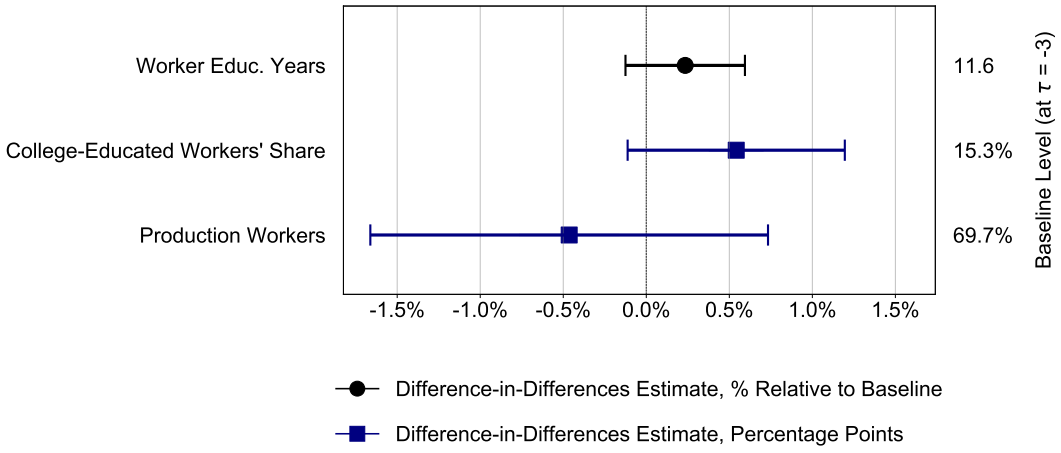


Figure B2: Matched Control Group: Skill Effects.

Notes: Difference-in-differences estimates from Equation 2. The estimates compare the main treatment group ("winners") to a matched control group (the matching procedure is described in Section III). The right-hand side reports outcome means at $\tau = -3$. Back to Section V.

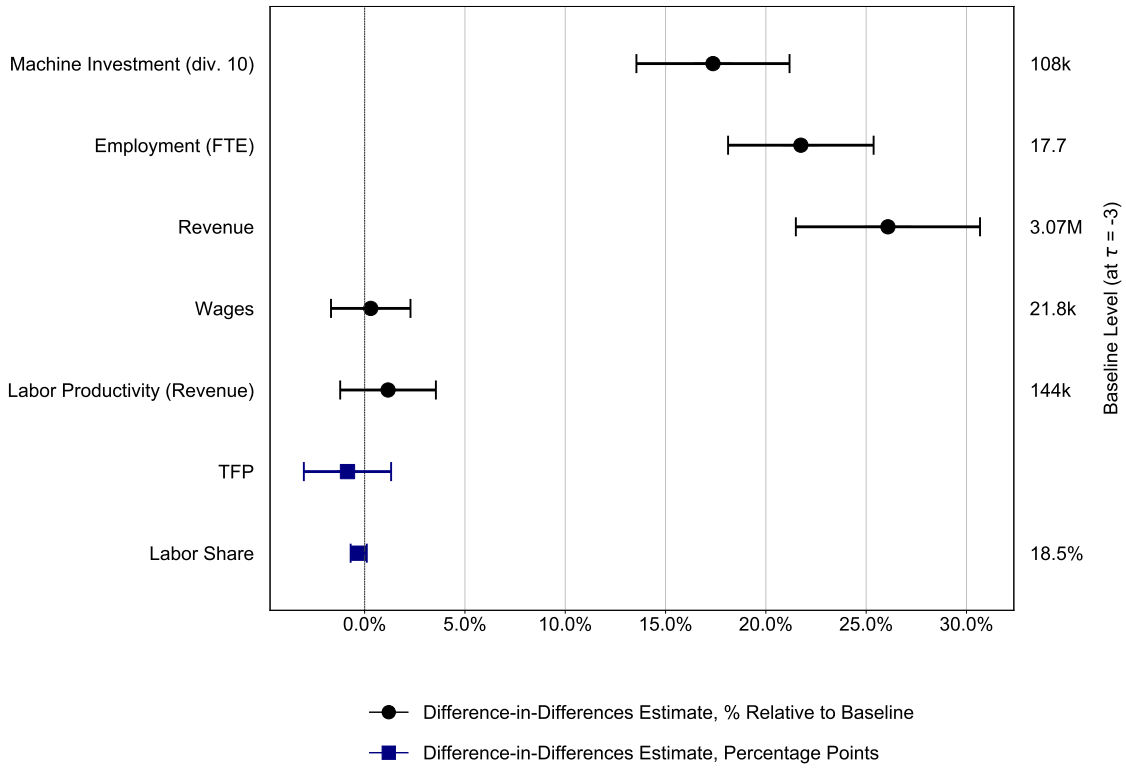


Figure B3: Matched Control Group: Firm-Level Effects.

Notes: Difference-in-differences estimates from Equation 2. The estimates compare the main treatment group ("winners") to a matched control group (the matching procedure is described in Section III). The right-hand side reports outcome means at $\tau = -3$. Back to Section V.

Table B1: Matched Control Group: Balance Table A (Winners vs. Matched Control).

	Treatment Group		**Control Group**		**Both**			**Tests**
Variable	Mean	Std. Dev.	Mean	Std. Dev.	10p	Median	90p	p-value
Revenue (EUR M)	3.06	26.57	3.09	9.15	0.17	0.96	6.26	0.963
Employment	17.46	46.27	18.03	38.79	1.60	8.20	37.70	0.704
Wages (EUR K)	21.60	8.08	22.06	8.36	12.15	22.43	30.56	0.109
Subsidy Applied (EUR K)	108.52	126.79	0.00	0.00	0.00	0.86	172.15	0.000
Subsidy Granted (EUR K)	78.62	100.55	0.00	0.00	0.00	0.49	122.38	0.000
Educ. Years	11.68	0.98	11.56	1.04	10.50	11.67	12.63	0.001
College Share (%)	15.24	16.84	15.39	18.45	0.00	12.50	34.62	0.820
Production Worker Share (%)	70.96	21.53	68.43	25.11	37.50	72.73	100.00	0.060
F-test								0.000
Observations	1600		1600			3200		3200

Notes: All variables measured at $\tau = -3$ expect for subsidies applied and received which are sums over τ= 0–2. The variables are not winsorized prior to calculating the summary statistics. Back to Sections III and IX.

Table B2: Matched Control Group: Balance Table B (Losers vs. Matched Control).

	Treatment Group		**Control Group**		**Both**			**Tests**
Variable	Mean	Std. Dev.	Mean	Std. Dev.	10p	Median	90p	p-score
Revenue (EUR M)	1.62	5.52	1.27	2.71	0.10	0.43	2.71	0.527
Employment	9.02	18.56	8.81	15.12	1.00	3.90	20.00	0.921
Wages (EUR K)	17.81	7.95	18.01	8.79	5.50	18.80	27.82	0.853
Subsidy Applied (EUR K)	47.47	76.19	0.00	0.00	0.00	0.00	65.59	0.00
Subsidy Granted (EUR K)	0.00	0.00	0.00	0.00	0.00	0.00	0.00	N/A
Educ. Years	11.34	1.12	11.42	1.23	10.00	11.50	12.56	0.641
College Share (%)	10.50	15.47	15.41	21.76	0.00	6.90	33.33	0.050
Production Worker Share (%)	74.25	25.39	70.77	27.93	30.95	79.63	100.00	0.647
F-test								0.062
Observations	123		123			246		246

Notes: All variables measured at $\tau = -3$ expect for subsidies applied and received which are sums over τ= 0–2. The variables are not winsorized prior to calculating the summary statistics. Back to Section III.

C Spikes Design

To explore external validity, we consider technology adoption without the subsidy program. This design exploits the precise timing of technology investment events, which we call spikes, to analyze technologies' short-term effects at the firm level. The second design is valuable because the subsidy-based design is subject to two external validity concerns: (1) subsidy program as variation source, (2) program participants' representativeness. The spikes design complements the subsidy design by using a different variation source and a different sample. It is similar to a mass-layoff design (Jacobson et al., 1993) as it uses the precise event timing for identification and builds on the work of Nilsen et al. (2009), Hawkins et al. (2015) and Bessen et al. (2025). The design detects distinct events because technology investments tend to be temporally concentrated (e.g., Doms and Dunne, 1998; Caballero and Engel, 1999; Cooper et al., 1999; Nilsen and Schiantarelli, 2003).

Treatment Group We define the technology investment event, the spike, as an indicator that equals one when a firm's technology expenditures are significantly above average for the firm:

$$D_{jt} = \mathbf{1}\left\{\text{Technology Expenditure}_{jt} > \text{Threshold} \cdot \overline{\text{Technology Expenditure}}_{jT \notin t}\right\}$$

The average expenditure is computed over timeline T leaving out the current year t. For our main specification, we use the threshold of four (robust to different thresholds). We measure technology expenditure as investment in machinery and equipment from the Financial Statement Register.

The sample design is the following. We consider years 1994–2018 and restrict the sample to manufacturing, warehouse and retail, transportation industries, and firms with full-time equivalent employees (FTE) between 10 and 750 at time $\tau = -1$ relative to the event. We focus on a balanced sample and require that the firms operate at least starting from time $\tau = -9$. With these restrictions, we can exclude new rapidly growing firms that are not relevant to our research questions and event definition and ensure comparability with the subsidies design. Very large firms tend to have several units or plants, which obscures the evaluation of the spike.

The treatment group is the firms that experience a technology investment event and satisfy the sample-design criteria. In the case of multiple spikes, we choose the largest spike and require no other spikes in window $\tau \in [-5, 8]$. While this is needed for defining unique events, this requirement selects firms that do not have overly positive post-event outcomes. Figure C2 shows the treatment group's average technology expenditure by year. The event time is normalized around the event ($\tau = 0$). There is a clear investment spike: A significant fraction of technology investment at the firm level is associated with significant variations (Figure C1).

Matched Control Group To construct the control group, we match the spiking firms to non-spiking firms. The matched control group serves as a counterfactual for what would have happened in the short term had the spiking firms not invested. We provide a theoretical basis for this comparison in Appendix J. We use coarsened exact matching (CEM). We match by revenue, employment, and wages at $\tau = -3$ and industries' main sectors (letter classes). The CEM percentiles are 10, 25, 50,

75, and 90. The final caliper match is the propensity score based on the same CEM variables. The match is up to 1:5 with replacement. Table C1 shows the covariate balance for the matched samples. We match only in the pre-period cross-section to ensure that the pre-trend comparison between the treatment and control is informative.

Estimation The empirical strategy contrasts the treatment group with a spike to the matched control group that did not have a spike within the same 5-year window using a dynamic difference-in-differences design. To do so, we estimate Equations 1 and 2 from Section III.B.

The First Stage Figure C2 shows the first stage. The outcome is technology investment. Treatment group firms invested EUR 2 million more in technologies than the control firms in the event year. Before and after it, the groups invested similar amounts and were on parallel trends.

Variation We outline a dynamic model that clarifies the source of variation in Appendix J, adapted from Cooper et al. (1999). The same model provides the basis also for the subsidies design, and we refer to it in Section III.A. The main result of the model is that with adjustment costs, firms may experience low technology-investment activity periods followed by bursts of investment activity.

This result clarifies that the treatment and the matched control group could be comparable in the short run because minor initial differences may lead to significant variations in technology investment. For example, in the model, one reason a firm invests and the other similar firm does not is that they have a different replacement cycle. Our estimates from the spikes design exploit the precise timing of technology investment events.

Robustness The estimates are robust to excluding firms that simultaneously start exporting, change their management, make significant investments in buildings and property, or open a new plant before the event, and to different controls (not reported).

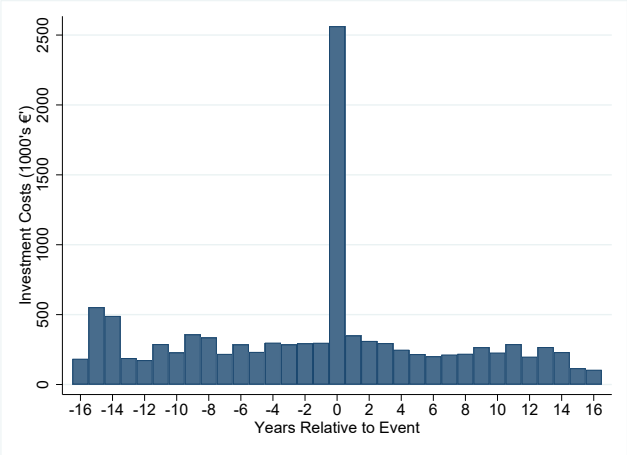


Figure C1: Spikes Design: Machinery Investment.

Notes: Machinery investment in EUR K. Event time is normalized to zero in the year of the largest machinery investment. The sample is restricted to manufacturing, retail, transportation industries and firms with employment 10–750 for comparability with the subsidies design. Consistent with the theoretical framework in Appendix J, technology investment is typically a spiky activity. Back to Section C.

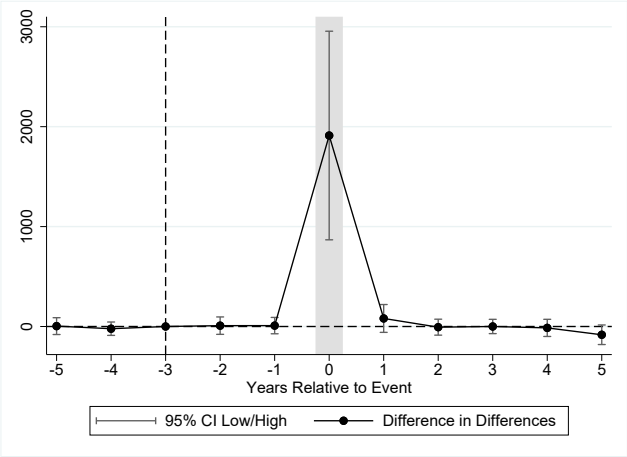
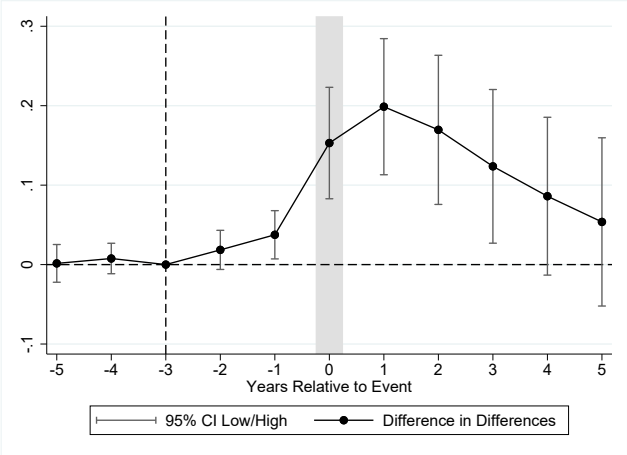
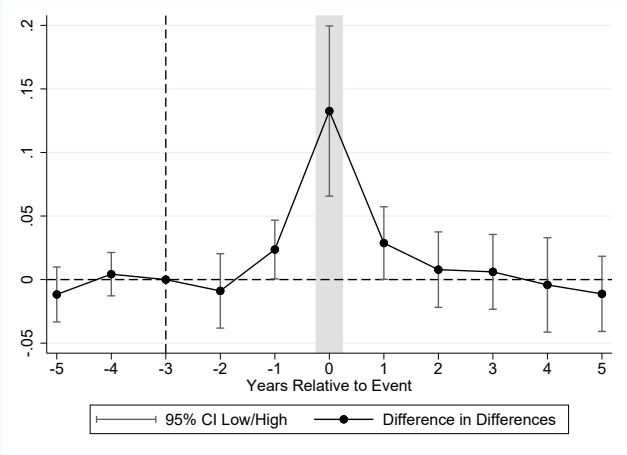


Figure C2: Spikes Design: First Stage on Machinery Investment.

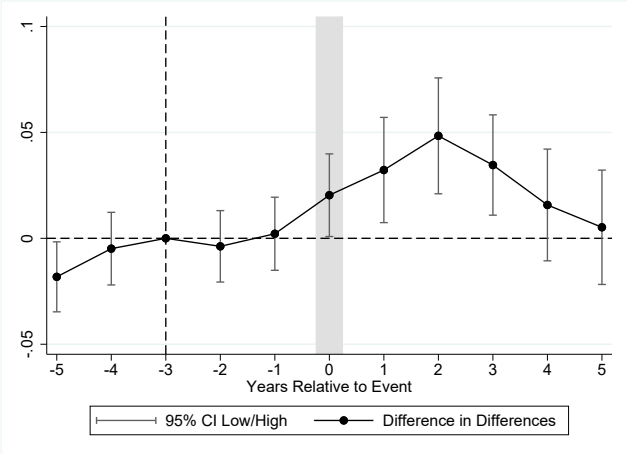
Notes: Event-study estimates from Equation 1. The estimates compare the spikes treatment group to a matched control group. The outcome is machinery investment in EUR K. Event time is normalized to zero in the year of the largest machinery investment. Back to Section C.



(A) Employment (%).



(B) Worker Entry Rate.



(C) Worker Exit Rate.

Figure C3: Spikes Design: Employment Effects.

Notes: Event-study estimates from Equation 1. The estimates compare the spikes treatment group to a matched control group. Event time is normalized to zero in the year of the largest machinery investment. Employment is in % relative to the base year $\tau = -3$. Entry (exit) rate is defined as the number of entering (exiting) workers divided by employment in the base year $\tau = -3$. Back to Sections V and VI.

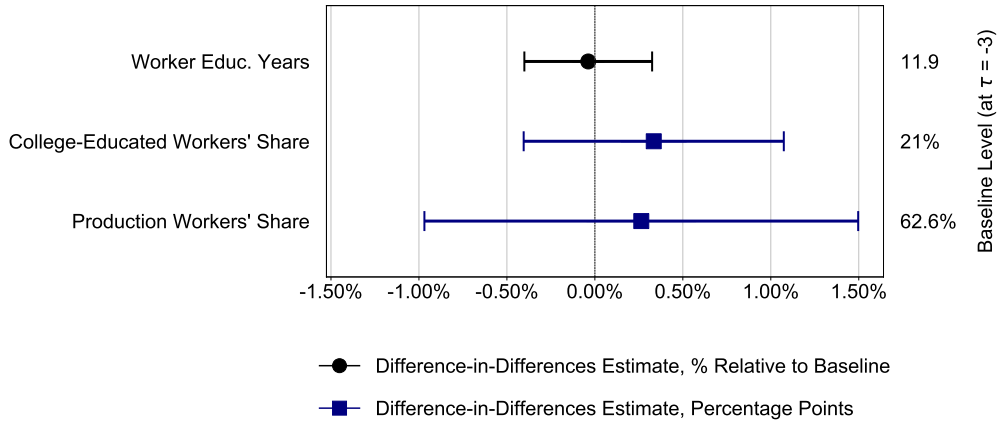


Figure C4: Spikes Design: Skill Effects.

Notes: Difference-in-differences estimates from Equation 2. The estimates compare the spikes treatment group to a matched control group. The right-hand side reports outcome means at $\tau = -3$. Back to Section C.

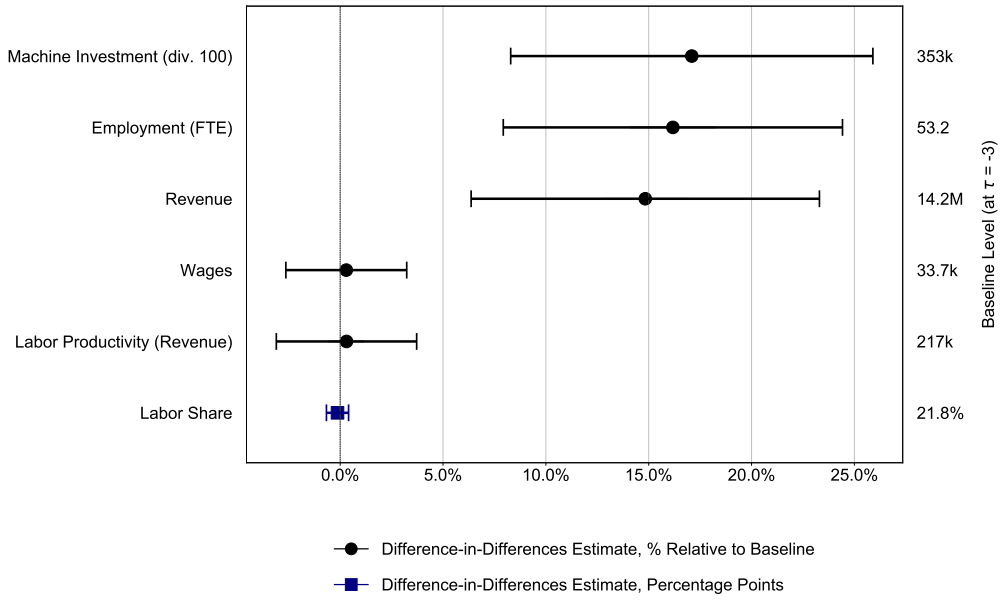


Figure C5: Spikes Design: Firm-Level Effects.

Notes: Difference-in-differences estimates from Equation 2. The estimates compare the spikes treatment group to a matched control group. The right-hand side reports outcome means at $\tau = -3$. The machinery investment estimate is divided by 100 to avoid scale issues in the figure. Back to Section V.

Table C1: Spikes Design: Summary Statistics.

	Treatment Group		**Control Group**		**Both**			**Tests**
Variable	Mean	Std. Dev.	Mean	Std. Dev.	10p	Median	90p	p-value
Machine Investment (EUR K)	271.21	858.93	376.70	999.26	6.96	109.55	770.54	0.042
Revenue (EUR M)	14.48	30.10	14.12	69.98	1.29	4.85	26.97	0.915
Employment	51.66	68.29	53.67	71.11	11.10	28.30	119.20	0.593
Wages (EUR K)	33.68	9.26	33.75	8.21	25.12	32.56	43.20	0.882
Labor Share (%)	21.37	11.49	21.92	9.89	8.69	21.45	34.90	0.317
Labor Productivity (EUR K)	231.95	197.06	213.41	273.42	84.92	150.98	421.43	0.179
Subsidy Applied (EUR K)	265.14	757.13	200.85	543.70	0.00	0.00	565.49	0.044
Subsidy Granted (EUR K)	169.49	454.79	123.68	334.04	0.00	0.00	388.35	0.019
Educ. Years	11.89	0.91	11.86	0.87	10.88	11.78	12.94	0.476
College Share (%)	21.24	16.70	20.90	14.95	5.56	17.65	40.91	0.674
Production Worker Share (%)	58.90	30.95	63.68	25.79	14.29	71.43	88.89	0.014
F-test								0.013
Observations	450		1593		2043			2043

Notes: All variables measured at $\tau = -3$ (three years before the investment spike). We use 1:5 coarsened exact matching (CEM) with replacement (details in Section C). Back to Section C.

D Regression Discontinuity Design

Design We use a regression discontinuity (RD) design generated by a change in the rules used to evaluate the applications as one tool to address internal validity. Einiö and Buri (2020) discuss the policy change and the RD strategy, and present related results. The advantage of the RD design is that the estimates are likely to reflect a causal relationship and satisfy Assumption 1. The disadvantages of the RD design in this context are statistical power, that the treatment is less precisely defined, and that it does not allow a natural way to use the text data to measure different types and uses of technology.

The EU expanded the definition of a small firm in 2005. Our RD design uses the fact that firms just below the new threshold were prioritized for subsidies but were otherwise similar to those just above it. Before the policy change, upper thresholds for small firms were 50 for employment, EUR 5M for the balance sheet, and 7M for turnover. The EU raised the thresholds for balance sheet and turnover to 10M. We use the balance sheet's total value as our running variable because it measured most precisely and had the most significant change; this gives us the statistical power to conduct the analysis.

The critical part is that the new rule was applied using retrospective data for firms. Thus firms could not immediately manipulate their size. However, as shown in Figure D1, firms adjusted their size later. This evidence leads us to focus only on the first year of the policy change when manipulation at the threshold was unlikely. Finland implemented the change in 2007 but considered retrospective data from 2004–2006. Our estimates use 2004 data as the running variable to avoid selection bias.

The policy change potentially affected firms' self-selection into the program, the likelihood of winning the subsidy, and the levels of subsidies. While being a small firm is not a strict criterion for receiving subsidies, the ELY Centers prioritize small firms (e.g., Takalo et al., 2013). The firms know this and are potentially more likely to apply for subsidies when the expected benefits are more significant. These facts and statistical precision lead us to focus on the reduced-form effects. There were no simultaneous policy changes at the same margin.

To produce the RD estimates, we use the following specification:

$$Y_i = \alpha + \beta E_i + f(z_{i,2004}) + \varepsilon_i \tag{5}$$

where Y_i is outcome for firm i, $f(z_{i,2004})$ is a function of the running variable (balance sheet in 2004) and E_i is the cut-off indicator (balance sheet under 10M in 2004). We use the bandwidth of five million, triangular kernel, and first-order polynomial (Gelman and Imbens, 2019) in our main specification. We cluster the standard errors at the 3-digit industry.

Results Table D1 shows the summary statistics for the RD sample firms.[8] As expected, the RD sample firms are larger than in the main design because, by definition, their revenue is around

[8] We exclude agriculture and forestry, the public sector, transportation, and finance since these sectors are generally not eligible for these ELY Center subsidies.

EUR 10M. Figure D1 documents firms starting to bunch around the new threshold after the change comes into effect. Figure D2 formally shows by a McCrary test (McCrary, 2008; implemented as in Cattaneo et al., 2018) that this is not yet the case in the pre-change year of 2004, which is the relevant year for our identification. Table D2 tests whether firms are different on different sides of the cutoff before the treatment and finds no statistically significant differences.

Next, we describe the first stage. Figure D3 shows a jump in the received subsidies at the new cutoff of EUR 10M. The running variable (x-axis) is the balance sheet in 2004 and the outcome variable (y-axis) is the total received subsidies in EUR 10K. The received subsidies are larger on the left side of the cutoff, likely because those firms became small under the new classification. Figure D3 also shows that these subsidies stimulated new investments: The linear graphs show a clear jump at the cutoff. Table D3 quantifies the same jumps using Equation 5 for subsidies received and investments made in 2007. Becoming a small firm increased the subsidies by EUR 38K and investments by EUR 188K. Both estimates are significant at the 5% level.

Table D4 presents the primary outcomes of the RD design. These results broadly confirm our main results of firm growth in employment and revenue but no skill bias. Being re-classified as a small firm increased employment by 9% and revenue by 25%. We see no changes in average wages, years of education, or the share of college-educated workers or production workers. The estimation is done by setting the average of 2003–2006 as a baseline value and comparing each observation from 2010 to 2015 separately to the baseline to increase statistical power. These differences are the outcomes in the estimation. Figure D4 visualizes a similar estimation for each year separately. We observe an increase of 8–10 employees from 2010 onwards.

We run multiple robustness and placebo tests for our estimates. Figure D5 explores robustness to the choice of bandwidth: Our results are not sensitive to it. Figure D6 runs our main specification with different thresholds: We cannot replicate our results with the placebo thresholds. Figure D7 run the estimation with placebo years' balance sheets: We observe no effect.

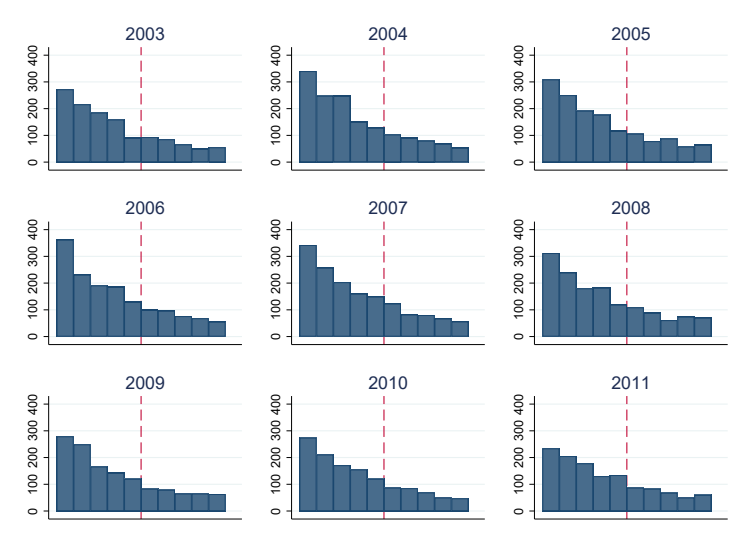


Figure D1: RD Design: Number of Firms at the Balance-Sheet Threshold.

Notes: The number of firms around the balance-sheet threshold for small firms announced in 2003, which came into effect in 2007. The distributions are plotted separately for each year 2003–2011. Back to Section D.

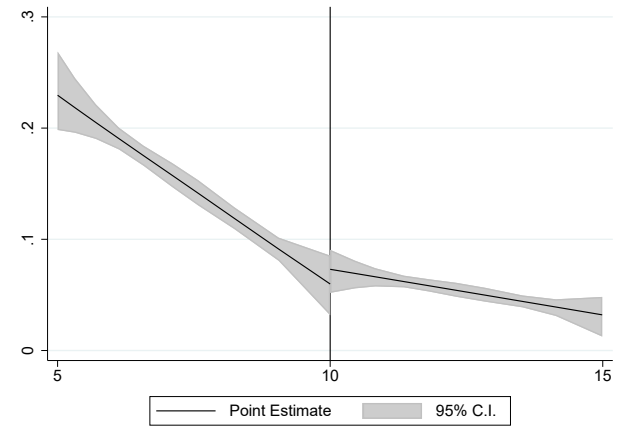


Figure D2: RD Design: Density of Firms at the Balance-Sheet Threshold.

Notes: McCrary test for the RD design (implemented as in Cattaneo et al., 2018). The horizontal axis is the firms' balance sheet in 2004 in EUR M. The vertical axis denotes the density of observations. Back to Section D.

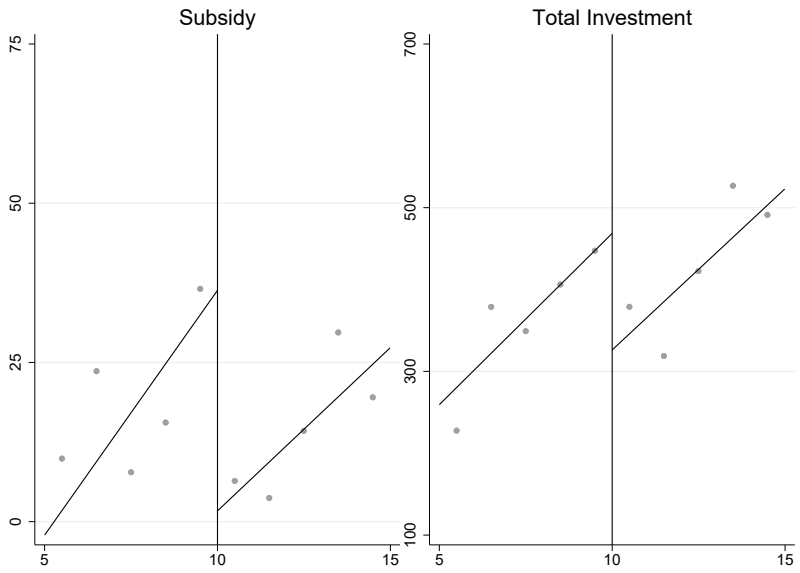


Figure D3: RD Design: The First Stage.

Notes: Discontinuity at the balance-sheet threshold for 2007 investment subsidies (left) and total investment (right). The vertical axis is in EUR K, and the horizontal axis is in EUR M. Back to Section D.

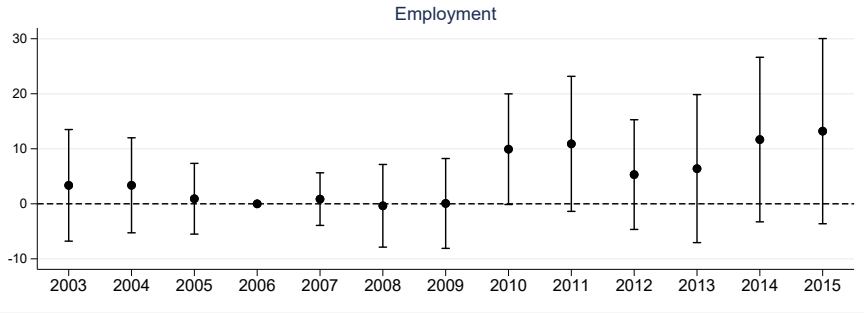


Figure D4: RD Design: Employment.

Notes: RD estimates from Equation 5. The outcome is the employment difference to base year 2006. The explanatory variable is the balance-sheet RD threshold indicator. We cluster the standard errors by three-digit industry, the kernel function is triangular, and the polynomial order is one. Back to Sections V and D.

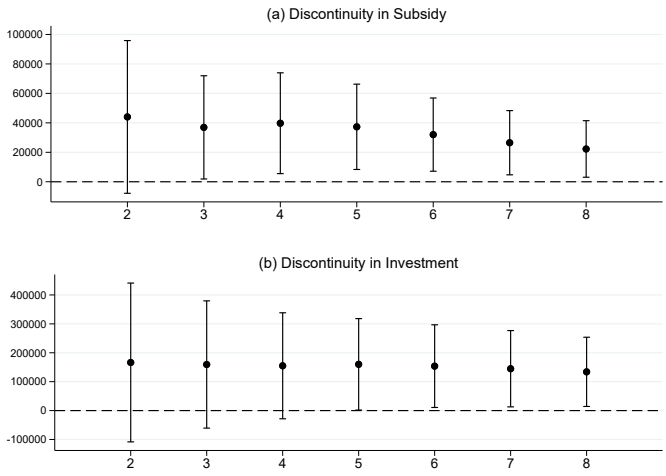


Figure D5: RD Design: Different Bandwidths.

Notes: RD estimates from Equation 5. The outcome is investment subsidies in the upper panel and total investment in the lower panel. The horizontal axis indicates the bandwidth of the estimation window. In all regressions, we cluster the standard errors by three-digit industry, the kernel function is triangular, and the polynomial order is one. Back to Section D.

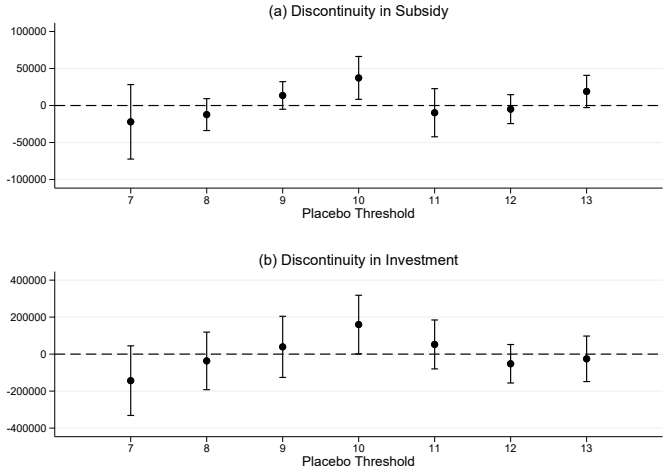
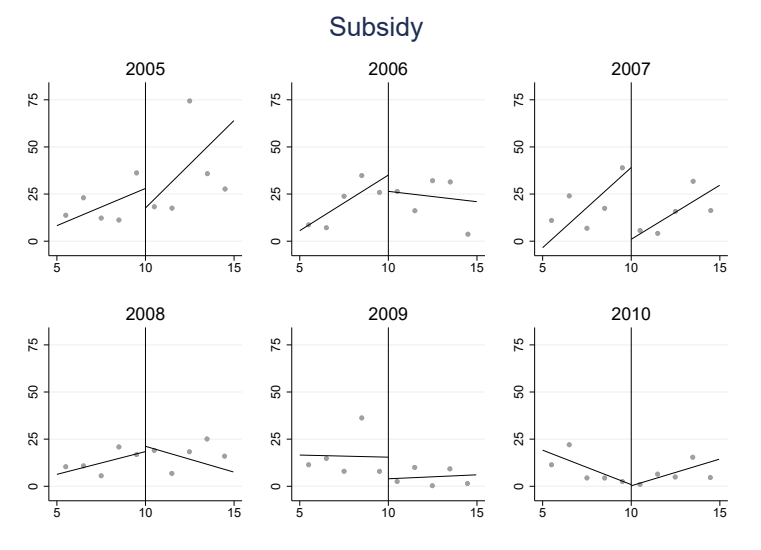
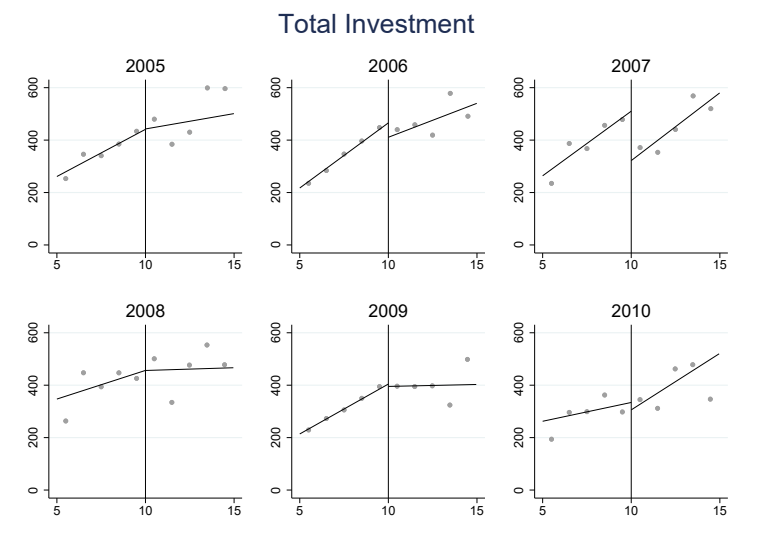


Figure D6: RD Design: Placebo Cutoffs.

Notes: RD estimates from Equation 5. The outcome is investment subsidies in the upper panel and total investment in the lower panel. The explanatory variable is the balance-sheet threshold indicator. The indicator equals one if the balance sheet is lower than the number indicated on the horizontal axis. The real cutoff of 10. In all regressions, we cluster the standard errors by three-digit industry, the kernel function is triangular, and the polynomial order is one. Back to Section D.



(A) Subsidies.



(B) Investment.

Figure D7: RD Design: Placebo Years.

Notes: Discontinuity at the balance-sheet threshold for investment subsidies (A) and total investment (B). The vertical axis is in EUR K, and the horizontal axis is in EUR M. In all versions, we consider the 2004 balance sheet. The discontinuity should be exactly in 2007. Before 2007, there should not be a discontinuity since the new balance-sheet criterion was not yet in place. After 2007, there should not be a discontinuity since the balance sheet 2004 value was no longer relevant. Back to Section D.

Table D1: RD Design: Summary Statistics.

	Mean	Std. Dev	N
Employment	65.75	76.93	1269
Revenue (EUR M)	16.7	16.5	1273
Wages (EUR K)	34.7	16.9	1269
Production Worker Share	0.40	0.32	1271
College Share	0.37	0.26	1273
Total Investment (EUR K)	377.6	579.0	1273
Investment Subsidies (EUR K)	16.2	127.6	1273
Total Subsidies (EUR K)	23.9	124.6	1273
Subsidized Loans (EUR K)	168.5	1,055.5	1273

Notes: Summary statistics for the RD sample, with balance sheet between 5 to 15 million euros. Back to Section D.

Table D2: RD Design: Pre-Treatment Covariate Balance.

	Investment	Subsidy	Revenue	Employment
Small 2004	5.771	16.17	-4.296	-7.745
	(88.22)	(19.03)	(2.849)	(10.37)
N	1273	1273	1273	1270

Standard errors in parentheses, clustered by 3-digit industry.
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Notes: RD estimates from Equation 5. The outcomes are pre-period averages over years 2000–2004. Back to Section D.

Table D3: RD Design: First Stage Estimates.

	(1) Subsidy	(2) Investment
Small 2004	38.07*	188.5*
	(16.44)	(86.53)
N	1273	1273

Standard errors in parentheses, clustered by 3-digit industry.
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Notes: RD estimates from Equation 5. The outcomes are 2007 investment subsidies (Column 1) and 2007 total investment (Column 2). The values are in EUR K. Back to Section D.

Table D4: RD Design: Reduced-Form Estimates.

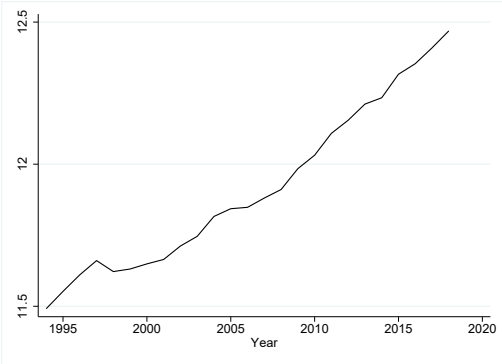
	(1) Employment	(2) Revenue	(3) Wages	(4) College Share	(5) Educ. Years	(6) Production Worker Share
Small 2004	0.0899*	0.251***	0.0214	-0.00108	-0.00902	0.00613
	(0.0417)	(0.0435)	(0.0208)	(0.0106)	(0.0625)	(0.0119)
N	6005	6006	6003	6012	6012	6012

Standard errors in parentheses, clustered by 3-digit industry.

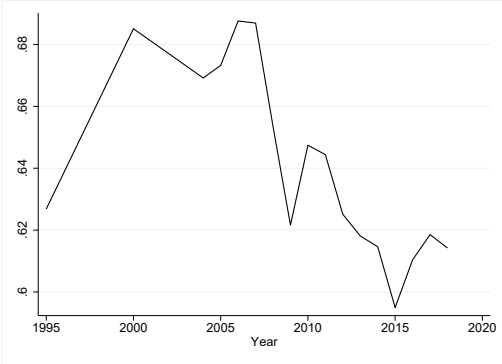
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Notes: RD estimates from Equation 5. The outcomes are defined in first differences. The pre-period level for the estimations is the average over 2003–2006. Employment, revenue, and wages are in relative changes, e.g., 0.20 would refer to a 20% increase. Education years is in years. The college and production worker shares are in percentage points. Back to Sections V and D.

E Descriptive Evidence



(A) Average Years of Education.



(B) Production Worker Employment Share.



(C) College vs. Non-College Wage Ratio.

Figure E1: Skill Trends in Finnish Manufacturing.

Notes: These figures document trends in Finnish manufacturing over 1994–2018. We restrict to firms with at least three workers. Note that occupation data, and thus also production worker employment share, is missing for years 1996–1999 and 2001–2003. We compute the year-level averages from firm-level observations. The numbers are unweighted to match our research design. The employment-weighted numbers are similar. Back to Section IX.

Table E1: Contrasting Robot and IT Use in Survey and Customs Data.

	(1)		(2)		(3)		(4)		
	CIS		**ICT**		**Etla**		**Customs**		
	Robots' Importance		*Robot-User*		*Robot-User*		*Robot-Importer*		
A: Robots	High	Low	Yes	No	Yes	No	Yes	No	No*
College Share	0.325	0.294	0.312	0.355	0.294	0.340	0.316	0.145	0.208
	(0.022)		(0.003)		(0.015)		(< 0.001)		
Production Workers' Share	0.600	0.555	0.633	0.549	0.600	0.529	0.565	0.694	0.634
	(0.0129)		(< 0.001)		(0.002)		(< 0.001)		
Labor Share	0.199	0.246	0.200	0.230	0.235	0.258	0.172	0.232	0.205
	(< 0.001)		(< 0.001)		(0.037)		(< 0.001)		
Productivity (€K)	311	246	324	275	231	212	209	140	173
	(< 0.001)		(0.009)		(0.258)		(< 0.001)		
N	271	1,195	357	521	298	306	760	260,434	91,880
Share	0.185	0.815	0.407	0.593	0.493	0.507	0.003	0.994	0.351
Years	2014–2018		2018		2015		2000–2018		

	CIS		**ICT**		**Etla**	
	Digitalization's Importance		*Computer Users' Share vs. Median*		*Big Data and Analytics*	
B: IT	High	Low	Above	Below	Yes	No
College Share	0.397	0.273	0.428	0.248	0.383	0.291
	(< 0.001)		(< 0.001)		(< 0.001)	
Production Workers' Share	0.473	0.623	0.481	0.685	0.506	0.599
	(0.046)		(< 0.001)		(< 0.001)	
Labor Share	0.240	0.234	0.210	0.225	0.254	0.245
	(0.527)		(0.097)		(0.500)	
Productivity (€K)	263	257	342	249	232	213
	(0.734)		(< 0.001)		(0.324)	
N	192	1263	436	443	137	493
Share (%)	0.132	0.868	0.496	0.504	0.217	0.783
Years	2014–2018		2018		2015	

Notes: Group means by technology use in the CIS, ICT, and Etla surveys and the Finnish Customs data. The text in italics refers to the question from each survey; the survey questions are listed in Appendix F. The p-values in parentheses are based on a t-test for the difference between the groups. The data contain Finnish manufacturing firms with more than two full-time equivalent employees (FTE) overlapping with the surveys and customs records. In Column 4, No* refers to firms that did not import robots but imported something. Productivity is measured as revenue per worker. More details in the main text and Appendix F. Back to Section IX.

Table E2: Industry-Level Evidence on Labor Share, Productivity, and Production Workers' Share.

	(1)		(2)		(3)	
	Labor Share		Productivity		Production Workers' Share	
A: Machinery Investment	-0.0012*	-0.0013*	0.0045	0.0040	0.0033***	0.0036***
	(0.0005)	(0.0005)	(0.0030)	(0.0033)	(0.0009)	(0.0008)
Pre-Period College Share		0.0528		0.341		-0.193
		(0.0599)		(0.307)		(0.159)
B: IT Expenditure	-0.0010	-0.0024*	0.0160***	0.0183***	-0.0043*	-0.0041
	(0.0006)	(0.0011)	(0.0019)	(0.0030)	(0.00196)	(0.0029)
Pre-Period College Share		0.113		-0.195		-0.0121
		(0.0765)		(0.184)		(0.170)
N	49	49	49	49	49	49
Mean Pre-Period	0.166	0.166	0.259	0.259	0.615	0.615
Mean Change	0.0435	0.0435	0.0802	0.0802	-0.073	-0.073
R^2, Machinery	0.150	0.185	0.130	0.216	0.168	0.236
R^2, IT	0.038	0.138	0.526	0.544	0.093	0.093
	Mean Total	10p	90p			
Machinery	9.236	2.560	21.75			
IT	2.306	0.610	4.011			

Standard errors in parentheses.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Notes: Industry-level long-difference estimates on predicting labor share, productivity (EUR M), and production workers' share changes separately with total machinery investment (Panel A) and IT expenditure (Panel B) between 1999–2018 in Finnish manufacturing. The technology variables are measured in EUR K per worker-years (FTE). The outcome variables are in long-changes over the timeline. The estimates and means are weighted by pre-period industry employment sum between 1999–2002. Pre-period values otherwise measured in 1999, expect for production workers' share which is available for 2000. The shares are measured between 0 and 1, and productivity is measured in EUR M (revenue per worker). N refers to 2-digit manufacturing industries. Back to Section IX.

F Data

This appendix describes our data on workers, firms, and subsidies, as well as our fieldwork. To ensure consistent measurement, we developed new crosswalks to harmonize Finnish occupation, industry, and geography classifications.

F.A Workers

We directly measure individual workers' employment, wages, education, school grades, occupations, cognitive performance, and personality. Our data track all individuals in Finland over time independently of their labor-market status.

Employment and Wages We obtain employment and wage data from the registers maintained by Statistics Finland. The data contain the employment status, wages, other income, and a link to the firm. The data allow us to track all persons in Finland over time, independent of their labor-market status. The data are combined from multiple government sources (including the social security system and tax authorities) and direct data collection by Statistics Finland. These registers also record the individuals' age, gender, and home county. At the firm level, we define the labor share as the wage bill divided by the revenue.

Education We measure education and school grades. Education is measured from The Register of Completed Education and Degrees. It provides exact information on the educational degrees the individual has obtained. We measure the level of education in four categories: (1) very low (no recorded degree), (2) low (high school), (3) medium (BA or equivalent), and (4) high (MA or PhD). High school contains both academic and vocational tracks. We measure the type of education also in four categories: (1) STEM (science, technology, engineering, and mathematics), (2) HASS (humanities, arts, and social sciences), (3) business and law, and (4) other types. We map degrees to years of education based on their official length. School grades are measured from the Secondary Education Application Register and the Finnish Matriculation Examination Board Register. We focus on the 9th-grade GPA and the standardized scores in the national high-school exit exam (12th grade). We normalize both grade measures to have a mean of zero and a standard deviation of one within cohorts.

Occupations and Tasks We measure occupations from the employment registers at the 3-digit level in the ISCO classification system. We manually harmonize the occupation classifications, resulting in 48 consistently defined occupations. For most analyses, we focus on three broad occupational categories: production workers (craft workers, operators, assemblers, and elementary occupations), non-production workers in lower-level positions (clerical, service, and sales workers), non-production workers in higher-level positions (technicians, associate professionals, professionals, and managers). The occupational data are available for 1995, 2000, and 2004 onward yearly. Because of these gaps in the data, we define the pre-period for occupational outcomes as the mean of non-missing observations over $\tau \in [-5, -1]$.

We use the European Working Conditions Survey (EWCS) to measure the task content of the occupations. The survey provides information on the tasks workers perform in their jobs. The data are collected through face-to-face interviews every five years. The advantages of the EWCS data are that it is based on workers' descriptions of their work, is available for a specific country and time, and is consistent with the European occupational classification. Using these data, we construct occupation-level measures of task intensity for routine, manual, abstract, and social tasks, and cognitive skill intensity (see also Autor et al., 2003, and

Kauhanen and Riukula, 2024). For example, an occupation is classified as highly routine if the workers describe they often perform repetitive and monotonous tasks and highly social if they often work with others. We use combined data from the Nordic countries (Finland, Sweden, Denmark, Norway) and Germany to trade off statistical precision and relevance to our context. The results are essentially the same using only data for Finland. Our exact classification rubric is available as part of the replication package.

Cognitive Performance and Personality We obtain data for cognitive performance and personality from the Finnish Defence Forces (FDF). The data cover 79% of Finnish men born 1962–1979, and are measured because of universal conscription. The cognitive-performance measures are visuospatial, arithmetic, and verbal reasoning. The visuospatial test is similar to Raven's Progressive Matrices (Raven, 1938). The personality-trait measures are sociability, activity-energy, self-confidence, leadership motivation, achievement motivation, dutifulness, deliberation, and masculinity. The personality test is based on the Minnesota Multiphasic Personality Inventory (MMPI). We normalize all measures to have a mean of zero and a standard deviation of one within cohorts. These are all available cognitive performance and personality measures; we do not exclude any. The FDF data are described in Izadi and Tuhkuri (2024, 2021).

F.B Firms

Our extensive data on technologies and firm performance track all firms over time.

Financial Statement Register These data (FIRM_FSS) contain information on machinery and equipment investment, IT expenditure, and overall capital. Machinery and equipment investment measures the value of the additions in machinery and equipment. This includes, for example, CNC machines, laser cutters, and robots. IT expenditure captures IT costs billed on outside providers. It includes software, programming, and computer-design expenditure, and also IT consulting services. Statistics Finland collects the investment and expenditure data directly and combines direct inquiry with estimation for capital. The register also includes data on revenue, profits, and outside R&D and marketing expenditure. We deflate all monetary values in this paper to 2017 euros using the Statistics Finland CPI and winsorize them at the 5% level, except for marketing costs, which are not winsorized, as only a small fraction of the sample reports nonzero marketing costs. Our sample includes only seven publicly traded firms. The data cover all Finnish firms in almost all industries and our analysis years 1994–2018.

Customs Register The Finnish Customs Register (TULLI_COMMOD and TULLI_ENTER) records firms' exports, export destinations, and imports. Export status is measured using the definition by Statistics Finland: A firm is defined as an exporter in a given year if its total export value is over EUR 12K during the calendar year spread over at least two different months or if a single export event is over EUR 120K in value. We measure firms' exported products at the 6-digit CN classification; we focus on the number of products and product turnover: introduced and discontinued products. We harmonize the product categories to be consistent over time. We do not apply any minimum level of imports or exports to the customs data. Price data for the exported products also come from the Customs Register. We focus on firm-level average prices computed as an unweighted average. We winsorize price data at the 10% level within product and year. The customs data also record 621 technologies that firms import in the 6-digit CN classification system. Together we refer to these as machinery imports. We manually classify these imported technologies into automated vs. non-automated technologies building on Acemoglu and Restrepo (2022). Sections VI and especially F.C describe this classification in more detail together with text data.

Patent Register The Finnish patent register (FIRM_PAT) covers the patents of all Finnish firms. We focus on the indicator whether the firm has patented in a given year. These data are available until 2013.

Industrial Production Statistics These data (FIRM_COMMOD) provide production statistics for a yearly sample of 3000–3500 plants. Based on our experience, the data appear imprecise for our primarily SME sample (the data focus on larger firms). For robustness, however, we compute prices from the product data, defining product-level prices as the revenue divided by the number of units sold. We winsorize price data at the 10% level within product and year. We also track the value of inputs from these data.

Industry Classifications We measure industries at a harmonized 2-digit level classification (based on NACE Rev. 2). We approximate the industry's scope for quality differentiation using the Rauch (1999) index (for robustness, with the Gollop and Monahan, 1991, and Sutton, 1998, indices). We gauge industries' tradability via Mian and Sufi (2014).

CIS Survey The Community Innovation Survey (FIRM_RDINNO) of the EU is implemented in Finland by Statistics Finland. It provides direct firm-level information on several relevant aspects for our study.

Motivations: CIS provides data on the importance of different objectives for product and process innovations as described in Section VI. The question that captures these aspects is worded typically: "How important were each of the following objectives for your activities to develop product or process innovations during the three years — to —? " or "During the three years – to –, how important were each of the following goals for your enterprise?" If multiple survey rounds match the firm, we choose the closest to the subsidy application year, and if two surveys are equally close, we choose the one before the application. Note that the option "Customer-Specific solutions" is only available in 2016.

Robots and digitalization: CIS also measures the importance of robots and digitalization as perceived by the firms. The question that captures these technologies is: "How important were the following digitization-related factors to your company's business in –year–?" We focus on "the utilization of robotics in production processes" and "the importance of digitization in producing goods and/or services." High importance means the answer is "large," and low importance means it is either "small" or "does not apply." Data on robots and digitalization are included in the survey rounds 2014–2016 and 2016–2018, and we pool them.

Constraints: CIS factors hampering innovation activities, including lack of credit, internal finance, or skilled employees, and the cost of innovation. The question was of the type "During — to —, how important were the following factors as obstacles to meeting your enterprise's goals?"

The survey is conducted every two years and we use data from years 1996, 2000, -04, -06, -08, -10, -12, -14, -16, and -18, covering the timeline of our study. The surveys used in the paper have the limitation that they cannot be tightly linked to the technology subsidy event, only to the firm. The (unrestricted) sample size varies between N = 2,000–4,000 firms and typical response rate is approximately 70%. The overlap with our main sample is 708 firms.

ICT Survey The ICT survey of Statistics Finland (FIRM_ICT) asked in 2018 directly whether the firm uses robots and what share of the workers use a computer. The robot question is: "Does your company use the following types of robotics?" The option we focus on is industrial robots, and there are two answer options: yes and no. The computer question is, "How much of your company's personnel use a computer in their work?" We discretize the answer into above/below the median in manufacturing. The overall sample size in 2018 for the survey is N = 3,028, and the response is 71%.

Etla Survey Our own Etla survey (also called the "Mittelstand survey" as it mainly focuses on SMEs) provides information on the firms' objectives and strategy. Etla stands for ETLA Economic Research, the research institute that implemented the survey in 2015.

Motivations and context: The questions cover firms' differentiation strategies and past and future changes. They probe differentiation of products/services and anticipated shifts over three years, including new products, methods, and customers. The questions also assess recent product improvements, operational changes in response to customer feedback, and firms' communication with their customers. The answer options are strongly, moderately, to some extent, and no.

Robots and digitalization: The survey asks firms whether they use or plan to use robots or several forms of IT, including big data and analytics. We focus on the question: "Will the following issues related to digitization impact your industry in the next 3 years?" and the two subtopics: "Using robotics to produce products/services" and "Big data, i.e., vast data masses and their analytics." The answer options we focus on are "Yes and we invest" and "No."

The target population had 33,390 active or closing/discontinued companies. 6,268 (18.8%) companies started the survey and N = 4,673 (14.0%) finished it. The overlap with our main sample is 202 firms.

SFINNO Data SFINNO is an innovation database that tracks firms' innovations from Finnish technical and trade journal articles from 1985 to 2020. These professional journals chronicle industry trends and firm-level innovation and technology investments. The SFINNO project systematically reviewed 15 technical and trade journals, identified innovations (including technology investments), and hand-coded variables describing these innovations. The current data identify a total of N = 5,481 innovations from approximately 1,700 firms. Our main sample firm IDs were linked to SFINNO by VTT, Technical Research Centre of Finland, resulting in 213 matches.

The data coding follows the Literature-Based Innovation Output (LBIO) method (Coombs et al., 1996) and adheres to the OECD's Oslo Manual guidelines (OECD, 2018). We use two variables coded from the journal text data: whether innovation was primarily a process or product innovation (following the definition of the Oslo Manual) and the complexity of the innovation. The data and variables are described in Palmberg et al. (1999) and (2000).

SFINNO complements the journal data by surveying the firms featured in the articles. The survey contains information on: how significant different factors have been for the commencement of the innovation's development (e.g., the realization of a market niche, the role of the customers, a new scientific breakthrough, new technologies) and the primary technological content of the innovation (e.g., development and integration of components and modules, development of production methods). The survey respondent in the firm was typically a person directly involved with the innovation project, identified from the article.

A key requirement for the firm to appear in the data is that the journal found the innovation newsworthy. While the editors of the journals included in SFINNO have confirmed in interviews that they aim to cover all important innovations in their industry (Torregrosa-Hetland et al., 2019), SFINNO is likely biased toward some forms of innovation—especially those that have relevance outside the firm. For example, product innovations might be more likely to be captured in the data (the early years until 1998 contain solely product innovations). However, sponsored material, advertisements, and simple new product listings were not included in the data. One advantage of SFINNO for us is that it oversamples SMEs (Palmberg et al., 1999). Lehtoranta (2005) and our own analysis document that SFINNO is consistent with CIS.

F.C Subsidies

Our main data sources for firm subsidies are two centralized systems, TUKI2000 and TUKI2014 (also referred to as Yrtti 1 and 2), which record all applications to the ELY Center subsidy program. These previously unstudied, confidential datasets track the full application process—from submission to decision—for every applicant. We also use the Statistics on Business Subsidies (FIRM_SUBSID) dataset to measure other types of firm subsidies. This section provides details on the text analysis: the categories, the coding process, and how we processed the raw text data.

F.C.1 Text Categories

We code the application texts into a set of categories that can be grouped into three classes: intentions, technologies, and robustness. Intentions describe how firms planned to use the technology—for example, to expand, develop new products, or improve productivity. Technologies capture the specific tools firms intended to adopt, such as machinery, IT, robots, CNC machines, or laser cutters. Robustness flags rejected applications with firm-specific issues or employment-related concerns.

Below, we outline the coding criteria for each category. We also include anonymized examples from the application texts, with key passages underlined. These examples are translated and anonymized to preserve confidentiality. The coding team has reviewed the descriptions and examples to ensure they reflect the actual coding decisions.

We begin with the categories describing firms' intended use of the technology. Each category is hand-coded independently, so an application may fall into multiple categories. Table 5 summarizes these categories.

Expansion This category includes firms that aim to expand firm scale, production capacity, output, or revenue. Expansion must be a stated goal—not just a possible byproduct of other objectives like improved quality or lower costs. Firms may express this goal indirectly—for example, by adding production lines, building new facilities, or easing capacity constraints. These cases are included. The underlying motive is often to meet increased demand or respond to new customers. Examples:

- The new production line increases the company's capacity by approximately 30-40%.
- The company believes that the project will enable it to increase revenue and expand its export activities.
- The company invests in a new production facility (including office spaces) for the manufacture of sheet and metal structures, expanding the company's operations.
- A new computer-controlled large format printer is needed to improve the company's competitiveness and business growth.

New Customers/Markets This category includes cases where firms aim to reach new customers or enter new markets. The motivation must be explicit: applications must mention targeting new clients, contracts, partnerships, or market segments. These may include local or international markets. Examples:

- The company's operations are expanding with a new customer, necessitating investments and automation as capacity grows.
- Modern equipment will allow the company to meet current customer needs and establish new customer relationships.

- With the investment, the company has the opportunity to enter the surface materials market, offering new business opportunities.
- Developing internationalization and exploring new material procurement channels is crucial for the company to find new customers and enhance its operational opportunities.

Exports This category includes cases where firms aim to initiate or expand exports or adjust their international operations. To be included, the application must mention goals such as entering foreign markets, increasing export volumes, improving export readiness, or forming international partnerships. Both the extensive (whether to export) and intensive (how to export) margins are included. This category is closely related to New Customers/Markets, but here the focus is specifically on foreign markets. Examples:

- [...] create opportunities to start exporting more suitable products for the international market, while increasing production capacity.
- A significant portion of new machine's production will be exported. The projects specifically aim to counter Asian competition, [...], especially in Central European markets.
- Entry into the market will be investigated with an external market study and participation in trade fairs in Sweden, Germany, France, and Russia. The project improves the conditions for the company's growth and internationalization.
- To reduce dependence on the domestic industry, the company is undergoing a heavy product development project with internationalization efforts, focusing on the development of products for export markets.

Vertical/Horizontal Integration This category includes firms that plan to shift their position in the value chain. Most firms in our sample produce intermediate goods, and integration often means bringing tasks in-house from suppliers and subcontractors or adding functions previously handled by customers. The application must describe this shift explicitly. We also include investments related to mergers and acquisitions in this category. These mergers can be vertical or horizontal integration. Generic references to "increasing value added" are not sufficient for classification. Examples:

- After the investment, it is possible to increase work done in-house and, correspondingly, reduce the use of subcontractors.
- The machine being acquired completes a piece in one clamping. Precision improves and the complexity increases. The firm has outsourced these tasks from abroad until now.
- Subcontracting machining increases production costs and extends delivery schedules. With the acquisition, the company can be self-sufficient and even start its own subcontracting offerings.
- The company's operations are expanding from manufacturing components to becoming a supplier of complete saunas.

New Products This category covers cases where the firm plans to introduce or develop new products or services. Common points of emphasis are expanding product portfolio, entering new business areas, or moving from product plans into production. Examples:

- The project contributes to improving the company's competitiveness as it expands the product range.
- The company has decided to expand its services to special electric motors. The new business area requires the acquisition of advanced equipment and expertise.
- The new machine significantly increases production capacity, speeds up production, and diversifies the product range.
- The applicant's strategy is to create a new business area alongside traditional activities.

Quality This category includes cases where firms aim to improve product, service, or production quality. Applications often cite the goal of meeting or upgrading quality standards. We include all cases where the application clearly references quality as a stated objective. Examples:

- The quality of services does not meet the required standards for exported products. Therefore, the applicant company is investing in its own surface treatment facility. The goal is to elevate the quality of products, especially since the company aims for growth through exports.
- The new machining center ensures consistent quality and precise dimensions for the products.
- The company is upgrading its existing paint shop to a state-of-the-art one, ensuring that the quality of painting meets export market standards, and simultaneously improving the quality of all products.
- The main goal is to maintain and further develop the company's business and meet the market's quality requirements. The new laser cutter provides reliability, as well as increased quality and precision in cutting work.

Capabilities/Flexibility This category includes cases where firms seek to expand their production capabilities or improve flexibility, often in response to varied or customized client needs. Unlike the New Products category, which requires a product, service, or business area, Capabilities/Flexibility captures potential—firms signal the ability to offer more diverse, tailored, or technically complex outputs without naming a product. This distinction is especially relevant for job shops and contract manufacturers, many of which do not produce their own branded goods. In these cases, investments aim to improve versatility—for example, by enabling more complex, customized, or flexible production. Examples:

- The firm is investing in modern production technology to meet the growing demand from customers for individual, tailored products.
- The firm is investing in its sewing line for filter bags. This will make it easier to sell customized solutions to customers.
- The new machining center allows for the processing of larger and more diverse products. The machine can handle more challenging pieces, with a larger spark size and better positioning and accuracy.
- Investment in a new cutting machine is necessary as the company's current machinery does not allow for the production of complex components without specialized procedures.

Precision This category includes cases where firms adopt technology to achieve greater precision, accuracy, or tighter tolerances in production. While precision can be viewed as a component of quality, we classify it separately because the term is used with consistent and concrete meaning in the application texts. Examples:

- The investment is needed because the company's two main customers demand good dimensional accuracy from their products and adherence to delivery times.
- The goal is to acquire a fast, efficient, ergonomic, and precise machining center and a measurement tool to ensure high precision and quality of components.
- The purchase of the machine aims for more efficient, higher quality, and error-free production through new technology.
- Current customers in timber sorting demand more precise length and quality. This production needs to be promoted to reduce dependence on basic product manufacturing. Therefore, the plan is to change the current timber sorter to one that provides efficiency and, above all, precision in the sorting process.

Productivity This category includes cases where firms state an intention to improve productivity, reduce costs, or enhance efficiency. The product's features remain unchanged from the customer's perspective (aside from price, potentially). The focus is on producing more with the same or fewer inputs. Productivity must be stated explicitly as a goal. As with all categories, applications may belong to multiple groups if multiple goals are stated. Examples:

- Through the project, the company seeks significantly more efficient production, especially in welding, which has been a production bottleneck for an extended period.
- The new equipment significantly speeds up packaging operations, being approximately five times faster than the old equipment. Accelerated production increases capacity and reduces production costs.
- A highly automated tube bending machine increases efficiency and productivity. The new machine allows for growth and the creation of new jobs.
- A more spacious operating environment allows for significant improvements in material management, workflow, and productivity.

Work This category includes cases where the application links to labor-related objectives—such as reducing labor costs, altering tasks, or improving labor productivity. To avoid undercounting, we apply deliberately inclusive criteria. Mentions of workforce impacts are broadly interpreted, and we include both automation and labor-augmenting technologies, since the texts do not consistently distinguish between them. The technology may also be part of a broader reorganization, as long as the intended effect on labor costs or productivity is clear. To be included, the text must refer directly to labor—for example, by mentioning the workforce, personnel use, task changes, labor productivity, or labor cost savings. Generic references to "efficiency" or "competitiveness" are insufficient. Examples:[9]

[9] We highlight two choices we make regarding classification in this category:

1. "Automation" language is interpreted with care. In the application texts, the term does not necessarily refer to the automation of labor tasks. It often describes automation in a narrow part of the production process—such as automatic defect sensing—or refers to a machine operating autonomously. Still, where the application also refers to labor, we include it.

- The machine requires fewer personnel resources than before. The sanding machine is a new acquisition; previously, sanding was done manually. The benefits of the investment include improved production efficiency, increased quality, and expanded capacity.
- Automation in production increases significantly, transforming the company's production methods from largely manual to nearly fully automated.
- The company achieves significant cost savings with the investment compared to its previous operations, especially in personnel and energy costs.
- A new machining center will be operated and programmed by a new hire, bringing new expertise to the company and reducing dependence on labor overall. The aim of the bandsaw is to speed up lead times, reduce waste of sheet material, and lighten the workload for the saw operator.

Delivery Firms in this category aim to improve delivery. Typical cases refer to delivery speed, reliability, and logistics. The improvement must affect delivery to customers—internal logistics alone is not enough. Examples:

- The project improves the company's delivery reliability and ensures that its capacity is sufficient, as the company increasingly subcontracts to others.
- The company is acquiring a modern machining center, enabling faster throughput. The project improves the firm's technical readiness, delivery, and especially quality assurance.
- The project enhances the company's competitiveness, saves on costs, accelerates deliveries to customers, improves inventory turnover rates, and enhances product quality through a quality monitoring system.
- Delivery reliability and speed improve, strengthening the company's competitiveness.

Brand In this category, firms aim to enhance their branding, corporate image, or public perception. These goals may include improving their reputation and being perceived as advanced, innovative, or industry leaders. The intention must be stated explicitly in the application. Examples:

- Improving the perceived quality image among customers is a significant factor in the project. Overall, the investments in the project increase customer trust in production quality and significantly enhances the company's competitiveness.
- The focus is on streamlining business operations and improving the environmental awareness that can be demonstrated to customers.
- Updating the machinery is essential to attract young employees to join the company.
- Obtaining the demanding 4th class 1090-CE certification is expected to increase market share in subcontracting. The ability to provide customers with required certificates for quality is a significant advantage.

2. Labor outside the firm is excluded. Changes that reduce labor inputs in the value chain—for example, shifting a task from a supplier to an in-house automated process, or improving product quality to reduce downstream labor at the client firm—are not coded as "work."

Adaptation This category includes cases where firms respond to changes—such as new customer demands or shifts in the factor market. The change must be external from the firm's perspective: it cannot be caused by the firm or the project itself. Applications often describe the need to adjust to maintain operations, retain clients, or respond to shifting conditions. General optimism about growth or rising demand is not included. The core idea is that the investment enables the firm to adapt to circumstances outside its control. Examples:

- The new machine's features meet growing customer demands. Customer requirements have evolved, particularly for industrial customers.
- By automating processes, the company can better respond to increased international competition in terms of quality and delivery reliability.
- The project aims to develop windows and doors to comply with new and potential future energy regulations.
- Spare parts are no longer available for the old machines.

Next, we discuss the categories of technology. We begin with the overall technology category, which is also discussed in the main text and Table A3.

Technology Our main design includes only subsidies that are coded as technologies. The criterion for this classification is an explicit mention of a technology or technological advancement. Technology is broadly defined in the everyday sense of the word. Typical cases are acquiring hardware or software. The texts from the application system provide information on firms' plans because the plan is binding: firms only receive subsidies against verifiable costs mentioned in the application. Ambiguous cases, though uncommon, were cross-checked by multiple coders.

Our design captures technological changes mainly involving new machinery and equipment, reflecting the focus of the subsidy program. But not all technological advances fit this category. Our coding and the program itself exclude cases like productivity gains not linked to technology adoption, R&D, organizational changes, and technologies in new firms (since our event study needs a pre-period). We use a conservative approach, classifying a case as technology only when it clearly meets the decision rule. Non-technologies include export promotion subsidies ("starting to export in the Nordics"), R&D projects that do not involve adopting new technology ("a new rifle support device"), and starting new businesses ("starting to mine nickel ore"). Table A3, referred to in the main text, also summarizes cases that are coded as technology vs. not technology. Examples:

- A significant investment effort by a micro-enterprise in a new 5-axis linear motor machining center and the tools required for its operation, as well as CAM software.
- Acquisition and implementation of an ERP (Enterprise Resource Planning) system.
- Expansion of operations and automation with a CNC-controlled press brake.
- Modernization of production machinery and equipment.
- The company is acquiring a new production line.
- Improving productivity with a "technology leap" through the acquisition of NC-based machines.

Negative Examples:

- The conditions for the company's market entry are explored with a market survey and by participating in trade fairs.
- Hiring of expert services related to product development.
- Development of a more powerful, energy-efficient, and user-friendly product and related production methods, along with a market survey.

Machinery The Machinery category covers investments in physical technologies, and constitutes a majority of the technology subsidy applications. The key criterion is the technology being tangible (something we can touch). Typical cases include CNC machines, welding robots, laser cutters, bending presses, surface-treatment technologies, robot arms, conveyor belts, sensors, and measurement devices. Examples:

- A CNC-controlled drilling rig enables more accurate product quality, efficient production...
- The acquisition of welding robots and a painting line streamlines the production of the company's key products.
- The company is acquiring a fully automatic laser system with automatic loading.
- The roller molding line enables more cost-effective production.
- Acquisition of a CNC tube bending center.
- The new edge banding machine allows for the production of popular rounded edge trims.
- The new grinding line can process birch veneer and provide the finishing touch on pine veneer, meeting customer demands for pre-coated panels.

IT The IT (Information Technology) category includes investments in IT, software, and digital technologies, such as computers, computer programs, databases, and digital tools. Typical examples include software for enterprise resource planning (ERP), computer-aided design (CAD), production control, and warehouse management. The applications may simultaneously include machinery and IT components and belong to both categories. Examples:

- The company is acquiring a laser cutting machine and a press brake, along with the CAD/CAM software needed for control the machines.
- The new production control system allows for the management of an increasingly diverse customer and product base.
- In addition to machinery and equipment, the project includes software investments to integrate the company's production control and financial management systems.
- The company is investing in a new laser, an ERP system, a press brake, a machining center, and a press. With a need for increased capacity, the ERP system significantly improves the company's ability to respond by providing real-time information on business conditions.
- Development of the production control system.

- The project includes acquiring a modern edge banding machine and utilizing expert services for CAD/CAM program modeling and training, as well as complementary equipment acquisitions for the control of production machines.

Specific Technologies We use keywords to identify specific technologies. The application belongs to the category if the keyword is present. We also use these keywords to standardize different spellings of common technology-related words when preprocessing the text data to reduce dimensionality (see Section H).

- CNC: *cnc*
- Robot: *robo*
- Laser: *laser, plasma*
- Machining: *työstökeskus, työkeskus, koneistuskesk, monitoimisorvi, koneistus*
- Welding: *hitsau, hitsi*
- Painting: *maalaam, maalaaj, maalaus, värit, pintakäsit, pinnoitt, raepuhalt*
- Driving (logistics): *kulj, traktor, kuormaus, kuormaaj, nostolav, kulkuneuv, truk, kaivinkon*
- Hoisting (logistics): *nostolait, nostin, nosturi, vinssi, vintturi*
- Automation: *automaat, automat, automoi, automisa*

Automated We classify machinery into two broad groups: automated and non-automated. This classification is developed for both the text and customs data, building on the approach in Acemoglu and Restrepo (2022).

The automated category includes technologies that operate without active human intervention. The classification is made ex ante, based on the nature of the technology—not on observed workplace effects. In other words, we do not require that the technology replaces human tasks to be classified as automated.

This category includes robots, CNC machines, conveyor belts, automatic packaging and sorting systems, and other equipment designed to function autonomously. It also covers investments in automated production lines and cases where firms explicitly mention automation. Our approach is conservative: it must be clear from the application that the technology is automated or contains an automated component.

Examples from the text data:

- Purchase of a used profiling machine and an automatic packaging line.
- Key acquisitions include welding robot with accessories, radial drilling machine, hydraulic press, and band saw.
- Automatic blade control device for a road grader, enabling the machine to perform even more demanding work. This investment aims to raise the quality level of the offered service. Considering the increased demands from customers; not all of them can be served without automation.
- Construction of new facilities (1,200 m^2) and acquisition of a welding robot, saw, drilling machine, and computer-aided design software.

- Acquisition of a punching and laser machine integrated with a robotic press brake. This forms a cell that automatically produces components from a stack of sheets, stacked on pallets without manual intervention.
- Automation and streamlining of production across three different production lines.
- Acquisition of 3D design software, lumber pre-processing line, automatic wall panel manufacturing line, base element assembly line, and wall panel coating line.
- Increase production efficiency with the acquisition of a more automated machining center.
- Enhancing efficiency through production automation. The manufacture of aluminum doors and windows involves a significant amount of manual labor, but with this investment, production can shift more towards mass production, as already initiated with the machining center acquisition.

We complement the text data with firm-level import records from Finnish Customs, covering technologies at the 6-digit CN-code level. Our classification of automated technologies closely follows Acemoglu and Restrepo (2022) and is hand-coded to the Finnish data. We focus on imported products within CN codes 82 (tools), 84 (mechanical machinery), 85 (electrical equipment), 87 (tractors and work trucks), and 90 (instruments and apparatus). The main categories are listed below; the full product code list is available upon request.

- Industrial robots (CN Code 847950 in the Finnish data)
- Numerically controlled machines, such as CNC machines
- Automated industrial machinery (equipment used in dedicated, streamlined, or automated production processes, such as packaging, sorting, and grading machines)
- Automatic lifts and conveyors
- Automatic tools (non-hand-operated tools, for example, used in metal processing)
- Automatic welding machines
- Automatic textile machinery
- Automatic agricultural equipment
- Regulation and control instruments

Non-Automated This category includes technology investments that are not automated. In the text data, where many applications offer limited technical detail, this category functions as a residual: if a technology is not classified as automated, it is assigned to the non-automated group.

In the customs data, non-automated technologies are identified directly based on CN codes. Our classification follows the distinction proposed in Acemoglu and Restrepo (2022). These are machines or tools that are manually operated or require active human control—or at least more human input than their automated counterparts. The following broad categories are included:

- Non-automated industrial machinery (e.g., hydraulic presses or cutting machines without automation)
- Machines labeled as not numerically controlled (e.g., "grinding machines; non-NC")

- Manual tools
- Manual welding equipment
- Non-automatic conveyors and production vehicles, such as roller-tables and forklifts.
- Traditional textile machines, such as manually-operated sewing machines.
- Non-automated agricultural machinery
- Other heavy capital goods, such as pumps, freezers, refrigerators, heaters, and fans.
- Laundry equipment, including washers, dryers, and ironing machines
- Vending machines

To support the research design, we define two additional text-based categories that capture the reasons evaluators cited for rejecting applications. The administrative records from ELY Centers allow us to compare accepted and rejected applications side by side. This comparison lets us manually identify features that contributed to rejection and code them accordingly.

No Good This category includes applications with notable issues or concerns, rendering them ineligible for subsidies even under different particular circumstances. To assess whether rejected applications provided valid counterfactuals, we reviewed all rejections in the analysis sample. Our investigation identified a small set of ten applications that were clearly unsuitable. These cases involved entrepreneurs with troubling histories or firms with unstable financial positions. Our findings remain robust after excluding these applications. Examples:

- The company falls into credit rating C (credit not recommended). Financial statements have not been received, even though they were requested during the company visit. The entrepreneur arrived one hour late for the meeting. Applicant's credit record is poor, the current financial position weak, and other financing for the project is pending.
- The applicant has not provided the necessary financial statements. Based on the available information, the company's financial position, and accumulated tax arrears burden the business to an extent that granting support is not justified.
- The requested information has not been provided. The operations have been unprofitable, and the financial condition is weak, with negative equity. The project does not have a significant impact on the company's growth.

No Jobs This category includes applications that were rejected at least partly due to job creation concerns. We read rejected applications, and found that none had employment-related issues as the primary reason for rejection. While five reports mentioned employment, the primary focus was on the potentially low initial stage on technology investment, with employment being a secondary consideration. Our findings are robust to excluding these applications. Note that, at the same time, winning applications do sometimes mention positive employment effects. Examples:

- The project involves the renewal of existing assets, but it does not contribute significantly to the company's technological capabilities, job creation, or value-added. Also, the impact of the support in implementing the project is not significant, as required by regulations.
- The growth potential of the firm is uncertain, and additional jobs are not expected to be created for many years. The value added does not increase sufficiently.
- The project is small and a routine acquisition for the company, and it is not anticipated to have such effects on revenue growth or company employment that supporting the project would be justified.

F.C.2 Manual Coding

Next, we describe the manual coding process applied to the text data.

Step 1: Defining the Categories. Our text categories, developed from fieldwork and application texts, cover a range of motivations and technologies. Interpretations do not rely on a single category but reflect several measures. We describe the specific, concrete technologies and goals that firms state in their subsidy applications.

Step 2: Training the Panel. Two teams of research assistants (RAs) were trained for different classification tasks. The first team (three RAs) focused on technologies and rejection reasons, while the second team (four RAs) classified intentions. Training involved explaining categories and application contexts, discussing fieldwork observations, and how to find contextual information.

The first team and authors classified 500 random applications together. The second team and authors classified 50 applications as there were more categories to be coded. Decision rules were refined iteratively based on similarities and differences, updating category descriptions and examples during this training period. Disagreements arose mainly from cases with limited classification information. We advised a conservative approach: classify only when a category clearly fit, and avoid uncertain classifications. We discussed best practices to ensure accuracy and consistency, aiming for replicability by an external team.

Step 3: Coding the Categories. In the main classification step, RAs determined category fit in texts, recording specific parts of the text that supported the classification. Some classes essentially relied on keywords/phrases (e.g., new products, exports), while others were more nuanced (e.g., capabilities/flexibility).

The authors maintained close contact with RAs, addressing ambiguous cases and clarifying categories. RAs were encouraged to discuss cases together. Breaks were scheduled to maintain concentration and quality, with 30-minute classification sessions instructed.

G Fieldwork

We conducted fieldwork as background research to understand the changes we document at the level of specific firms and workers. We visited manufacturing plants and interviewed CEOs, technology managers, production workers, and subsidy administrators.

Firm Visits and Interviews We chose five manufacturing firms for in-depth case studies. The primary purpose of the case studies was to observe the technologies, production, and work firsthand. We spent on average four hours at each manufacturing plant observing the production and conducting interviews. We also conducted five separate firm interviews (a total of ten firms). With one firm we met twice.

Our qualitative research method was open-ended interviews, building on prior qualitative research on technologies in firms (e.g., Piore, 1979; Dertouzos et al., 1989; Piore, 2006; Berger, 2013). This method is helpful because it allows us to identify the prevalence of mechanisms we had postulated ex ante and uncover new mechanisms that we had not anticipated. We asked the firm representatives about their production, technology adoption, motivations behind adopting technologies, the observed effects, and government subsidies.

We selected the firms to be illustrative of the sample and different from each other. We visited and interviewed firms with employment from approximately 30 to 20,000 workers; subsidy winners, subsidy losers, and non-applicants; firms in rural and urban areas; privately owned and publicly traded firms; firms with high levels of own capital and firms in the corporate restructuring. All firms were in the fabricated metal product, machinery, and wood product industries.

Worker Interviews We separately interviewed five production workers using similar in-depth interviews as in our firm visits. In all interviews, we asked the respondents broadly about their work and skills, technologies they use at work, other technologies at their workplace, and the effects of technologies they had observed. Our qualitative methods draw from a long social sciences tradition to directly ask the respondents how they perceive the cause and effect. We used a semi-structured approach to interviewing that uses open-ended questions to allow a wide range of responses to emerge (see, e.g., Piore, 1979; Boyd and DeLuca, 2017; Bergman et al., 2024). We recruited the interview respondents in collaboration with the Finnish Industrial Union, the largest Finnish union representing industrial workers.

Subsidy Program Interviews and Text Data To understand the subsidy program, we interviewed (1) officers in all four main ELY Centers, (2) program administrators at the Ministry of Economic Affairs and Employment, (3) an external program auditor at the Ministry of Finance, and (4) a consulting firm that assists firms in subsidy applications (a total of 18 interviewees in seven groups). We also used text records from the administrative system of the subsidy program to track the applications and qualitatively understand how the subsidy program works.

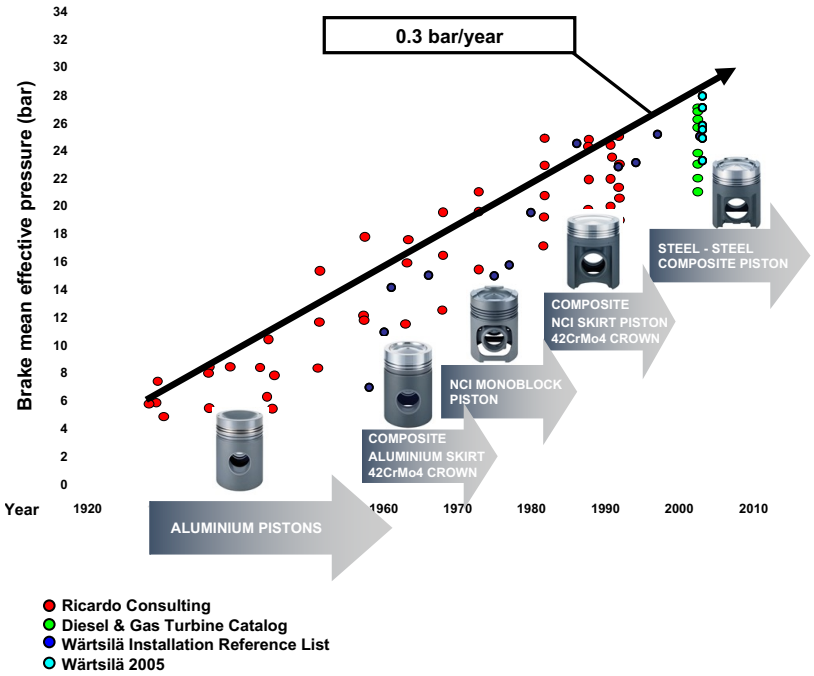
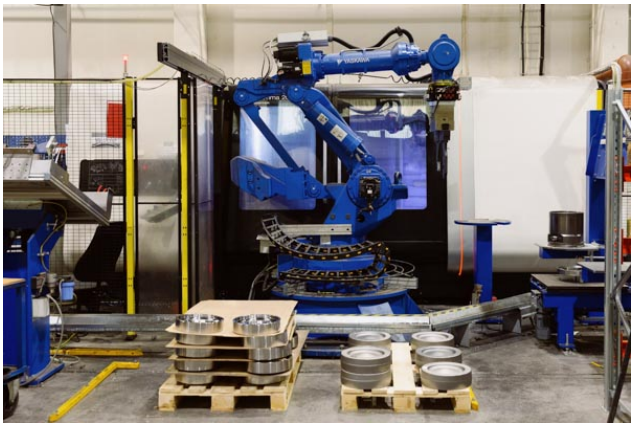
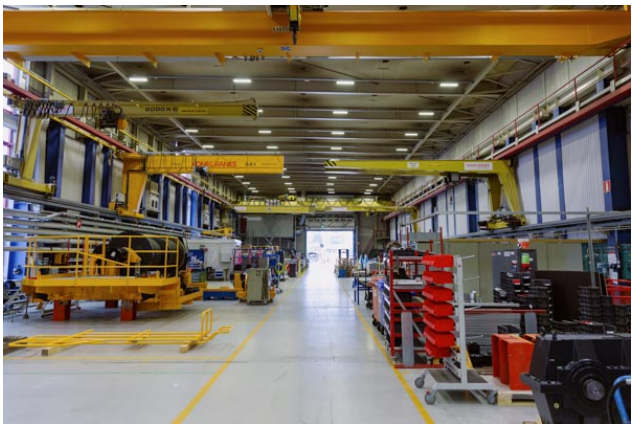


Figure G1: Moore's Law for Pistons.

Notes: The development trend of piston materials over the last 100 years. Back to Section II.



(A) A CNC Machine and a Robot.



(B) Inside an Industrial Manufacturing Plant.



(C) Machine Operators and a Milling Machine.

Figure G2: Fieldwork: Documenting the Context.

Notes: Photographs from fieldwork. Back to Section III.

H Text Processing

We describe the text processing for classification and matching.

Pre-Processing[10] We apply similar preprocessing steps both to the application summary texts (used in classification) and the evaluation texts (used in text matching). These steps are:

1. All text is converted to lowercase.
2. Non-letter symbols are removed.
3. Common "stop words" are removed using the Finnish corpus in the Natural Language Toolkit (Bird et al., 2009).
4. Words are returned to their base form (also known as lemmatization) using Voikko. Voikko performs a variety of NLP preprocessing tasks for Finnish text (`https://voikko.puimula.org/`).
5. Single-character words are removed, as there are none in Finnish.
6. Words indicating firm type are removed (such as "Oy", which translates to "LLC").

We apply the following two final steps to the application summary texts, but not to the evaluation texts. The reason for this is simple: the two additional steps are helpful in removing details from the text that are unrelated to classification, but the same details can be important in predicting the approval decision.

7. Countries, cities, municipalities, and publicly available firm names are changed to generalized versions. The generalized versions of the words are inside the symbols "<" and ">". For example, the word "Helsinki" (the capital of Finland) is changed to "<City>". This reduces dimensionality in the data, as a mention of a foreign country, for example, receives a one variable in the bag-of-words representation instead of one for each country.
8. Different versions of words associated with technology are replaced with generalized versions of those words. This is mainly to generalize compound words, which are common in Finnish. For example, after prior preprocessing steps, the word "hitsausrobotti" (welding robot in English) is a distinct word from both "welding" and "robot". After the last preprocessing step, the word is replaced with "<Weld><Robot>" to capture its similarity to the words "<Robot>" and "<Weld>".

Word Vectors We use Word Vectors (a case of word embedding) for the text matching. Word Vectors are an increasingly popular method of transforming text into numerical form to use in natural language processing tasks, as they have been shown to outperform other text presentation models in multiple different applications (Pennington et al., 2014). Word vectors represent individual words as high-dimensional vectors, and the similarity of different words can therefore be measured as the distance of their respective vectors using a suitable metric. They are also capable of capturing word semantics, something that simpler transformation methods are unable to do.

We construct word vectors using FastText (Bojanowski et al., 2016). FastText builds on the approach of Mikolov et al. (2013) which creates vector representations of words by predicting "context words" (e.g. words appearing before or after a given word). A key feature that makes FastText attractive for our purposes is its

[10] All text-related data work is done using Python 3.7.

subword approach to building word vectors: the model creates vector representations from combinations of characters that also appear inside words. This allows the model to better capture the semantic similarity of different forms of the same word. For example, the words "technology" and "technological" often have similar meaning in context, but simpler models treat them as entirely separate words and therefore require sufficient training data on both forms to estimate each vector accurately. FastText helps overcome this limitation by constructing vectors for shared character n-grams in both words, such as "technolog," which contribute to the representations of all words containing those sequences. Hence, words that share character sequences capturing much of their semantic content tend to have similar vector representations. This aspect is especially important in morphologically rich languages such as Finnish, where various case suffixes are common (Finnish generally does not use prepositions in the same way, but instead alters word forms to express grammatical meaning). We train the model using the skip-gram method.

In our application, the words appearing in the evaluation texts (i.e., in our corpus) are first transformed into 100-dimensional word vectors. We highlight the fact that we use the corpus of subsidy application texts to train the model, rather than using pre-trained models of the Finnish dictionary, for example. The reason for this is that words appearing in the subsidy application texts are likely to hold different semantic meanings than the same words in more general contexts. After constructing the initial word vectors, each of them is weighted by the term frequency-inverse document frequency (TF-IDF). Finally, another 100-dimensional vector is built for each application text by taking the average over each of the TF-IDF weighted word vectors in the text, leaving us with one 100-dimensional "sentence vector" for each application (and firm). We use these sentence vectors to build propensity score measures and match recipients to non-recipients with replacement.

Text Matching The text matching procedure is outlined in the main text. We use the CalibratedClassifierCV estimator in the scikit-learn library to calibrate a linear SGDClassifier with modified Huber loss on the word vectors. The calibration uses Platt scaling (sigmoid method) on the classifier scores with five-fold cross-validation (Zhang et al., 2002). The sentence vectors are used as features and the model outputs the estimated probability of the application being successful (i.e. probability of treatment assignment). We calibrate the propensity scores with all possible applications, including exports and R&D. The propensity scores are robust to fully out-of-sample calibration but less precise. Standard errors and confidence intervals for the calibration results are obtained via bootstrap resampling.

The propensity score summarizes the evaluation text into a single number between 0 and 1, reflecting the likelihood of winning a grant. Because we construct the propensity scores from evaluation reports, they capture critical information managers use when deciding whether to award the subsidy. Recall that all applications in our sample have gone through pre-screening and thus have a realistic opportunity to win a grant. Adjusting for the scores then aims to control for factors that are not directly observable (as numbers) or known ex ante but predict the acceptance decision and potentially the outcomes of interest.

Researchers often face a trade-off between prediction accuracy and model interpretability when using machine learning (James et al., 2021). Here text matching aims to create accurately predicted probabilities. When using word vectors, there are limited methods to pinpoint the text features driving specific propensity scores—an issue that ongoing research (e.g., Ludwig and Mullainathan, 2024) is beginning to address. However, we can build intuition by returning to the text data: investigating specific cases, and systematically analyzing features that predict winning the grant. Examining the text data also complements our interviews, allowing us to construct a more comprehensive understanding of the program.

Table H1 shows examples of successful and unsuccessful applications with identical propensity scores. The text data are confidential: These cases are translated, anonymized, and chosen to be representative, but

we caution against overinterpreting specific cases.

Examining application descriptions (not used for matching), we find that applications with the same propensity scores are relatively similar. The descriptions typically list technologies and some contextual details. For example, the winning and losing cases include "investments include a flatbed laser, welding robot, deep-drawing equipment, etc." and replacing "old machinery with a new computer-controlled engraver." Descriptions are also relatively similar across different propensity scores.

Analyzing evaluation texts (used for matching), we find that evaluations with similar propensity scores are similar, but those with different propensity scores are different. At the higher end, evaluation texts are consistently positive. For example, the 0.94 winning case details: "the flatbed laser enhances the competitiveness [...]" The 0.94 losing case similarly states: "The acquisition of a new engraving machine will speed up delivery times and increase quality, delivery reliability, and capacity." At the lower end, winning and losing applications often discuss the project's pros and cons. For example, a 0.50 winning case describes that while the project is modest, it could still be eligible because it could be a prerequisite for further investments.

More systematically analyzing these patterns, Figure H1 documents that words reflecting the project's significance ('significant' and 'high-quality') and technical terms ('statement' and 'report') predict subsidy approval. Conversely, words linked to rejection cover the same themes in negative terms (for example, 'minor,' 'conventional,' and (likely missing) 'information'). Reassuringly, the word 'approve' strongly predicts grant approval, and 'negative' predicts rejection. Longer texts weakly correlate with higher approval rates, but text length does not predict propensity scores, as shown in Table H4.

Table H2 displays decision texts that, while not factored into propensity score calculations, provide information about grant applications. Positive decisions often echo their evaluation reports. For instance, a decision for an application with a 0.94 score explicitly endorses it: "The project raises the technological level of the company's production and improves its long-term competitiveness." In contrast, rejections often articulate specific deficiencies. For example, a decision rejects a 0.94 score application by stating the second-hand machine has minor value and lacks sufficient importance. For a 0.78 score, the rejection specifies the ineligibility of the industry and the preemptive start of the project as disqualifying factors exacerbated by depleted funds at that point. A decision with a 0.50 score cites industry restrictions, competition effects, grant effectiveness, and procedural flaws as reasons for application failure.

The alternative approach, cosine similarity, gives a similarity metric between two vectors. Unlike the propensity score, this method mechanically creates matches with similar description texts. We calculate this metric for each winner-loser pair in our main analysis sample using the sentence vectors. The match is 1:1 with replacement, so we keep only the matched loser firm with the highest similarity with a given winning firm. After manual inspection of the match quality, we discard all matched pairs where the similarity metric between the texts is less than 0.85, where unity reflects identical documents. The matched texts share a substantial amount of semantic content. Examples of matched text pairs:

1. 'Acquiring a CNC machine.' and 'Acquiring a CNC milling machine.' Similarity: 0.992.

2. 'Purchase of CAD-Cam software.' and 'Financial management software, computer, fax.' Similarity: 0.940.

3. 'Expanding operation. Constructing new facilities and acquiring machinery and equipment.' and 'Production hall construction and machinery and equipment investment.' Similarity: 0.854.

The cosine similarity approach matches applications on the short version of the description texts available in the text data instead of the longer evaluation reports (the first sentence in the description columns in Table H1).

ML-Classification We next explain how we detect the technology subsidies using ML classification. After the pre-processing, and equipped with our training data, we turn to scikit-learn (Pedregosa et al., 2011) to perform the classification into technology vs. non-technology subsidies.

We first transform the texts into a Bag of Words (BoW) representation, where each application text corresponds to a vector of the length of the corpus (containing all the words appearing in any texts). Then, the corresponding indices of the vector mark the number of occurrences of each word appearing in the application's text. The vectors are then transformed using term frequency-inverse document frequency (TF-IDF) weights (Salton and Buckley, 1988). The general idea of TF-IDF is to give higher weights to informative words appearing often within an application text.

These weighted vectors are finally used in training a support vector machine (SVM) classifier.[11] We also performed the classification using other classifiers than SVMs, such as boosting algorithms and neural network variants. SVMs performed the best for our purposes, but the differences were relatively minor.

To cross-validate the classifier's performance, we use K-fold cross-validation with five splits. We search the grid for optimal hyperparameters in the learning rate (or alpha), the penalty function, and several other parameters used in vectorization.[12] The score to be optimized is the F_1-score, which gives equal importance to minimizing both false negatives and false positives, as neither one is more crucial in our classification problem. The optimized parameters are:[13]

1. Regularization parameter (alpha) set to 0.00001.
2. Penalty function set to elastic net.
3. Words appearing in more than 50% of application texts are removed.
4. In addition to single words, combinations of two and three words are also used as elements in the training vectors.

This training procedure attains around 90% F-score and 95% accuracy for both technology and automation classification in our out-of-sample tests, as shown in Table H3.

As an alternative coding approach, we employed k-means clustering to see whether an unsupervised learning method provides a similar classification. The unsupervised learning method with k=4 uncovers four clusters of texts related to technology, R&D, exports, and starting a new business. This observation, combined with interviews with subsidy officers, increases confidence that our supervised classification captures meaningful categories.

[11] SVMs divide the n-dimensional space of vectors (where n is equal to the length of the corpus) with a $(n-1)$-dimensional hyperplane. In the case of a binary classification problem, points on one side of the hyperplane are classified as belonging to one category, and points on the other side to the other category.

[12] These include the n-gram range and the threshold for corpus-specific stop words.

[13] All other parameters are set to default ones in the SGDClassifier estimator in the scikit-learn library. We tested optimizing other parameters as well using randomized search, but find virtually no improvements in accuracy.

Table H1: Propensity Score Text Examples: Descriptions and Evaluations.

	Description		Evaluation	
P-Score	**Winner**	**Loser**	**Winner**	**Loser**
0.94	Machine and equipment investments include a flatbed laser, welding robot, deep-drawing equipment, etc. The applicant is investing in new production equipment, with the flatbed laser (*sum*) being the most significant. Additionally, the company will purchase a robot, deep-drawing equipment, and other production tools. Modifications worth [*sum*] will be made to the facilities.	Purchase of an engraving machine. [*Firm name*] specializes in engraving products, often featuring logos and individual text sections on various materials. With a steady customer base and typically small, quick orders, the company emphasizes flexible service. They are replacing old machinery with a new computer-controlled engraver, bought used, for more efficient, higher-quality, and less error-prone production. It costs [*sum*] euros and is financed by a [*bank name*] loan, with a 15% subsidy requested. No prior national or EU subsidies were claimed on the used [*technology name*] engraving machine, as certified by the seller [*firm name*]. A new machine of similar type currently costs about [*sum*] euros.	The project signifies a shift in the company's production toward utilizing technologically more advanced production machinery. In particular, the flatbed laser enhances the competitiveness of the company's production. Additionally, the implementation of robotics can significantly improve productivity.	The acquisition of a new engraving machine will speed up delivery times and increase quality, delivery reliability, and capacity. The purchase, which includes new technology, is essential for the uninterrupted and high-quality continuation of the company's business operations. It is expected to raise the technological level of production/products.
0.78	Hall and machine investments: Hall expansion and machinery. The project includes expanding a hall by 100 square meters at a cost of [*sum*], purchasing a washing machine for [*sum*], a teflon machine for [*sum*], a vehicle for [*sum*], and a grinding machine for [*sum*]. Other acquisitions are expansion purchases, except for the grinding machine, which replaces an old machine. The application seeks support only for the price difference between the old and the new machine. The new machine's technology is significantly more advanced than before. The project is scheduled for completion in the winter of [*year*]-[*year*].	Purchase of a pipe bursting machine. The company is acquiring a pipe bursting machine along with tools and equipment for use in sewer renovations.	The company has the financial capabilities and expertise to implement the project. The project will have a positive impact on profitability and the company's finances, and the timing for its execution is well chosen. The project will improve the company's competitive position and will not disruptively affect the industry.	The project involves a machinery investment by a successful company with solid financing. The project further enhances the company's competitiveness in its sewer renovation contracting work.
0.50	Operational development: Construction of a timber warehouse, acquisition of a cutting line. The company's operational framework has remained unchanged, leading to cramped production and storage spaces due to expansion. This has complicated production, with the immediate issue being length-specific cutting needing additional capacity. Investment and development needs have been identified with the help of external experts and training programs focused on generational change. The project is divided into two phases: the first for a lumber warehouse and cutting line, and the second for a new machining hall and efficient log joinery line. The company seeks financing for the first phase, with construction by an external supplier. The project had not started when the company visit took place. No conflicts of interest are known. [Discusses the use of training programs, ongoing generational change, budget, and timeline for the two phases]	ERP system for building services engineering. The company is seeking to establish a manufacturing partnership with house factories, which requires investment in the development of their enterprise resource planning system. The application includes an offer from [*firm name*] that covers the estimation of new construction and renovation projects, the preparation of bids and contracts, integration with inventory management, and user interfaces for potential contract partners' systems.	The project improves competitiveness through the enhancement of production facilities and, to a lesser extent, product quality. The project can be considered justified because storing goods in production spaces and unsuitable temporary spaces currently hinders labor productivity, causing unnecessary movement of goods multiple times. The investment in the first phase is modest, but it is a prerequisite for implementing the second phase. In this sense, the project can be considered eligible for financing, even though facilities investments in the [*region*] region have been considered secondary investment targets. Financing combines grants, bank loans, and revenue. The plan can be considered realistic. The growth of the project comes mainly from an increase in value added. The subsidy can be considered to have an impact on the project.	From the perspective of business development, the project is favorable and could provide the company with the competitive edge needed to initiate collaboration with house factories. However, due to the competitive situation in the industry, granting support for the project is not justified.

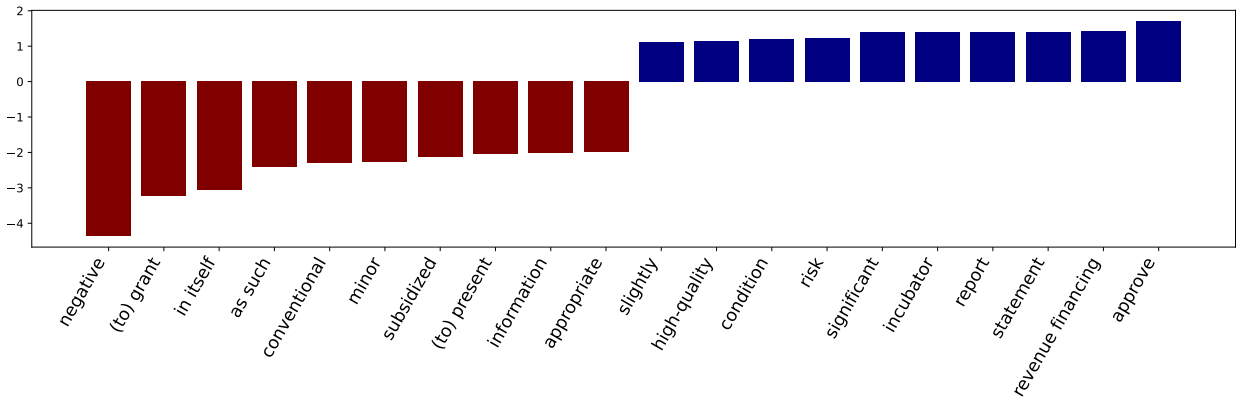
Notes: The table shows examples of description and evaluation texts of winning and losing applications with the same propensity scores. The propensity scores are estimated based on the evaluation texts. The texts have been translated to English and longer texts have been slightly shortened without removing important details. We cannot share any identifying information on individual cases, so certain parts of the texts have been replaced with generic terms in brackets. The application pairs are the same as in Table H2. Back to Sections III and IV.

Table H2: Propensity Score Text Examples: Decisions.

	Decision	
P-Score	**Winner**	**Loser**
0.94	I propose granting the applicant an investment subsidy of [*sum*]. The project raises the technological level of the company's production and improves its long-term competitiveness.	A used machine of relatively small total value, acquired to replace an older device, cannot be considered to have sufficient importance for the overall operation or growth of the company. The amount of [*sum*] euros cannot also be regarded as significant for the company's operations. The presenter of the case does not support granting the aid.
0.78	This project report replaces the previous one dated [*date*], ref. no. [*number*]. It corrects the breakdown of expenses and the granted aid according to the payment decisions (dated [*date*]).	The application is recommended to be rejected. The company in question primarily contracts sewer renovations for municipalities and cities. According to the financing guidelines of the ELY Center, development assistance for companies has not generally been granted for contracting activities due to the distorting effects on competition. The project has already started, and according to the evaluator's assessment, the subsidy would not have a significant impact on the project's implementation. Furthermore, supporting the project would not be possible as the budget is all but exhausted. [Discusses reasons for why the project is not suitable for rural funding either.]
0.50	The project improves competitiveness through the enhancement of production conditions. At this stage, the project is relatively small in size. Growth in the first phase of the investment comes from an increase in value added. Conditions for profitable operations exist. The support has an impact on the project. The level of support follows the department's guidelines.	Referring to the reasons presented in sections 5.1 and 5.2, the application is recommended for rejection. The business in question operates in contracting, and support for such projects has generally been approached with caution due to the distorting effects of grants on competition. The project is relatively small in terms of the company's operational scale and its financial resources. The company can easily implement the project without assistance. This assessment is reinforced by the fact that the company has not submitted additional information requested on [*date*] and [*date*], nor has it otherwise been in contact with the application manager, whose contact details were provided to the applicant. No response has been received to the requests, so the application has been processed with the information available (Administrative Law 434/2003, Section 33).

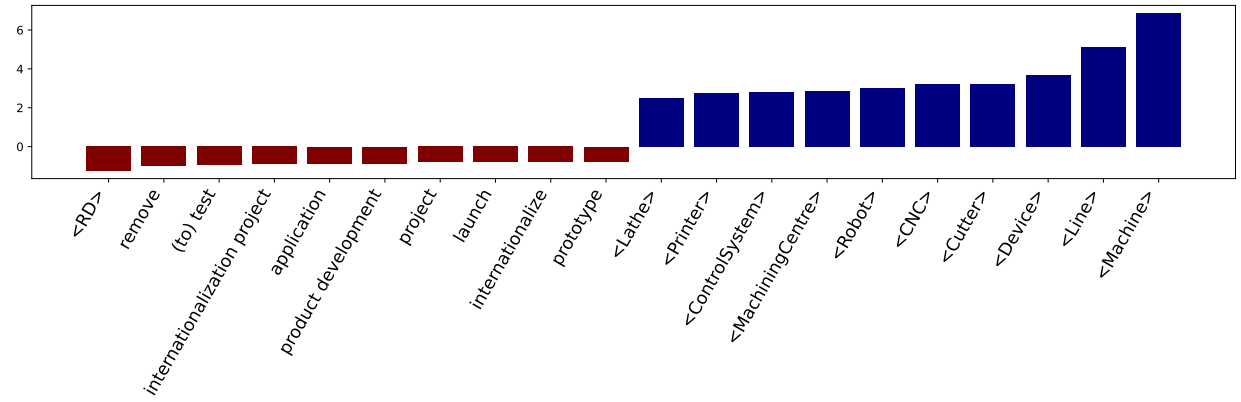
Notes: The table shows examples of decision texts of winning and losing applications with the same propensity scores. The propensity scores are estimated based on the evaluation texts, not the decisions. The texts have been translated to English and longer texts have been slightly shortened without removing important details. We cannot share any identifying information on individual cases, so certain parts of the texts have been replaced with generic terms in brackets. The application pairs are the same as in Table H1. Back to Section III.

Figure H1: Predictive Words for Winning a Subsidy in the Application Texts.



Notes: The features (words) are plotted from top and bottom linear SVM coefficients predicting treatment status in the evaluation texts of the applications. The texts are represented as TF-IDF-weighted bag-of-words vectors so that the coefficients for individual words are interpretable. The y-axis refers to the coefficient size and indicates the relative importance of each feature. Positive (negative) values indicate that the word is typically (not) associated with applications winning a subsidy. The features are translated from Finnish into English. Back to Section III.

Figure H2: Predictive Words for Technology in the Application Texts.



Notes: The features (words) are plotted from top and bottom linear SVM coefficients predicting technology in the short description texts of the applications. The texts are represented as TF-IDF-weighted bag-of-words vectors so that the coefficients for individual words are interpretable. The y-axis refers to the coefficient size and indicates the relative importance of each feature. Positive (negative) values indicate that the word is typically (not) associated with applications classified as technology. The sample contains all applications which we have hand-classified as technology vs. not technology. N = 21,699. The features are translated from Finnish into English. Features in <> refer to compound terms combining similar spelling versions of the same term. Back to Section IV.

Table H3: Predictive Accuracy: Coarse Text Category Predictions Using SVM.

Class	Precision	Recall	F1-score	Test Support	Number of Cases
Technology (1)	0.88	0.92	0.90	571	11887
Not Technology (0)	0.97	0.96	0.96	1550	31022
Accuracy			0.95	2121	42909
Balanced Accuracy			0.94	2121	42909
Macro Avg.	0.93	0.94	0.93	2121	42909
Weighted Avg.	0.95	0.95	0.95	2121	42909

Notes: Test Support refers to the 10% random out-of-sample of the applications classified by hand, from which accuracy measures are computed. The number of cases refers to the total number of subsidy applications with labels (both classified by hand and predicted). Precision is the ratio of correctly predicted positive observations to the total predicted positive observations. Recall (Sensitivity) is the ratio of correctly predicted positive observations to all observations in the category. F1 Score is the harmonic mean of Precision and Recall. Accuracy is the ratio of correctly predicted observations to the total observations. Back to Section IV and Appendix F.

Table H4: Predicting Treatment Status and Propensity Scores Using Text Lengths.

	(1)		(2)
A. Evaluation Text	Accepted		Propensity Score
Characters (K)	0.0478***	0.0386***	-0.0004
	(0.009)	(0.008)	(0.005)
B. Description Text			
Characters (K)	0.0413***	0.0309***	0.0061
	(0.008)	(0.007)	(0.004)
Propensity Score		✓	
Observations	1831	1812	1812

Standard errors in parentheses.
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Notes: The table shows the results from regressing the acceptance indicator (Column 1) and the propensity score (Column 2) on character counts in the evaluation (Panel A) and the description (Panel B) texts. The character counts are measured in thousands. There is a small positive correlation: Applications with longer description texts and evaluation reports are more likely to win a subsidy, even when controlling for propensity score. In contrast, no statistically significant correlation exists when regressing the propensity score on the lengths. Back to Section III.

I Conceptual Framework

We outline a simple framework that illustrates two types of technological change operating along two margins: *intensive* and *extensive*.[14] The intensive margin improves productivity within an existing variety, while the extensive margin introduces new varieties. This approach builds on established models (Dixit and Stiglitz, 1977; Melitz, 2003; Bustos, 2011), but applies them in a novel context. A key feature is the imperfect substitutability between varieties, which aligns with our context and allows the framework to capture different types of products.

The intuitive idea is that a firm may choose to adopt a new technology to cut costs within an existing variety or to generate a new one. We show that these distinct margins of technological change imply different observable outcomes. In practice, technologies may combine these elements or reflect entirely different mechanisms.[15] The framework demonstrates that some technological changes could enhance efficiency, while others may drive expansion through new products and services without necessarily affecting productivity—particularly in markets with differentiated products.

The key point is that the effects of technology subsidies are ex ante uncertain. Moreover, looking beyond the narrow link between technology and labor may help better understand the firm-level mechanisms of technology subsidies.

Before turning to the model, we note a few disclaimers. First, we do not model the allocation of labor inputs, so our framework does not directly address changes in employment or skill demand. Our fieldwork suggests these outcomes are often highly context-dependent and ultimately an empirical question. One contribution of this paper is to provide empirical evidence on the employment and skill effects in a context of flexible, specialized firms adopting advanced manufacturing technologies.

Second, for both types of technological advances, the model assumes zero effects on input prices, such as wages, because input prices are determined in the sectoral equilibrium, and firms are small relative to the market. While firms in many industries may possess market power on the input side, incorporating this into the model would produce wage predictions that, like the employment and skill effects, depend on specific assumptions. Since input prices lie outside our main scope, we do not incorporate such features here.

I.A Setup

Our basic setup is based on Melitz (2003) and Melitz and Redding (2014).[16] The market structure is monopolistic competition with product differentiation and increasing returns to scale at the firm level. The model specifies preferences and firm heterogeneity in a differentiated product market. This structure allows technology to have a role in creating new varieties—as in many standard growth models (e.g., Romer, 1990). We show that the view of new varieties has different implications than one emphasizing technology's role in allowing productivity improvements within a variety.

[14] Researchers describe closely related ideas as product vs. process (Utterback and Abernathy, 1975), cost vs. differentiation (Porter, 1985), automation vs. augmentation (Autor et al., 2024), secondary vs. primary (Saint-Paul, 2002), and defensive vs. enterprise (Boone, 2000).

[15] For example, using AI for increased worker monitoring may not fit neatly into either margin.

[16] We aim to introduce the simplest model necessary to explain our findings, which captures the essence of a broad class of models featuring the two types of technological change. The Melitz (2003) framework allows for a simple way of introducing imperfect substitutability between varieties. We specifically build on the version by Melitz and Redding (2014). Related approaches include Hopenhayn (1992), Ericson and Pakes (1995), Klette and Kortum (2004), Acemoglu et al. (2018), Akcigit and Kerr (2018), and Hémous and Olsen (2022).

Preferences over sectors $j \in \{0, 1, ..., J\}$ take the Cobb-Douglas form:

$$U = \sum_{j=0}^{J} \beta_j \log Q_j, \quad \sum_{j=0}^{J} \beta_j = 1, \beta_j \geq 0. \tag{6}$$

There is a continuum of differentiated varieties within each $j \geq 1$ sector, and these preferences take the Constant Elasticity of Substitution (CES) Dixit and Stiglitz (1977) form:[17]

$$Q_j = \left[\int_{\omega \in \Omega_j} q_j(\omega)^{(\sigma_j - 1)/\sigma_j} d\omega \right]^{\sigma_j/(\sigma_j - 1)}, \quad \sigma_j > 1, j \geq 1. \tag{7}$$

Sector $j = 0$ is a homogeneous numeraire good with a unit-input requirement for production.

The upper-tier Cobb-Douglas preferences imply that consumers spend $C_j = \beta_j Y$ in sector j, where Y denotes aggregate income. The lower-tier CES preferences imply that the demand for each differentiated variety within sector j is:

$$q_j(\omega) = A_j p_j(\omega)^{-\sigma_j}, \quad A_j = C_j P_j^{\sigma_j - 1}, \tag{8}$$

where P_j is the price index:

$$P_j = \left[\int_{\omega \in \Omega_j} p_j(\omega)^{1-\sigma_j} d\omega \right]^{1/(1-\sigma_j)}, \tag{9}$$

and A_j is a market demand index, determined by sector spending and the price index. There is a continuum of firms; each firm is of measure zero relative to the market, and takes A_j as given.

Firms produce varieties using a composite input X_j with unit cost w_j in sector j. They choose to supply a distinct differentiated variety. Production has a fixed cost f_j and a constant marginal cost, inversely proportional to productivity φ. The composite input needed to produce q_j units is:

$$x_j = f_j + \frac{q_j}{\varphi}. \tag{10}$$

We focus on the equilibrium within a sector (and drop the sector j subscript for clarity). The firms choose their prices to maximize profits subject to a residual demand curve with constant elasticity σ. The equilibrium price for each variety is a constant markup over marginal cost derived from the first-order condition for profit maximization:

$$p(\varphi) = \frac{\sigma}{\sigma - 1} \frac{w}{\varphi}. \tag{11}$$

This gives the equilibrium firm revenue:

$$r(\varphi) = Ap(\varphi)^{1-\sigma} = A \left(\frac{\sigma - 1}{\sigma} \right)^{\sigma - 1} w^{1-\sigma} \varphi^{\sigma - 1}, \tag{12}$$

and the equilibrium firm profit:

$$\pi(\varphi) = \frac{r(\varphi)}{\sigma} - wf = B\varphi^{\sigma - 1} - wf, \quad B = \frac{(\sigma - 1)^{\sigma - 1}}{\sigma^{\sigma}} w^{1-\sigma} A. \tag{13}$$

[17] This representation has two interpretations: (1) consumers demand differentiated consumption goods with "love-for-variety" preferences (e.g., Grossman and Helpman, 1991), or (2) final-good firms demand differentiated intermediate inputs, and a greater variety of inputs increases the "division of labor" (e.g., Romer, 1987, 1990). In our context, the typical firm is an intermediate-good producing firm that sells its output to final-good producing firms.

I.B Intensive Margin

Intensive margin advances improve firms' productivity within a variety: This allows firms to produce the same product more efficiently. The change is within the production function.[18]

We introduce intensive margin advances as in Bustos (2011). The firm has a constant marginal cost $1/\varphi$ within a variety. It can adopt a technology T_I that reduces that cost. This choice is a tradeoff between a fixed cost f_I and a productivity increase to $\iota\varphi$, where $\iota > 1$. The resulting total cost functions are:

$$x = \begin{cases} f + \frac{q}{\varphi} & \text{if } T_I = 0 \\ f + f_I + \frac{q}{\iota\varphi} & \text{if } T_I = 1. \end{cases} \tag{14}$$

This type of technology adoption is characterized by sorting according to firm productivity: There is a productivity cutoff φ_I^* above which the firm adopts the technology because the adoption choice involves a tradeoff between a fixed cost and a scaled productivity increase. We focus on predictions conditional on the firm finding it optimal to adopt an intensive margin technology.[19]

Firms with lower marginal costs produce more and earn higher revenues. The CES demand structure predicts that the relative outputs and revenues of firms depend on their relative productivities:

$$\frac{q(\varphi_1)}{q(\varphi_2)} = \left(\frac{\varphi_1}{\varphi_2}\right)^{\sigma}, \quad \frac{r(\varphi_1)}{r(\varphi_2)} = \left(\frac{\varphi_1}{\varphi_2}\right)^{\sigma-1}, \quad \varphi_1, \varphi_2 > 0 \tag{15}$$

Lower marginal costs imply higher revenue-based productivity because of the fixed production cost:

$$\frac{r(\varphi)}{x(\varphi)} = \frac{w\sigma}{\sigma - 1}\left[1 - \frac{f}{x(\varphi)}\right], \tag{16}$$

where input use $x(\varphi)$ is increasing in φ.

Since the price of the composite input is fixed, an increase in revenue-based productivity also implies a higher profit margin. Firms with lower marginal costs also earn higher profits. As shown earlier:

$$\pi(\varphi) = \frac{r(\varphi)}{\sigma} - wf = B\varphi^{\sigma-1} - wf, \quad B = \frac{(\sigma-1)^{\sigma-1}}{\sigma^{\sigma}} w^{1-\sigma} A. \tag{17}$$

Firms with lower costs charge lower prices because the equilibrium price for each variety is a constant markup over marginal cost:

$$p(\varphi) = \frac{\sigma}{\sigma - 1}\frac{w}{\varphi}. \tag{18}$$

In summary, technological change on the intensive margin predicts increases in revenue, productivity, and profit margin. The intuitive idea is that firms with lower marginal costs produce more and earn higher revenues due to the CES demand structure. Lower marginal costs imply higher measured productivity and profits due to the increasing returns to scale. The firms sell more with lower prices.

[18] The process efficiency motive is present in the models of specialization (Smith, 1776), labor-saving technologies (Marx, 1867), routine-replacement (Autor, Levy and Murnane, 2003), tasks (Acemoglu and Autor, 2011), automation (Acemoglu and Restrepo, 2018b), product and process (Utterback and Abernathy, 1975), and in the Schumpeterian models (Grossman and Helpman, 1991; Aghion and Howitt, 1992).

[19] If the productivity terms (φ) are correlated across products for a firm (as in Bernard et al., 2011), intensive-margin improvements may lead to product changes. The idea is that the intensive-margin innovation then reduces marginal costs in other products and makes firms more willing to pay the per-product fixed cost for a new product. This correlation between products effectively blurs the line between intensive and extensive margin advances.

Before presenting the model's predictions for technological changes on the extensive margin, we clarify the distinction between two directions of change on the product side: vertical (within the same variety) versus horizontal (a new, imperfectly substitutable variety). In this class of models, vertical cost reductions and quality improvements within the same variety are essentially equivalent. In other words, the productivity term can be interpreted in terms of cost or within-variety quality; these interpretations are isomorphic to a change in units of account (Kugler and Verhoogen, 2012). This implies that technological changes increasing quality within a variety have the same predictions as cost-reducing changes, except for their effect on prices—prices decrease if the technology is cost-reducing and increase if it is quality-improving. In practice, distinguishing between vertical and horizontal quality changes is often difficult.

I.C Extensive Margin

Advances on the extensive margin enable expansion through the introduction of new varieties, allowing firms to produce new products. Unlike intensive margin changes, which occur within the production function, extensive margin changes affect the product-market side. A defining characteristic of this type of technological change is that outputs with different types are imperfect substitutes. This dynamic was frequently highlighted by firms during our interviews.[20] [21]

We introduce the extensive margin advances by adapting from Melitz (2003). The firm can introduce a new variety by adopting a technology T_E. The technology requires a fixed entry cost f_E. Potential entrants to the new variety, both existing and new firms, face uncertainty about their productivity in the new variety. After the firm pays the entry cost, it observes its productivity φ for the new variety, drawn from a distribution $g(\varphi)$, with cumulative distribution $G(\varphi)$. The firm then decides whether to produce or exit the project. Melitz (2003) shows this decision is characterized by a cutoff productivity φ_E^* where the firm makes zero profits:

$$\pi\left(\varphi_E^*\right) = \frac{r\left(\varphi_E^*\right)}{\sigma} - wf = B\left(\varphi_E^*\right)^{\sigma-1} - wf = 0. \tag{19}$$

In equilibrium, the expected ex ante net profits equal zero due to free entry:

$$\int_0^\infty \pi(\varphi) dG(\varphi) = \int_{\varphi_E^*}^\infty \left[B\varphi^{\sigma-1} - wf\right] dG(\varphi) = wf_E. \tag{20}$$

Figure I1 visualizes the relationship between profits and productivity. Firms with $\varphi < \varphi_E^*$ would lose if they produced. They exit the project, receive $\pi(\varphi) = 0$ in that new variety, and cannot recover their entry cost. The subset of the firms that produce and have $\pi(\varphi) > wf_E$ make positive profits after the entry cost.

Firms that draw a high enough productivity after adopting an extensive margin technology introduce a new variety. This leads them to produce more and earn higher revenues. For clarity, we consider a simplified case of two products, conditional on producing the new variety ($\varphi \geq \varphi_E^*$):

$$q = \begin{cases} q\left(\varphi_0\right) \\ q\left(\varphi_0\right) + E\left[q\left(\varphi\right) \mid \varphi \geq \varphi_E^*\right] \end{cases} \qquad r = \begin{cases} r\left(\varphi_0\right) & \text{if } T_E = 0 \\ r\left(\varphi_0\right) + E\left[r\left(\varphi\right) \mid \varphi \geq \varphi_E^*\right] & \text{if } T_E = 1, \end{cases} \tag{21}$$

[20] A new variety has several interpretations: a new product, a quality change not perfectly substitutable with quantity, re-purposing production to respond to changing demand, expansion to new markets, capturing a larger share of the value chain, etc. A new variety may be the same product but with an improved process that provides more reliable scheduling or a faster response time to orders, changing the aspects customers receive.

[21] The expansion of variety in consumer and intermediate goods plays a central role in many theoretical models of growth (Romer, 1990; Grossman and Helpman, 1991). The product view is closely related to Porter (1985): gaining competitive advantage through a quality-differentiation strategy instead of a cost-leadership strategy.

and use more of the composite input:

$$x = \begin{cases} f + \frac{q(\varphi_0)}{\varphi_0} & \text{if } T_E = 0 \\ f + \frac{q(\varphi_0)}{\varphi_0} + f_E + f + E\left[\frac{q(\varphi)}{\varphi} \mid \varphi \geq \varphi_E^*\right] & \text{if } T_E = 1. \end{cases} \tag{22}$$

Intuitively, extensive-margin adoption widens scope rather than raises efficiency: it increases revenue and input use, but—holding per-variety technology fixed—does not shift the per-variety r/x mapping. The effect on firm-level measured productivity is therefore ambiguous, depending on whether the new variety's revenue-per-input exceeds that of existing varieties. By free entry, the expected net profit from the additional variety (net of the entry cost) is competed away ex ante. Holding sectoral demand A_j fixed, firms expand either by lowering marginal cost within a variety (intensive margin) or by adding a new variety (extensive margin); adding a variety is akin to adding a parallel product line whose scale and productivity depend on its productivity draw.

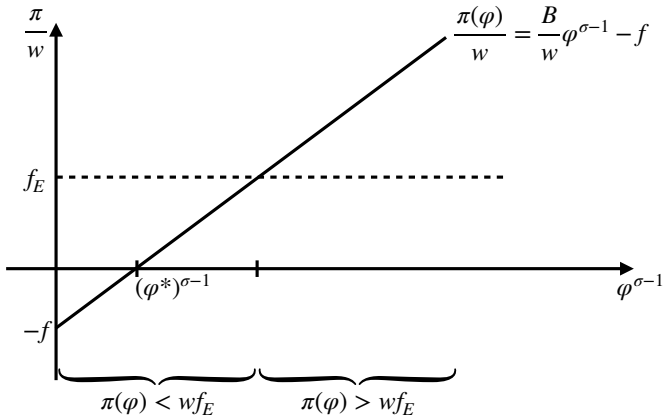


Figure I1: The Productivity Cutoff for the Extensive-Margin Advances.

Notes: Illustration of the productivity cutoff after introducing a new product variety. Firms that draw productivities above the cutoff make positive profits, while firms below the cutoff find it optimal to exit and incur a loss equal to the entry cost. In equilibrium, the ex ante expected profits are zero due to free entry. Adapted from Melitz and Redding (2014). Back to Section I.

J Dynamic Model

To clarify the source of variation in our identification strategies, we consider the forces that influence a firm's technology adoption and its factor demand. We proceed in two steps. In Step 1, we focus on the firm's technology-adoption decision. In Step 2, we consider the firm's conditional factor demand, treating the technology as a quasi-fixed factor; the idea is to show that we can trace the implications of the technology adoption problem for factors' relative demand. The framework is general to allow for the analysis of multiple types of technologies and factor inputs. The adoption model is adapted from Cooper et al. (1999).

J.A Step 1: Technology Adoption

In Step 1, we model the general technology-adoption problem of an individual firm. In the model, the firm makes the discrete choice between replacing existing technology with a new technology or continuing to use the old technology for another period. Consider a firm i that maximizes:

$$E_0 \sum_{t=0}^{\infty} B_t Y_t^i \tag{23}$$

subject to:

$$Y_t^i = A_t^i \theta_t^i F\left(T_t^i; X_t^i\right) - D_t^i \Theta_t^i \tag{24}$$

$$T_{t+1}^i = \begin{cases} (1-\delta) T_t^i & \text{if } D_t^i = 0 \\ \tau_{t+1}^i & \text{if } D_t^i = 1, \end{cases} \tag{25}$$

where $\tau_{t+1}^i = \mu_t^i \tau_t^i$ and $\mu_t^i \geq 1$ is the rate of exogenous technological progress.[22] The choice variable in this problem is D_t^i where $D_t^i = 1$ if the new technology T is adopted in period t.

The first equation (23) is the firm's objective function. The firm maximizes the discounted present value of profits, which are defined as output minus the adjustment costs. The discount rate is $B_t \in (0,1)$.

The second equation (24) describes the production process and the adjustment costs. The function $F(\cdot)$ is increasing and concave in the level of technology. The output also depends on the state of productivity A_t^i. We assume that A follows a first-order Markov process $\Phi(A'|A)$. The model has two types of adoption costs. The first is a fixed adjustment cost (Θ_t^i). If the firm adopts the new technology ($D_t^i = 1$), it has to incur a cost Θ_t^i. It reflects the direct cost of the technology, such as its installation costs and other fixed adjustment costs. We assume that Θ_t^i is i.i.d. The second is the opportunity cost that is proportional to the production volume. It is characterized by θ_t^i that equals $\lambda_t^i \leq 1$ during an adoption period and 1 otherwise.[23] The intuition is that investment temporarily diverts resources away from production.

The third equation (25) describes the time path of the given technology. The technology frontier is τ_t. The firm's actual technology that is in-use is T_t^i. The in-use technology is typically less productive than the latest version because technology depreciates at an exogenous rate δ and because the latest technologies improve at rate μ_t^i. The firm can decide to adopt the latest version of the technology ($D_t^i = 1$); in that case its technology will be equal to τ_{t+1}^i in the next period. The gains to adoption reflect both technological progress (μ_t^i) and the rate of depreciation (δ).

[22] We allow the technological progress to contain an idiosyncratic and a deterministic common component to clarify the potential mechanisms. That is, we assume $\mu_t^i = \mu_t + \varepsilon_t^i$, where $\varepsilon_t^i \geq 0$.

[23] This implies that adjustment costs are heterogeneous across firms even if $\lambda_t^i = \lambda < 1$, i.e., equal for all firms i and periods t.

Under this framework, the firm's technology adoption reflects several forces:

1. Replacement cycle: The underlying deterministic replacement cycle—driven by depreciation (or aging) of capital δ and the exogenous technological progress μ_t—will imply that the older vintage of the capital, the more likely is replacement.

2. Shocks to technologies' costs: Idiosyncratic shocks to costs Θ_t^i affect the investment in a straightforward way: lowering the costs and increasing the likelihood of the investment.

3. Shocks to technological progress: Idiosyncratic shocks to technological progress, that is, shocks to μ_t^i via ε_t^i, increase the benefits from the technology investment and increase the likelihood of the investment.

4. Shocks to productivity: The response of investment to A_t^i depends on both the nature of the adjustment costs (λ_t^i and Θ_t^i) and the persistence of the shock ($\Phi(A'|A)$). The firm would prefer to replace machinery during a period where inputs are not very productive (reflecting $\lambda_t^i < 1$) and would also prefer to have a new machine available when productivity is high. To build intuition, suppose that adjustment costs are fixed. If A is i.i.d., investment is independent of A. But if a shock to A is informative of similar shocks in the future, then the investment is more likely when A is high—the firm invests now to benefit from the high productivity in the future.

Proofs and further details are in Cooper et al. (1999).

In words, two forces determine a technology's productivity: the technology's age and a shock to total factor productivity. Given the state of productivity, the producer compares the discounted expected benefits of more productive technology relative to the current adoption costs. The gain to adoption is that a new version of the technology is more productive as it reflects some aspects of technological progress. There are two types of costs for replacement. First is the direct loss of output associated with the acquisition and installation of new capital goods. Second is that the process of installing the new machinery and retraining workers reduces productivity in the firm. The nature of the adjustment costs and the structure of the stochastic process governing the shocks jointly determine adoption timing.

The model assumes that small adjustments of technologies are either infeasible or undesirable. In particular, many technology-investment projects (e.g., the purchase of large machinery) are not possible in small quantities. In addition, the model assumes that the costs of adjusting the technologies stock may be nonconvex. Consequently, at the firm or plant level, we may see periods of low technology investment activity followed by bursts of investment activity, i.e., investment spikes. Empirical observations support this view of technology adoption: We find that a significant fraction of technology investment activity at the firm level is associated with large variations in the technology stock, i.e., technology investment is typically a lumpy activity.

J.B Step 2: Conditional Factor Demand

In Step 2, we consider the firm's conditional factor demand, treating the technology as a quasi-fixed factor. This approach is closely related to the work by Berman et al. (1994), who treat machinery investments as quasi-fixed and invoke Shephard's lemma to justify their empirical specification. Cost-function estimates with quasi-fixed capital trace back to Caves et al. (1981). Our aim is to trace the implications of the technology adoption problem for factors' relative demand. The intuition is that technology is relatively more costly to adjust than labor.[24]

[24] Hamermesh (1989) analyzes the costs firms face in adjusting labor demand to exogenous shocks. The study argues that adjustment costs could be viewed as fixed and documents that labor adjustment tends to be lumpy.

The firm's production function is written as:

$$Y = F(T; X), \tag{26}$$

where T is the technology of our focus and X is a vector of multiple other factors. An element X_i is the quantity of factor i used in the production of a quantity Y of output. We assume F is strictly increasing with each of its arguments and strictly concave. We denote the relative price of factor i by $p_i > 0$. For the purposes of this analysis, these relative prices reflect potential relative productivity effects from technology T. The conditional factor demands are characterized as solutions to the cost-minimization function:

$$\min_{(X_1 \ldots X_n)} \sum_{i=1}^{n} p_i X_i \quad \text{subject to} \quad F(T; X_1, ..., X_n) > Y. \tag{27}$$

The minimum value of the total cost is the cost function $C(p_1, ..., p_n, Y)$. Under this framework, it satisfies the standard properties of a cost function. It is increasing, homogeneous of degree 1, and concave in $(p_1, ..., p_n)$, and it satisfies the Shephard's lemma.

The Shephard's lemma gives us an analytical tool to interpret the relationship between factor demands and their prices. It states that:

$$\bar{X}_i = C_{p_i}(p_1, ..., p_n, Y), \tag{28}$$

where $\bar{X}_i$ denotes the factor demand for the factor X_i and C_{p_i} denotes the partial derivative of the cost function C with respect to price p_i. In other words, the cost function says that the conditional factor demands can be characterized through a shock to the price vector $(p_1, ..., p_n)$.

The expression (28) allows us to provide a theoretical basis for analyzing the effects of technology adoption on the demand for different types of labor. In this framework, technology's effect on labor demand is translated through its effect of the (potentially unobserved) prices of labor, which reflect the productivity of labor combined with the technology. For example, complementarity between technology and skills would mean that technology T would change the price vector $(p_1, ..., p_n)$ in a way that the factor demands $\bar{X}_i$ would shift toward high-skill labor $X_H \in X$.